

DIVISION 21 - FIRE SUPPRESSION INDEX

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SECTION 21 05 00 - COMMON WORK RESULTS FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 PREFACE

- A. The Work shall be performed using the General Contracting (GC) / Single-Prime Contractor project delivery method and the Design-Bid-Build process. References to Work to be performed by different Trades Contractors or Subcontractors are all subject to the Scope of Work by the Prime Contractor. Any conflicts regarding Scope of Work shall be directed to the Prime Contractor for resolution. THE PRIME CONTRACTOR FOR THIS PROJECT SHALL ALSO BE KNOWN AS THE LEAD CONTRACTOR.
- B. Terminology: The terms below will be used throughout Division 21, in accordance with Division 00 and Division 01.
1. Owner: Eureka Apostolic Christian Home
 - a. Contracting Authority: [REDACTED]
 - b. Using Agency (Facility): [REDACTED]
 - c. Owner's Agent: [REDACTED]
 2. Architect/Engineer (A/E): pH7 / Roger D. Fields Associates, Inc.
 3. Prime Contractor / Lead Contractor: [REDACTED]
 4. Contractors:
 - a. All Trade Contractors listed below, other than the Lead Contractor, are Subcontractors.
 - b. General Trades: Refer to Divisions 02 through 14.
 - c. Fire Suppression: Refer to Division 21.
 - d. Plumbing: Refer to Division 22.
 - e. HVAC: Refer to Division 23.
 - f. Electrical: Refer to Division 26.
 - g. Communications: Refer to Division 27.
 - h. Electronic Safety and Security: Refer to Division 28.
 - i. Site: Refer to Divisions 31 through 35.
 5. Authority Having Jurisdiction (AHJ) mentioned in Division 21:
 - a. Eureka-Goodfield Fire Department.

delete?

- b. City of Eureka Building Department.
- c. Woodford County Health Department.
- d. ~~State of Illinois Department of Commerce.~~
Illinois Department of Public Health

C. Scope of Work Summary for the Contractor Performing Fire Suppression Work on this Project:

- 1. Provide fire suppression systems noted in the project documents.

1.02 RELATED DOCUMENTS

- A. Refer to the additional Divisions below for specific requirements, responsibilities, and methods relating to Division 21 Work. If there is a conflict, comply with the more restrictive requirements.
 - 1. Division 00, "Procurement and Contracting Requirements": All sections.
 - a. The "General Conditions" will provide additional information on the Contractor's Scope of Work responsibilities.
 - 2. Division 01, "General Requirements": All sections.
- B. Read the entire Specification covering other branches of the Work. Contractor shall be held responsible for the coordination of its work with the work performed by other Trades.

1.03 DESCRIPTION

- A. Fire Suppression Work includes the Work described in Division 21, "Fire Suppression," Sections 21 00 00 through 21 99 99 (as included), unless otherwise noted.
- B. Fire Suppression Work also includes the Work described in the following Sections:
 - 1. Division 10, "Specialties," **Section 10 44 00**, "Fire Protection Specialties."
 - 2. Division 23, "Heating, Ventilating, and Air Conditioning," Section 23 38 13, "Commercial-Kitchen Hoods."
 - 3. Division 33, "Utilities," **Section 33 14 16**, "Site Water Utility Distribution Piping."
- C. In the Sections listed above, the term "Contractor" shall mean the Contractor performing Fire Suppression Work on this Project, unless otherwise noted.
- D. Fire Suppression Work includes, but is not limited to, the following:
 - 1. Unit Prices, Allowances, or Alternates, as shown, if any.
 - 2. Fire extinguishers and fire extinguisher cabinets. **Refer to Section 10 44 00.**
 - 3. Selective demolition.

4. Concrete work.
5. Underground fire-suppression systems.
6. Water-based fire-suppression systems:
 - a. Hydraulic design and pipe sizing.
 - b. Specialties
 - c. Standpipes.
 - d. Wet-pipe sprinkler system.
 - e. Dry-pipe sprinkler system.
7. Fire-extinguishing systems:
 - a. Gas-based fire-extinguishing system.
 - b. Dry-chemical fire-extinguishing system.
8. Commercial-kitchen hoods: Fire-suppression system. **Refer to Section 23 38 13.**
9. Site water utility distribution piping: Site fire-suppression work. **Refer to Section 33 14 16.**

~~E. Work Not Included:~~

- ~~1. The Owner will provide the following fire-suppression work directly or through a separately negotiated Contract:~~
 - ~~a. ~~
2. Site work: Other Contractor(s) shall be responsible for all site work not listed above, including the following work (includes all work from point of connection to existing utilities):
 - a. Combined fire-suppression / domestic water service to inside the building to 12 in. above the floor.
Make agree with P001, C301, FP501 detail 1, and P701
 - b. Underground fire-suppression water service mains.
 - c. Thrust blocking.
 - d. Backflow preventer.
 - e. Fire hydrants.
 - f. Fire line loop.
 - g. Metering.
 - ~~h. Post indicator valves with supervisory switches.~~
 - ~~i. Protection posts for fire hydrants.~~

*Need to make agree with
Civil C301 and P701.*

- j. Siamese connection. coordinate with C301, FP501, and FP001
- k. ~~_____~~

3. General Trades Work shall include the following work:

- a. Fire extinguishers and fire extinguisher cabinets.
- b. ~~Fire alarm valve enclosure.~~
- c. ~~_____~~

4. ~~Plumbing Work shall include the following work:~~

a. ~~_____~~

wet pipe according to 21
13 13

?

5. HVAC Work shall include the following work:

a. ~~_____~~

6. Electrical Work shall include the following work:

- a. Alarm and tamper switch wiring.
- b. Wiring of equipment and devices.
- c. ~~_____~~

Water heater flue
and outside air int

- F. Contractor is responsible for the more restrictive requirement between these general provisions and other requirements contained elsewhere in the Contract Documents.
- G. Furnish material, labor, tools, accessories, and equipment to complete and leave ready for operation all fire-suppression systems of this Project as described in these Specifications and as shown on the Drawings.
- H. Use sufficient skilled workers and competent supervisors in the execution of this portion of the Work to ensure proper and adequate installation throughout.
- I. Refer to Section 21 05 00.31, "Project Coordination for Fire Suppression," regarding final electrical connections.
- J. Work Sequence:
1. The Work shall be conducted in one phase.
- K. All fire-suppression systems shall be provided as complete and operational, and shall not be considered complete unless, and until, at a minimum, all Code and Governing Agency requirements are satisfied.

1.04 COORDINATION MEETINGS AND COORDINATION DRAWINGS

A. Terminology:

1. Coordination Meetings: Coordination Meetings shall be held to identify Coordination Areas and the need for a Coordination Drawing for any Coordination Area. The Lead Contractor shall schedule and conduct the Coordination Meetings.
2. Coordination Areas: Areas of the project that require special attention, i.e., where there are potential conflicts between Trades.
3. Coordination Drawings: Drawings and electronic files prepared by the Coordination Participants to demonstrate how multiple-system and interdisciplinary work shall be coordinated.
4. Coordination Participants: All Contractors shall participate in the process of developing the Coordination drawings, including, at a minimum, the following Trades:
 - a. General Trades.
 - b. Fire Suppression.
 - c. Plumbing.
 - d. HVAC.
 - e. Electrical.
 - f. Communications.
 - g. Electronic Safety and Security.

B. Refer to Section 23 05 00, "Common Work Results for HVAC," for additional requirements.

1.05 ADDITIONAL WORK CONSIDERATIONS

A. Available water flow:

1. Water flow test data was obtained from a private testing company, performed on fire hydrants located on Cruger Avenue on 2/25/2026 that indicated 66 PSI static and 64 PSI residual at 493 GPM. This unconfirmed data is rough and is presented only for the Contractor's general information.
2. Contractor shall certify that test results are correct and acceptable before performing pipe-sizing calculations, shall independently verify water flow data, and shall conduct its own testing if required.

B. Sprinkler Layout Drawings and Hydraulic Calculations:

1. Before any work is started, prepare dimensioned sprinkler layout drawings showing new pipe runs and sizes, sprinkler locations, valves, and accessories. Show luminaires, smoke detectors, speakers, diffusers/grilles, heating equipment, and

structural elements in light background on the reflected ceiling plans, with sprinklers and piping shown in dark/heavy lines or symbols. Verify the final hazard classification for each area even though the documents may call for certain hazard classifications.

2. Review complete building sprinkler system and correct any Code violations. Inform the Owner and A/E of any violations. Rework sprinkler system in remodeled areas to meet Code and function requirements.
3. Provide sprinkler coverage for all areas of the building, whether or not that area is indicated on the Drawings.
4. Submit (2) copies of sprinkler layout drawings and hydraulic calculations for review by the A/E.
5. Submit kitchen hood and duct fire-suppression layout drawings showing tank(s), piping locations, and remote manual pull station locations.
6. Resubmit layout drawings if changes are made as a result of the A/E's review of the layout drawings.
7. After receiving initial review by the A/E, submit (2) copies of sprinkler layout drawings and hydraulic calculations, signed by a State-Certified Fire Suppression Designer, for final approval by the AHJ.
8. Submit sprinkler layout drawings to affected Contractors for their review and coordination before fabricating or installing system.
9. Coordinate all changes in sprinkler layout with all other Contractors.
10. Submit a final copy of the approved sprinkler layout drawings and calculations to the A/E for record purposes.

1.06 QUALITY ASSURANCE

- A. Standards: Refer to individual Specification Sections for additional referenced standards. ~~If a standard is undated, the referenced edition or most recent version of the standard applies.~~
- B. Standards: The following is a list of referenced standards. Not all of these standards necessarily apply to this Project. Refer to individual Specification Sections. ~~If a standard is undated, the referenced edition or most recent version of the standard applies.~~
 1. American Institute of Architects Committee on Architecture for Health (AIA/CAH), Health Guidelines Revision Committee (HGRC): 2014 Guidelines for Design and Construction of Hospitals and Outpatient Facilities.
 2. American Concrete Institute (ACI).
 3. American National Standards Institute (ANSI).
 4. American Society for Testing and Materials (ASTM).

5. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
 - a. Standard 62.1, ~~Current Enforced Edition~~, "Ventilation for Acceptable Indoor Air Quality."
 - b. Standard 90.1, ~~Current Enforced Edition~~, "Energy Standard for Buildings Except Low-Rise Residential Buildings."
6. American Society of Mechanical Engineers (ASME):
 - a. Boiler and Pressure Vessel Code (BPVC), Section IX, "Welding and Brazing Qualifications."
7. Americans with Disabilities Act (ADA), 2010, Accessibility Guidelines, Code of Federal Regulations (CFR), 36 CFR 1191.
8. National Electrical Manufacturers Association (NEMA):
9. National Fire Protection Association (NFPA):
 - a. 2023 NFPA 70, "National Electrical Code® (NEC)."
 - b. 2021 NFPA 99, "Health Care Facilities Code."
10. The Joint Commission.
- ~~11. Underwriters Laboratories, Inc. (UL).~~
12. United States Department of Labor, 29 CFR 1910.7, Occupational Safety and Health Administration (OSHA), Part 1926, "Safety and Health Regulations for Construction."
13. United States Environmental Protection Agency (EPA).
- ~~14. United States Green Building Council (USGBC).~~
15. State of Illinois:
 - a. Illinois Administrative Code (IAC):
 - 1) IAC Title 35: Environmental Protection.
 - 2) IAC Title 41: Fire Protection.
 - 3) IAC Title 71: Public Buildings, Facilities, and Real Properties.
 - 4) IAC Title 77: Public Health.
 - 5) IAC Title 83: Public Utilities.

END OF SECTION 21 05 00

SECTION 21 05 00.31 - PROJECT COORDINATION FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 26, "Electrical."
- B. Section 26 05 00.31, "Project Coordination for Electrical."

1.02 DESCRIPTION

- A. Provide Fire Suppression Work associated with equipment being provided by other Trades.
- B. Electrical Contractor shall provide all line-voltage wiring, and all control, interlock, and power wiring and conduit, regardless of voltage, required to power, control, monitor, or interlock fire-suppression equipment, unless noted otherwise.
- C. Coordinate all wiring requirements for Fire Suppression Work with the Electrical Work.
- D. Electrical Contractor shall provide power wiring, rough-in, and final connection, for each piece of fire-suppression equipment requiring power or other wiring.
- E. Alternative Manufacturer Electrical Requirements: Contractor furnishing the equipment shall compensate the Electrical Contractor for the cost of any additional labor and materials required to wire and hook up equipment other than the Design Base Model and Manufacturer.
- F. All electrical equipment shall be UL-listed and shall be furnished in accordance with Division 26 requirements.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. National Fire Protection Association (NFPA): 2023 NFPA 70, "National Electrical Code® (NEC)."
 - 2. Underwriters Laboratories, Inc. (UL).

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.01 PREPARATION

- A. Consult Drawings that may affect location of equipment, conduit, or wiring, and make minor adjustments in locations to secure coordination.

3.02 INSTALLATION

- A. Installation shall comply with NEC and shall be inspected and approved by the AHJ.
- B. Assist in the check-out and start-up operations for fire-suppression equipment.
- C. Fire Suppression Contractor is responsible for the final operation of all equipment furnished under its Work.
- D. Provide the Electrical Contractor with a full detailed set of all required wiring diagrams for equipment furnished under the Fire Suppression Work.

END OF SECTION 21 05 00.31

SECTION 21 05 00.33 - SHOP DRAWING SUBMITTALS FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."
- B. Section 21 05 00, "Common Work Results for Fire Suppression."

1.02 DESCRIPTION

- A. Submittal dates:
 - 1. Within (14) calendar days after the Award of Contract, submit complete materials list of all items proposed to be provided on this Project.
 - 2. Submit shop drawings within (28) days after the Award of Contract.
- B. Materials and equipment installed under the Fire Suppression Work shall meet all the requirements of the Contract Documents. Do not order or install materials or equipment until submittals are reviewed by the A/E.
- C. Submit shop drawings for the installation of the Fire Suppression Work for review by the A/E before any Fire Suppression Work is installed, enlarged, or extended. Under NO circumstances shall any Fire Suppression Work be performed prior to receiving shop drawings reviewed by the A/E.
- D. The A/E's review is ONLY for conformance with the design concept of the Project and with the Contract Documents.
- E. Catalog data for equipment reviewed by the A/E shall not take precedence over the requirements of the Contract Documents. Review by the A/E shall not relieve the Contractor from the responsibility for deviations from Drawings or Specifications, nor from the responsibility for providing proper clearance and coordination with other Contractors.
- F. Color Selections: The A/E shall approve all color selections for fire suppression equipment.

1.03 SHOP DRAWINGS

- A. Submit shop drawings for each manufactured item of equipment and all components to be used in the Work. Refer to the Shop Drawing Log attached to this Section and to the Sections listed in Section 21 05 00, Article 1.03, for items.

- B. Submit shop drawings electronically in Portable Document Format (PDF). Submit number of copies indicated in Division 01. Contractor shall be responsible for making all copies for distribution and file.
- C. Include the following information:
 - 1. Complete original PDF or first-generation scanned PDF copies of the Manufacturer's catalog sheets, brochures, diagrams, schedules, performance charts, illustrations, and other standard descriptive product data.
 - 2. Brand name.
 - 3. Catalog number.
 - 4. Specific performance data.
 - 5. Material description.
 - 6. Characteristics.
 - 7. Ratings.
 - 8. Capacities.
 - 9. Working pressure.
 - 10. Dimensional data.
 - 11. Material gauge or thickness.
 - 12. Wiring diagrams and control diagrams.
 - 13. Arrangement of component parts, physical dimensions, clearances, materials, service requirements, colors (where required), controls, and accessories.
- D. Clearly mark or highlight each shop drawing copy to identify pertinent materials, products, models, and colors (where appropriate).
- E. Coordinate each submittal with requirements of the Work and the Contract Documents.
- F. When (2) or more items of the same material or equipment are required, they shall be of the same Manufacturer.
- G. Completely review shop drawings, product data, and samples prior to submission.
- H. When submitted for review, all shop drawings shall bear the Contractor's stamped and signed certification of the following:
 - 1. Contractor has reviewed, checked, and approved the shop drawings.
 - 2. Contractor has coordinated shop drawings with the requirements of the Project and with the provisions of the Contract Documents.

3. Contractor has ensured that the equipment shown on the shop drawings meets all the requirements of the Contract Documents.
4. Contractor has determined and verified all of the following:
 - a. Materials.
 - b. Field measurements.
 - c. Field construction criteria.
 - d. Catalog numbers and similar data.
 - e. Conformance with Specifications.
 - f. Hands of equipment.
 - g. Quantities and sizes.
- I. UNSIGNED COPIES WILL NOT BE REVIEWED.
- J. Notify the A/E, in writing, at the time of submission, of any deviations in the submittals from the requirements of the Contract Documents.
- K. Provide the following certifications information:
 1. Test Agency results verifying capacities, operating conditions, and power requirements at design conditions.
 2. Manufacturer's statement of compliance with Standards discussed in individual Specification Sections.
 3. Equipment labels indicating Certification requirements.
 4. Quality standard designations on each unit piece, e.g., each pipe length, pressure vessel, or valve.
 5. Typed verification that noted mixes, chemical compositions, and testing procedures were complied with.
 6. Other Certifications listed in other Sections of the Specifications.
- L. Contractor shall make submittals promptly in accordance with the approved schedule, and in such sequence as to cause no delay either in its Work or in the work of any other Contractor.
- M. If a submittal is rejected, immediately make any corrections or changes required by the A/E. Resubmit until accepted.
- N. If the Contractor orders equipment or materials, or begins installation, fabrication, or work prior to the return of approved submittals, it shall be "at the Contractor's own risk."

- O. Incorporate Shop Drawings into the Operating and Maintenance Manuals.

1.04 REQUIRED SUBMITTAL INFORMATION

- A. The form attached to this section contains a list of required submittals. This list may not be a complete list of all required submittals. Submit for approval all items to be provided, whether listed or not.

END OF SECTION 21 05 00.33

SHOP DRAWING LOG FOR FIRE SUPPRESSION

PROJECT NAME: EUREKA APOSTOLIC CHRISTIAN HOME

PROJECT NUMBER: RDFA #25059, PH7 #2025-11

Roger D. Fields Associates, Inc.
580 N. 4th ST., Suite 240
Columbus, Ohio 43215
614-451-2248

SUBMITTAL KEY

A	Catalog Cuts/Shop Drawings	G	Completed start-up test forms
B	Installation instructions	H	Coordination Drawings / Studies
C	Color samples	I	Calculations
D	Product samples	J	Schedule
E	Typed statement of material to be furnished	K	Sample of finish
F	Typed verification of compliance with certification requirements		

STATUS KEY

APP	Approved (conforms)
CAN	Corrections Are Noted (conforms as noted)
ICN	Incorporate Corrections Noted
R&R	Revise & Resubmit
R	Rejected

Fire Suppression Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Division 10										
Fire Extinguisher Cabinets	10 44 00	A								
Fire Extinguishers	10 44 00	A								
Division 21										
Certificate of Fire Suppression Approval from AHJ	21 05 00	F								
Fire Suppression / Sprinkler Layout Drawings	21 05 00	A								
Hydraulic Calculations	21 05 00	A								
Reflected Ceiling Plans and Sections	21 05 00	A								
Access Panels	21 05 00.73	A								
Valves	21 05 23	A								
Valves, Fire Pump Suction Control	21 05 23	A								
Valves, Wall Post Indicator	21 05 23	A								
Concrete Tests	21 05 29	A								
Hangers, Equipment	21 05 29	A								
Hangers, Piping	21 05 29	A								
Pipe-Coding Materials	21 05 53	D								
Valve-Tagging Chart	21 05 53	D								
Valve-Tagging Materials	21 05 53	D								
Automatic Ball Drip	21 10 00	A								
Automatic Drain Device	21 10 00	A								
Backflow Preventer	21 10 00	A								
Backflow Preventer Test Reports	21 10 00	F								

Fire Suppression Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Certificate of Completion of Water Line Chlorination	21 10 00	F								
Drains	21 10 00	A								
Electric Bell	21 10 00	A								
Fire Department (Siamese) Connection	21 10 00	A	-	-	-	-	-	-	-	-
Fire Department Connection	21 10 00	A								
Fire Hose Rack	21 10 00	A	-	-	-	-	-	-	-	-
Fire Hose Reel	21 10 00	A	-	-	-	-	-	-	-	-
Floor Control Valve Sectional Test and Drain Device	21 10 00	A								
Gauges, Pressure	21 10 00	A								
Hydrants, Roof	21 10 00	A								
Identification Signs	21 10 00	A								
Inspector's Test and Drain Assembly	21 10 00	A								
Pipe and Fittings	21 10 00	E								
Switches, Pressure Alarm	21 10 00	A								
Switches, Valve Supervisory	21 10 00	A								
Switches, Water Flow	21 10 00	A								
Water Motor Gong	21 10 00	A								
Fire Hose Cabinets	21 12 00	A								
Fire Valve Cabinets	21 12 00	A								
Contractor's Material and Test Certificate	21 13 13	F								
Escutcheon Plates	21 13 13	A								
Flexible Sprinkler Drops	21 13 13	A								
Sprinkler Cabinet	21 13 13	A								
Sprinklers, Wet-Pipe	21 13 13	A								
Valves, Alarm Check	21 13 13	A								
Valves, Riser Check	21 13 13	A								
Air Compressor, Dry-Pipe System	21 13 16	A								
Compressed Air Piping Test Report	21 13 16	F								
Contractor's Material and Test Certificate	21 13 16	F								
Escutcheon Plates	21 13 16	A								
Sprinklers, Dry-Pipe	21 13 16	A								
Valves, Dry-Pipe Alarm	21 13 16	A								
Air Compressor, Preaction System	21 13 19	A								
Compressed Air Piping Test Report	21 13 19	F								
Contractor's Material and Test Certificate	21 13 19	F								
Division 23										

Fire Suppression Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Kitchen Hood and Duct Fire Suppression System	23 38 13	A								
<i>Division 33</i>										
Automatic Ball Drip	33 14 16	A								
Fire Department (Siamese) Connection	33 14 16	A								
Hydrants, Fire	33 14 16	A								
Meter	33 14 16	A								
Meter Pit	33 14 16	A								
Strainers	33 14 16	A								
Sump Pump	33 14 16	A								
Switches, Valve Supervisory	33 14 16	A								
Valves, Post Indicator	33 14 16	A								
Valves, Watch	33 14 16	A								

SECTION 21 05 00.35 - SPECIAL PROCEDURES FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. ~~Section 21 05 00.81~~, "Indoor Air Quality Requirements for Fire Suppression."
- B. Section 21 05 43, "Underground Fire Suppression Systems."

1.02 QUALITY ASSURANCE

- A. Standard: Occupational Safety and Health Administration (OSHA), including 29 CFR 1910.147.

1.03 SAFETY

- A. Contractor shall be solely responsible for safety precautions and programs in connection with the Work. The Owner or A/E will not be responsible for the Contractor's failure to employ proper safety procedures.
- B. Maintain the Project in accordance with Federal, State, and Local Safety and Insurance Standards. All procedures shall comply with the latest OSHA regulations.
- C. Contractor shall provide head, eye, and ear protection for each of its employees in accordance with OSHA regulations.
- D. Exercise precaution for the protection of persons and property.
 - 1. Provide guard rails, barricades, enclosures, canopies, passageways, lanterns, warning lights, and other protective safety devices required by the AHJ, and required to protect persons and property against accidentally dropped materials or other construction hazards.
 - 2. Provide guard lights at all barricades, railings, obstructions in the streets or sidewalks, trenches, or pits.
- E. Provide protection as may be required to prevent glass breakage. Replace broken glass at no cost to the Owner.
- F. DO NOT USE POWDER-ACTUATED ANCHORS OR FASTENERS.
- G. Excavation Safety: Refer to Section 21 05 43.

1.04 ENVIRONMENTAL PROCEDURES

A. Hazards Control:

1. Do not dispose of volatile wastes, such as mineral spirits, oil, or paint thinner, in storm or sanitary drains. Dispose of such wastes properly.
2. Store volatile wastes in covered metal containers and remove from premises daily.
3. Prevent accumulation of waste that creates hazardous conditions.
4. Provide adequate ventilation during use of volatile or noxious substances.

END OF SECTION 21 05 00.35

SECTION 21 05 00.40 - QUALITY REQUIREMENTS FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide Work of the highest quality, complete in every respect and conforming to the accepted practices and standards of the Trades involved.
- B. Further definition of quality is given by reference to various Laws, Codes, Standards, and Regulations.

1.02 REGULATORY REQUIREMENTS

- A. All work on this Project shall conform to the currently enforced Edition of all applicable National, Federal, State, and Local Laws, Rules, Regulations, Standards, and Building Codes, and any local ordinances and requirements pertaining to this work, unless otherwise noted.
- B. Any Law, Regulation, Standard, or Code referred to in other Sections of Division 21 is included in its entirety as a part of these Specifications.
- C. Give notice and comply with all Laws, Rules, Regulations, Ordinances, and lawful orders of the AHJ bearing on the performance of the Fire Suppression Work.
 - 1. If the Contractor observes that any of the Contract Documents are at variance therewith in any respect, it shall promptly notify the A/E in writing, and any necessary changes will be arranged by the A/E.
 - 2. If the Contractor performs any Work knowing it to be contrary to such Laws, Rules, Regulations, and Ordinances and fails to give prior notice to the A/E, the Contractor shall assume full responsibility for, and shall bear all costs associated with, correcting the Work.
- D. Only a Contractor and craftsmen licensed by the AHJ shall provide this Fire Suppression Work.
- E. Methods and materials shall be certified where noted in the individual Specification Sections.
- F. All equipment and appliances installed on this Project shall bear the label of an Approved Testing Agency and shall be installed in accordance with the Manufacturer's instructions for the labeled equipment and appliances.

Don't know a specific reference, but this shouldn't be on the fire contractor anyways

- G. All structural steel used on this Project shall be manufactured in the United States, in accordance with the Illinois Administrative Code, unless specifically approved otherwise by the Owner.

1.03 APPLICABLE PERMITS, FEES, BONDS, LICENSES, AND CERTIFICATIONS

- A. The A/E will submit all Contract Drawings and Specifications and the plan approval application to the AHJ, pay the application fees, and obtain plan approval.
- B. Contractor shall secure and pay for any ADDITIONAL permits, governmental fees, bonds, and licenses required for the proper execution and completion of the Fire Suppression Work.
- C. Contractor shall pay for all other fees and other charges related to Fire Suppression Work and payable to Utility Companies or Code Enforcement Agencies.

1.04 REFERENCES

- A. Titles, Abbreviations, and Symbols: Titles and abbreviations may be used in these Specifications. Abbreviations may be shown on the Fire Suppression Drawings. Refer to the list of abbreviations attached to this Section. Refer also to the symbols lists shown on the Drawings for further abbreviations. All titles and abbreviations may not necessarily apply to this Work.
- B. Definitions:
1. Provide: To furnish, install, and connect to make completely ready for regular operation.
 2. Furnish: To supply or deliver to site complete with all required accessories and installation instructions.
 3. Install: To mount, erect, hang, or fasten in place, and connect to make ready for regular operation.
 4. Concealed: Either embedded in masonry or other construction, or installed below floor slab, behind wall furring, within chases or soffits, within walls, within double partitions, above ceilings, in trenches, in tunnels, or within crawl spaces.
 5. Exposed: In full or partial view; not "Concealed" as defined above.
 6. Accessible Ceiling: Lay-in ceiling with removable ceiling tiles.
 7. Piping: Pipe, fitting, flanges, valves, controls, specialties, hangers, bracing, insulation, and other items required or necessary.
 8. Refer to additional Definitions in Division 01 and in the AHJ's Building and Mechanical Codes.

TITLES, ABBREVIATIONS, AND SYMBOLS

#	Number	ANSI	American National Standards Institute
&	And		
∠	Angle	APPROX.	Approximately
@	At	ARCH.	Architect
Ø	Diameter	ARR'T.	Arrangement
Ø	Round OR Phase	ASA	Acoustical Society of America
A.	Compressed air	ASHRAE	American Society of Heating, Refrigerating, and Air Conditioning Engineers
A.D.	Access door OR Area drain		
A.F.D.	Acid floor drain	ASME	American Society of Mechanical Engineers
A.F.F.	Above finished floor	ASPE	American Society of Plumbing Engineers
A.P.	Access panel		
A.W.C.O.	Acid wall cleanout	ASSE	American Society of Sanitary Engineering
A/E	Architect/Engineer		
AABC	Associated Air Balance Council	ASTM	American Society for Testing & Materials
AB.	Above	AUTO.	Automatic
ACI	American Concrete Institute	AV	Acid vent
ACPA	American Concrete Pipe Association	AW	Acid waste
ADA-AG	Americans with Disabilities Act — Accessibility Guidelines	AWG	American wire gauge
ADC	Air Diffusion Council	AWS	American Welding Society
AFCI	Arc-fault circuit interrupter	AWWA	American Water Works Association
AH	Air handler	B.D.D.	Backdraft damper
AHJ	Authority Having Jurisdiction	B.T.	Bathtub
AHRI	Air-Conditioning, Heating, and Refrigeration Institute	B.V.	Backwater valve
ALT.	Alternate	BAL.	Balancing
ALUM.	Aluminum	BFP	Backflow preventer
AMCA	Air Movement and Control Association	BHP	Brake horsepower
amp	ampere	BICSI	Building Industry Consulting Services International
		BLDG.	Building
		BSBD.RAD.	Baseboard radiation

BSMT.	Basement	CSI	Construction Specifications Institute
BTM.	Bottom	CT	Current transformer
BTU	British thermal unit	CU IN.	Cubic inch
c.	Conduit	CU. FT.	Cubic feet
C.B.	Catch basin	CUH	Cabinet unit heater
C.I.	Cast iron	Cx	Commissioning Agent
C.O.	Clean out	D.	Deep
C/C	Center to center	D.F.	Drinking fountain
C/L	Center line	D.L.	Door louver
CAB.	Cabinet	D.M.	Damper motor
CAP.	Capacity	D.S.	Downspout
CEIL./CLG.	Ceiling	D-B	Design-Build OR Design-Builder
CFC	Chlorofluorocarbon	DB	Dry bulb
CFH	Cubic feet per hour	DBL.	Double
CFM	Cubic feet per minute	DCW	Domestic cold water
CFR	Code of Federal Regulations	DET.	Detail
CGA	Compressed Gas Association	DHW	Domestic hot water
CHEM.	Chemical	DHWR	Domestic hot water return
CISPI	Cast Iron Soil Pipe Institute	DIA. / Ø	Diameter
CMA	Construction Manager Agent/Advisor	DIFF.	Diffuser
CMR	Construction Manager at Risk	DIM.	Dimension
COL.	Column	DISCH.	Discharge
COMB.	Combination	DN.	Down
CONC.	Concrete	DOE	United States Department of Energy
COND.	Condensate OR Condenser	DPR.	Damper
CONN.	Connection OR Connect	DR.	Door
CONST.	Construction	DW	Distilled water
CONTR.	Contractor	DWG.	Drawing
CONV.	Convactor OR Converter	DWH	Domestic water heater
COORD.	Coordinate	DWV	Drain, waste and vent
CSA	CSA Group (formerly the Canadian Standards Association).	E.A.T.	Entering air temperature
		E.C.	Electrical Contractor

E.S.P.	External static pressure	FDA	Food and Drug Administration
E.T.	Expansion tank	FHA	Federal Housing Administration
E.W.T.	Entering water temperature	FIG.	Figure
EA.	Each	FIN.	Finish
EFF.	Efficiency	FIN.RAD.	Finned radiation
EIA	Electronic Industries Alliance	FLEX.	Flexible
ELEC.	Electric OR Electrical	FLR. / FL.	Floor
ELEM.	Element	FMG	Factory Mutual Global
ELEV.	Elevation	FOR	Fuel oil return
ENGR.	Engineer	FOS	Fuel oil supply
ENT.	Entering	FPC / FSC	Fire Protection / Fire Suppression Contractor
EPA	Environmental Protection Agency	FPM	Feet per minute
EQUIP.	Equipment	FT.	Feet
ETL	Electrical Testing Laboratories	FT.HD.	Feet of head
EW	Eye wash	FURN.	Furnish(ed)
EWC	Electric water cooler		
EX.	Existing	G	Gas
EXH.	Exhaust	G.C.	General Contracting OR General Contractor
EXIST.	Existing	G.I.	Grease interceptor
EXT.	Exterior	G.T.C.	General Trades Contractor
F & T	Float and thermostatic	gal.	Gallon
F.	Fan OR Fire	GALV.	Galvanized
F.D.	Fire damper	GEN.	General
F.E.	Fire extinguisher	GFI / GFCI	Ground-fault circuit interrupter
F.E.C.	Fire extinguisher cabinet OR Food Service Equipment Contractor	GPM	Gallons per minute
F.H.	Fire hydrant	GR	Grille
F.H.C.	Fire hose cabinet	GRAV.	Gravity
F.H.E.C.	Fire hose/extinguisher cabinet	Grd.	Ground
F.V.	Flush valve	H./HT.	Height
FCC	Federal Communications Commission	H.P.	High pressure
FD	Floor drain	H.S.	Hair strainer
		H'STAT	Humidistat

HAC	Heating and air conditioning	INT.	Interior
HB	Hose bibb	INV.	Invert
HCFC	Hydrochlorofluorocarbon	INV. ELEV.	Invert elevation
HHS	United States Department of Health and Human Services	IPS	Iron pipe size
HORIZ.	Horizontal	ISO	International Organization for Standardization
HP	Horsepower	J.R.	Janitor's receptor
HPC	High pressure steam condensate	JCAH	Joint Commission for Accreditation of Hospitals
HPS	High pressure steam		
hr.	Hour	kVA	Kilovolt-ampere
HTG.	Heating	L.	Length
HTR.	Heater	L.A.T.	Leaving air temperature
HUD	United States Department of Housing and Urban Development	L.W.T.	Leaving water temperature
HV	Heating and ventilating OR High velocity	LAB	Laboratory
HVAC	Heating, ventilating, and air conditioning	LAV	Lavatory
		lb.	pound
		LEED	Leadership in Energy and Environmental Design
I.D.	Inside diameter	LH	Left hand
IAQ	Indoor air quality	LP	Liquefied petroleum
IBC	International Building Code	LPC	Low pressure steam condensate
ICC	International Code Council	LPS	Low pressure steam
ICEA	Insulated Cable Engineers Association, Inc.	LV'G.	Leaving
IDF	Intermediate distribution frame	M.A.	Mixed air
IEC	International Electrotechnical Commission	M.O.	Motor operated
IEEE	Institute of Electrical and Electronics Engineers, Inc.	MAN. DPR.	Manual damper
IESNA	Illuminating Engineering Society of North America	MAT.	Material
IN.	Inside OR Inches	MAX.	Maximum
IND.	Indirect	MBH	1,000 British thermal units/hour
IND. U.	Induction unit	MECH.	Mechanical
INSUL.	Insulation	MERV	Minimum efficiency reporting value
		MET./MTL.	Metal
		MEZZ.	Mezzanine

MFR.	Manufacturer	NO. OR #	Number
MH	Manhole	NOM.	Nominal
MIN.	Minimum	NP	Not permitted
MISC.	Miscellaneous	NPT	National Pipe Thread Tapered
MP	Multiple Prime	NSF	National Sanitation Foundation
MPS	Medium pressure steam	NSPE	National Society of Professional Engineers
MS	Motor starter		
MSS	Manufacturers Standardization Society of the Valve and Fittings Industry Inc.	O.	Oxygen
		O.A.	Outside air
		O.D.	Outside diameter
MTD./MT.	Mounted OR Mount	O.I.	Oil interceptor
N.I.C.	Not in contract	O.V.	Outlet velocity
N.T.S.	Not to scale	O/C	On center
N2	Nitrogen	OAC	Ohio Administrative Code
N2O	Nitrous oxide	ODH	Ohio Department of Health
NADCA	National Air Duct Cleaners Association	ODOE	Ohio Department of Energy
NAECA	National Appliance Energy Conservation Act	ODOT	Ohio Department of Transportation
NB	National Board of Boiler and Pressure Vessel Inspectors	OPG.	Opening
NCPI	National Clay Pipe Institute	OPP.	Opposite
NEBB	National Environmental Balancing Bureau	ORC	Ohio Revised Code
NEC	National Electrical Code	OSHA	Occupational Safety and Health Administration
NECA	National Electrical Contractors Association	OU./OZ.	Ounce
NEMA	National Electrical Manufacturers Association	P.D.	Pressure drop
NESHAPs	National Emissions Standards for Hazardous Air Pollutants	P.I.V.	Post indicator valve
NFPA	National Fire Protection Association	P/L	Property line
NIOSH	National Institute for Occupational Safety and Health	PDI	Plumbing & Drainage Institute
NIST	National Institute of Standards and Technology	PEI	Petroleum Equipment Institute
		PLBG.	Plumbing
		PNEU.	Pneumatic
		PRESS.	Pressure
		PROP.	Propeller
		PRV	Pressure-reducing valve

PSF	Pounds per square foot	SAN.	Sanitary sewer
PSI	Pound-force per square inch	SB	Shampoo bowl
PSIG	Pound-force per square inch gauge	SD	Shower drain OR Smoke detector
PT	Plaster trap OR Potential transformer	SECT.	Section
PVC	Polyvinyl chloride	SHR.	Shower
R	Register	SHT.	Sheet
R.A.	Return air	SHT. MT'L.	Sheet metal
R.D.	Roof drain	SMACNA	Sheet Metal and Air Conditioning Contractors National Association
R/W	Right of way	SPD	Surge protection device
RAD.	Radius OR Radiation OR Radiator	SPEC.	Specification
RECIRC.	Recirculating	SQ.	Square
REG.	Register	SQ. FT.	Square feet
REINF.	Reinforced	SQ. IN.	Square inch
REL.	Relief	ST	Sound trap
REQ'D.	Required	STAT	Thermostat
RH	Right hand	STD.	Standard
RHC	Reheat coil	STL.	Steel
RM.	Room	STM.	Storm
RPM	Revolutions per minute	SUCT./S	Suction
RTA/C	Rooftop air-conditioning unit	SW	Softened water
RW	Raw water	SW.	Switch
RWC	Rainwater conductor	T.W.	Tempered water
S	Sink	T'STAT	Thermostat
S & R	Supply and return	TD	Temperature difference
S. DPR.	Smoke damper	TEMP.	Temperature
S.A.	Shock absorber OR Supply air	THERM.	Thermometer OR Thermostat
S.F.	Square feet	TIA	Telecommunications Industry Association
S.M., S/M	Sheet metal	TYP.	Typical
S.P.	Static pressure	UC	Undercut
S.S.	Service sink OR Storm sewer	UH	Unit heater
S.S., S/S	Stainless steel	UL	Underwriters Laboratories, Inc.

UR	Urinal	W.G.	Water gauge
USGBC	United States Green Building Council	W/	With
UV	Unit ventilator	W/O	Without
V.	Vent	W/W	Wall to wall
V.T.R.	Vent through roof	WAT.	Water
VAC.	Vacuum	WB	Wet bulb
VC	Vacuum cleaning	WC	Water closet OR Water column
VCP	Vitrified clay pipe	WH	Wall hydrant OR Water heater
VERT.	Vertical	WO	Waste oil
VIB. ISO.	Vibration isolator	WOG	Water/oil/gas
VOC	Volatile organic compound	XFMR	Transformer
VSP	Vitrified sewer pipe	YD.	Yard
W.	Width OR Water	YH	Yard hydrant

END OF SECTION 21 05 00.40

SECTION 21 05 00.45 - TESTING AND INSPECTING SERVICES FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 21 05 43, "Underground Fire Suppression Systems."

1.02 INSPECTIONS, TESTS, AND APPROVALS

- A. The Contract Documents, Laws, Ordinances, Rules, Regulations, or Orders of any AHJ may require the Work to be inspected, tested, or approved.
- B. Contractor shall bear all costs of such inspections, tests, or approvals.
- C. Contractor shall pay all professional engineering fees involved in required testing of equipment.
- D. Contractor shall arrange and pay for all inspections required to obtain a Certificate of Occupancy. Contractor shall include this cost in its bid.
- E. Arrange for inspection of the Work by the AHJ.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.01 TESTS AND INSPECTIONS

- A. Notify the A/E of all scheduled tests and adjustments at least (48) hours before they are scheduled, so that the A/E may witness same. If the Contractor performs any test or adjustment without the A/E present, or without proper notification, the Contractor shall perform the test or adjustment a second time, in the presence of the A/E. Coordinate all test schedules with the Owner to minimize inconvenience.
- B. Provide all required testing and obtain approvals. Secure required certificates of inspection, testing, or approval, and include them in the Operating and Maintenance Manual.
- C. After Work has been completed, but before pipe covering has been applied, test each system in accordance with the requirements of other Sections of this Specification. At the test pressures, the circulation shall be free and the piping shall be proven free of leaks.

- D. Concealed lines shall be tested and approved before being concealed. If a leak appears during the final test, repair the line and any damage resulting from the leak.
- E. Coordinate the enclosure or concealment of the Work with the required inspections and tests so as to minimize the necessity of uncovering work for that purpose.
- F. Should any of the Work be covered up or enclosed prior to completion of all required inspections, tests, and approvals, uncover the Work and, after it has been completely inspected, tested, and approved, make all repairs and replacements with such materials and workmanship as are necessary to secure the approval of the A/E, and at no additional cost to the Owner.
- G. Furnish all test pumps, gauges, equipment, and personnel required, and test to demonstrate the integrity of the finished installation to the approval of the AHJ and the A/E.
- H. Check each piece of equipment for defects and verify that all parts are properly furnished and installed, that all items function properly, and that all adjustments have been made.
- I. Conduct such tests and adjustments of equipment to verify performance requirements or to meet the requirements of the A/E or AHJ. Submit data taken during such tests to the A/E.
- J. Test and check all control valves, drain valves, gauges, alarm devices, and waterflow and supervisory switches for proper operation. Submit a written record of these tests to the A/E.
- K. Demonstrate to the Owner that all Fire Suppression systems are operating properly.

3.02 FINAL OPERATING TESTS AND PROCEDURES

- A. Prior to Contract completion, conduct system operational tests for a period of at least (5) days, not necessarily consecutive, as scheduled by the Owner, to demonstrate fulfillment of the requirements of the Contract. During this time, adjust equipment so that it will perform as the Manufacturer intended, and so that systems will function as designed.
- B. Operate each system in every mode of operation, and check the position of valves, switches, and other devices for proper closure, operation, and switching.

3.03 CERTIFICATE

- A. Furnish approved Certificate of Final Inspection.

3.04 FOLLOW-UP INSPECTIONS

- A. Make an inspection within (90) days after occupancy of the building to make minor adjustments as needed to ensure that all equipment is operating properly and that

satisfactory conditions are being maintained throughout, and to satisfy any unusual conditions. Schedule with the A/E. Consult with the Owner.

- B. A minimum of (1) month before the end of the guarantee period, contact the A/E and Owner to discuss system operation, and to plan for the future care and maintenance of the system.
- C. (1) month before the end of the guarantee period, contact the Owner and perform an inspection to review any items needing correction.
- D. Refer to Section 21 05 43 for additional inspection requirements related to excavation and backfill.

END OF SECTION 21 05 00.45

SECTION 21 05 00.50 - TEMPORARY FACILITIES AND CONTROLS FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide and maintain temporary facilities and controls required for construction as long as needed for the safe and proper completion of the Work and remove upon completion of the Work.

1.02 QUALITY ASSURANCE

- A. Standards: Comply with the provisions of the following Codes:
 - 1. United States Department of Labor, Occupational Safety and Health Administration, Part 1926, "Safety and Health Regulations for Construction."
 - 2. Applicable Local Building Codes.

1.03 TEMPORARY CONSTRUCTION

- A. Materials shall be adequate in capacity for the required usage, shall not create unsafe conditions, and shall not violate applicable Codes and Standards.
- B. Contractor shall be solely responsible for the structural design of all temporary items that it uses in the construction of the building, or that become a permanent part of the building, including, but not limited to:
 - 1. Hoisting.
 - 2. Temporary bracing for structural steel.
 - 3. Shoring for concrete and masonry work.
 - 4. Shoring for cut earth banks.
 - 5. Suspended ceilings, equipment, walls, etc.
- C. Provide hoists, staging, rigging, scaffolding, runways, and necessary appurtenances required for the performance of the Work.
- D. Provide necessary tarps or other appropriate cover material to protect openings.
- E. Provide effective means of dust control. Dust enclosures shall extend tight to the structure and extend around all pipes, ducts, and similar items. Enclosures shall confine dust and dirt to the area of construction and completely seal off and isolate the area of construction from adjacent Owner-occupied areas.

- F. Completely remove temporary materials and equipment when their use is no longer required.
- G. Clean and repair damage caused by temporary installation or use of temporary facilities.

END OF SECTION 21 05 00.50

SECTION 21 05 00.60 - PRODUCT REQUIREMENTS FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 00, "Procurement and Contracting Requirements."

1.02 DESIGN BASE MANUFACTURER STANDARD

- A. The Drawings and Specifications reflect a design based on the specific equipment requirements and configuration for a Design Base Manufacturer. Design coordination of equipment with the building and with other Trades has been made for this specific Model and Manufacturer of equipment. Where several Manufacturers are listed for an item of equipment or material, the first-named shall be considered the Design Base Manufacturer Standard.

1.03 OTHER ACCEPTABLE COMPONENTS (MANUFACTURERS)

- A. Other Manufacturers listed in the Specifications shall be acceptable; however, whenever the Contractor elects to furnish specified equipment or material manufactured by other than the Design Base Manufacturer, the Contractor shall be responsible for the cost and coordination of all modifications required to accommodate the selected equipment or material, including any Work of other Trades that might be affected. Where changes to other Trades' Work are required, the Contractor shall include the additional costs of all such Work in its bid.
- B. Any specified Equipment Manufacturer furnished by the Contractor, other than the Design Base Manufacturer, shall be, in the opinion of the A/E, equivalent in quality, design, features, performance, arrangement, and appearance to that of the Design Base equipment or material, including any special features or requirements.
- C. The inclusion of other Manufacturers in these Specifications shall not be construed to infer that each Manufacturer's product has been reviewed by the A/E for this application, or that it will provide all the features or capabilities required, or that its physical dimensions can be accommodated in the available space. Verify that the proposed equipment will provide all features and accommodations provided by the Design Base Manufacturer, and that it will fit in the available space.
- D. The A/E will not give consideration to any Manufacturer that the Contractor proposes as a Substitution, unless the A/E approves the Manufacturer during the bid period and specifically names the Manufacturer in the Contract Documents or Addenda thereto.
- E. Where deemed necessary by the A/E, the Contractor shall, at no additional cost to the Owner, prepare layouts for these other brands of equipment that may have different

dimensional or service requirements from the Design Base Manufacturer Standard. Submit these layouts to the A/E for review.

- F. Contractor shall reimburse the Owner for the cost of any design changes incurred by the A/E in the preparation of revised Drawings or Specifications to accommodate the use of any Manufacturer other than the Design Base Manufacturer.

1.04 SUBSTITUTIONS

- A. Refer to the Bid Documents for substitution requirements. If this Specification Section conflicts with the Bid Documents, comply with the Bid Documents.
- B. Contractor shall submit information for review no later than (10) days prior to the bid opening to the A/E on any other proposed equipment or material that the Contractor desires to use.
- C. Contractor shall be responsible for the same costs of coordination and modifications, etc., listed above in Article 1.03, "Other Acceptable Components (Manufacturers)."
- D. If the A/E determines that the proposed Manufacturer is acceptable, the A/E will issue an Addendum adding that Manufacturer to the Specifications and/or the Drawings prior to the bid opening.

1.05 EQUIPMENT SUITABILITY

- A. All equipment provided shall perform as intended. All items listed shall function properly, and as the Manufacturer intended. Install equipment according to the Manufacturer's recommendations. Properly attach equipment to the floor, wall, or structure. Each item of equipment shall be compatible with all other accessories or hook-ups, including piping, controls, wiring, and other equipment not furnished by the equipment Manufacturer, but required to achieve the intended function.

1.06 MISCELLANEOUS ACCESSORIES

- A. Provide any additional adapters, fittings, trim, structural steel angles, channels, strut channels, brackets, or other appurtenances to securely install all items of equipment specified or shown on the Drawings.
- B. All steel installed outside or exposed to moisture shall be hot-dipped galvanized.
- C. These accessories are required even though they may not be shown or detailed on the Drawings.
- D. Installation shall be compatible with the building construction on which the item is to be located.

- E. Verify the type of construction prior to ordering the equipment item, so that all required accessories are included.

1.07 QUANTITIES

- A. Equipment may be referred to in these Specifications, or on the Drawings, as either singular or plural; the Contractor shall verify the exact number of items required to complete its Work.

END OF SECTION 21 05 00.60

SECTION 21 05 00.71 - EXAMINATION AND PREPARATION FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 EXAMINATION OF SITE

- A. Before submitting its Bid, it is recommended that the Contractor visit the site of the proposed Project.
- B. After receipt of Bids, no adjustments to Bids will be permitted for lack of knowledge of Project conditions.
- C. Verify and reconcile Work required by the Contract Documents with existing conditions at the Site.
- D. Should the Contractor note any discrepancies during the Bidding Period, it shall notify the A/E immediately, in writing, to permit issuance of an Addendum to prevent misunderstandings at a later date.

1.02 PREPARATION

- A. Provide a Superintendent, on site, whose duties include the directing and supervising of work. Inform the A/E of the Superintendent's name and phone number, and the method of contact when the Superintendent is not at the site.
- B. As soon as possible after the Award of the Contract and the approval of shop drawings, the Contractor shall place orders for materials and equipment required for its Work. Immediately inform the A/E in writing as to any materials and equipment that cannot be obtained within the required time period, either due to conditions of the market or other governing factors.
- C. Inspect each item of materials or equipment immediately prior to installation. Reject damaged and defective items.
- D. Contractor shall determine erection procedures and sequence the construction to keep its work on schedule and to ensure the safety of the building and its occupants. This includes supplying any temporary bracing, guys, or tie-downs that might be required. Such materials shall remain the Contractor's property after completion of the Project.
- E. Provide attachment and connection devices and methods for securing work. Secure all work true to line and level and within recognized industry tolerances. Allow for expansion and building movement.

- F. Recheck measurements and dimensions of the Work as an integral step of starting each installation.
- G. Where mounting heights are not indicated, individual units generally can be mounted at industry-recognized standard mounting heights for the particular application indicated. However, all mounting heights shall be approved by the A/E.

1.03 COORDINATION OF WORK

- A. Contractor shall coordinate its Work carefully with all other Contractors.
- B. Do not install work without first coordinating the layout, routing, etc., with other Contractors whose work may be affected.
- C. Coordinate location of all work with other Contractors and equipment.
- D. Consult all Contract Drawings that may affect the location of any equipment, apparatus, and piping, and make any other adjustments in location to secure coordination.
- E. Coordinate locations of all Fire Suppression Work with hangers, piping, ductwork, diffusers, grilles, registers, cable trays, occupancy/vacancy sensors, bus ducts, luminaires, speakers, smoke and heat detectors, conduits, and other equipment, and with ceilings, soffits, structural members, joists, purlins, metal decks, bracing, etc.
- F. Notify the A/E immediately of any changes in location or other conflicts that may affect the work of other Contractors.
- G. Contractor shall be responsible for the cost of additional engineering work required for changes to the work as shown or described, due to the relocation of items requested by the Contractor.
- H. Review all equipment nameplate ratings and advise the A/E immediately of any system design changes required to wire the equipment properly.

END OF SECTION 21 05 00.71

SECTION 21 05 00.73 - EXECUTION OF FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 00, "Procurement and Contracting Requirements."
- B. Division 01, "General Requirements."
- C. Division 09, "Finishes."

1.02 DESCRIPTION

- A. Contractor shall be solely responsible for construction means, methods, techniques, sequences, and procedures in connection with the Work.

1.03 QUALITY ASSURANCE

- A. Standard: National Fire Protection Association (NFPA). 2023 NFPA 262, "Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces."

1.04 CONTRACT DOCUMENTS

- A. The Contract Drawings are schematic and show approximate locations, general arrangement, and general extent of Work. Not all items of work required for a complete and operating system are shown on the Drawings. Refer also to the Specifications for additional work that shall be provided. If a conflict occurs, the Contractor is responsible for the more restrictive requirement.
- B. The Contract Drawings are not intended to show every vertical or horizontal offset that may be necessary to complete the systems. Having piping and fittings fabricated and delivered in advance of making actual measurements shall not be sufficient cause to avoid making offsets or other changes as may be necessary to install piping and equipment.
- C. Verify exact locations in the field, and coordinate with all other Contractors.
- D. The A/E shall approve, in writing, significant deviations from the Drawings.
- E. The A/E reserves the right to make minor changes in location that do not require additional labor or material, up to the time of roughing-in, without additional cost. The A/E reserves the right to determine what is "significant" and what is "minor."
- F. If a conflict occurs between the Drawings and Specifications, or an overlap of work among the Trades become evident, immediately submit a written request for an interpretation or clarification from the A/E, who shall determine which interpretation has precedence. Refer to Division 00 General Conditions. In such event, none of the Contractors shall assume that it is relieved of the Work that is specified under its branch unless instructions are received, in writing, from the A/E.

1.05 UTILITIES AND OUTAGES

- A. Locate and identify all existing utilities prior to construction.

- B. Where necessary, make minor relocations to permit installation of the Fire Suppression Work.
- C. Advise the A/E immediately of major conflicts to permit modification of the Contract Documents prior to bidding.
- D. Conform to all Utility Company requirements.
- E. Closely coordinate any utility service shutdowns or outages, interruptions, or downtime with the A/E and Owner.
- F. Provide temporary service during the curtailment, if required by the A/E.
- G. Avoid inconveniencing the Owner.
- H. Notify the Owner at least (10) full working days in advance of commencing work in the area of existing utilities.
- I. Contractor shall alert occupants of nearby premises of any emergency conditions that arise as a result of its work in connection with existing utilities.

1.06 PROTECTION

- A. Unless equipment and material can be protectively stored in a manner acceptable to the A/E, they shall not be delivered to the site until the Work is ready to receive them.
- B. Protect all equipment and materials during construction from damage by weather, water, dirt, paint droppings, welding and cutting spatters, and other construction activities.
- C. All materials or equipment stored outside shall be elevated and protectively covered in a secured and locked area.
- D. Store materials and equipment sensitive to weather or construction conditions inside. Where necessary, store sensitive equipment in a heated area.
- E. During construction, cover all equipment and other items that are susceptible to damage until they can be installed in place.
- F. Immediately repair or replace damaged equipment or materials to the satisfaction of the A/E and at no additional cost to the Owner.
- G. Contractor shall protect the building and other Contractors' material and equipment from damage caused by its Work. Protect floors from cutting oil and chips.
- H. Use all means necessary to protect materials before, during, and after installation.
- I. Refer also to individual Specification Sections for specialized protection.

1.07 FIRE SUPPRESSION LUBRICATION AND PACKING

- A. Contractor shall lubricate all the equipment that it furnishes with the correct grade, type, and quantity of lubrication before placing equipment and valves in service.
- B. Refer to the Equipment Manufacturer's recommended lubricants. Furnish equal lubricants as supplied by Castrol, Mobil, Shell, or the Owner's present supplier.

- C. Follow the Equipment Manufacturer's recommendations for specific lubricants and application schedule.
- D. Check each shaft or valve stem containing a packing gland for condition and examine for proper packing.
- E. When filling systems initially for hydrostatic pressure tests, adjust valve packing glands to finger-tight and allow packing to absorb water for (5) minutes prior to tightening packing nuts.
- F. Maintain lubrication and packing seals during construction and ensure that, at the time of final acceptance, all are operating properly.
- G. Contractor shall repair any damage caused by improper lubrication of the equipment that it furnished, at no expense to the Owner.
- H. Incorporate lubrication instructions in the Operating and Maintenance Manual.

1.08 EQUIPMENT ACCESS

- A. Locate all equipment to provide sufficient access to operate equipment or to service items requiring periodic maintenance. Coordinate with the structure and with all other Contractors' work.
- B. Maintain Code-required setbacks from the edge of the roof, minimum 10 ft., for equipment.

1.09 ACCESS PANELS

- A. Lead Contractor shall be responsible, unless otherwise noted, for all access panels shown on the Drawings, or where otherwise required, to permit convenient access to concealed fire suppression equipment, boxes, connections, controls, or other devices requiring service, adjustment, or maintenance. Refer to the Architectural Drawings for location.
- B. Acceptable Manufacturers: Inryco-Milcor, Babcock-Davis, Bilco, Boico, Cesco, Inland-Ryerson, JL, Milco, Karp, Larsen's Manufacturing Co., Mifab, Mitco, or Zurn.
 - 1. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number. Other Acceptable Manufacturers will be permitted, unless specifically excluded.
- C. Access Panels:
 - 1. Panels shall be 18 in. x 18 in., unless larger panels are shown or required. Panels shall be steel construction (except where aluminum or stainless steel is specified) with overlapping edges and concealed hinge door with screwdriver lock. Mounting frames shall be compatible with the building construction material in which they are installed. Each access panel in fire-rated shaft walls and in fire-rated ceilings shall be B Label.
 - 2. Ceiling access panels:
 - a. Gypsum: Inryco-Milcor Style DW. Prime-painted for finish painting with ceiling.

- b. Plaster:
 - 1) Inryco-Milcor Style AP. With finish material same as ceiling material.
 - 2) Inryco-Milcor Style K. Prime-painted for finish painting with ceiling.
 - c. Acoustical concealed spline: Inryco-Milcor Style AT.
 - d. Fire-rated: Inryco-Milcor Style ATR.
 - 3. Wall access panels:
 - a. Gypsum: Inryco-Milcor Style DW. Prime-painted for finish painting with wall.
 - b. Plaster: Inryco-Milcor Style K. Prime-painted for finish painting with wall.
 - c. Masonry and tile:
 - 1) Inryco-Milcor Style M Standard. Prime-painted for finish painting with wall.
 - 2) Inryco-Milcor Style M. Stainless steel.
 - d. Fire-rated: Inryco-Milcor. Fire-rated access door, prime-painted for finish painting with wall.
- D. Coordination:
- 1. Coordinate closely with other Contractors to ensure that all panels furnished by the various Contractors shall be identical.
 - 2. Coordinate the location and installation of each access panel with equipment, the Lead Contractor, and other Contractors.
 - 3. Advise other Contractors of specific requirements relating to access panel installation, including sizes of openings to receive access panel and frame, as well as locations of supports, inserts, and anchoring devices.
- E. Determine specific locations and sizes for access panels needed, but not indicated on the Drawings, to gain access to concealed equipment.
- F. Plan panel locations to clear ceiling luminaires, grids, and obstructions, and to allow, wherever possible, full shoulder clearance beside the device to be inspected, adjusted, or repaired.
- 1.10 AIR-HANDLING PLENUMS
- A. Where space is used for air handling, such as above ceilings and elsewhere, do not install combustible or noxious materials.
 - B. All materials shall be listed for use in air-handling plenums. All wiring shall be listed with 2023 NFPA 262.

1.11 BLOCK COURSE COORDINATION

- A. Mounting heights are called out on the Drawings for many items. Adjust equipment mounting heights to accommodate brick or block coursing.
- B. Coordinate installation of all items in a masonry wall with the Lead Contractor and A/E.

1.12 CUTTING AND PATCHING

- A. All cutting and patching shall be accomplished in a neat and workmanlike manner, acceptable to the A/E.
- B. Cutting:
 - 1. Contractor shall perform all cutting required for installing its own Work unless otherwise noted.
 - 2. Cutting shall be done with such tools and methods so as to prevent unnecessary damage to surrounding areas and equipment.
 - 3. Use rotary drills to cut holes through concrete, brick, plaster, or tile.
 - 4. No cutting shall be done that will, in any way, reduce the structural strength of the building. Cutting of structural support beams, joists, plates, or other structural members is strictly prohibited without the specific written consent of the A/E and the Structural Engineer. Should such cutting be necessary, consult the A/E and do not proceed further without written approval of the A/E.
- C. Patching:
 - 1. Lead Contractor shall be responsible for patching except as otherwise noted.
 - 2. Only a qualified Finish Tradesperson, skilled in the respective craft required, shall perform patching.
 - 3. Patching shall match adjacent surface construction.
 - 4. Materials and equipment used in the patching work shall comply with requirements of other Sections of this Specification.
- D. Perform all cutting and patching promptly. Make all repairs to leave the entire Work in good condition, including all cutting, fitting, and drilling of masonry, concrete, metal, wood, plaster, or other materials, as specified for proper assembly, fabrication, installation, and completion of the Work.
- E. Coordinate drilling, welding, etc., and method of attachment to columns, joists, beams, purlins, etc., with the Lead Contractor and A/E before proceeding.
- F. If a Contractor finds it necessary to cut into the Work of another Contractor, the cutting and patching shall be performed by such other Contractor at this Contractor's expense. With the written consent of the A/E, the Contractor shall be allowed to perform the cutting. Patching shall be similarly executed.

- G. **Lead** Contractor, or other Tradesperson approved by the A/E, shall replace floors, ceilings, roofs, etc., damaged during construction. The cost of repairs shall be borne by the Contractor causing damage.
- H. **Lead Contractor shall provide** openings, with lintels if needed, for sleeves, piping, and other similar openings in new interior masonry walls and floors. Locate and dimension the opening prior to construction. Refer to the Fire Suppression Drawings.
- I. Remove and replace masonry, wall tile, drywall, ceilings, and ceiling tiles in order to install Fire Suppression Work.
- J. Contractor shall be responsible for all cutting of the floor slab related to its own Work.
- K. Contractor shall be responsible for construction that is disturbed during the installation of piping, sprinklers, or equipment.
- L. Cutting and patching includes remodeling and repairing of previously graveled or paved areas, walks, curbs, sod, floors, etc. Saw cuts shall be done in neat, straight lines.
- M. Contractor shall repair or replace any roads, sidewalks, or other items that its employees may damage during the performance of the Fire Suppression Work.
- N. Refer to Division 01 for additional requirements.

1.13 PAINTING AND RELATED WORK

- A. ALL PAINTING SHALL BE DONE BY A QUALIFIED TRADESPERSON SKILLED IN THE CRAFT.
- B. Lead Contractor shall be responsible for finish painting of walls, ceilings, and other Architectural items in the areas of new construction.
- C. Previously Finished Areas: Contractor shall repaint any previously finished areas disturbed by its own cutting and patching. Painting of the patched area shall match the color of the adjacent construction in the general area of the patch. Repaint the entire wall-to-wall and floor-to-ceiling surface if, in the opinion of the A/E, a uniform appearance cannot be accomplished by touch-up.
- D. Factory-Finished Equipment: Clean, spot-prime with zinc chromate, and finish, equal to the original quality, any new factory-finished equipment that has rusted, has been damaged, or has deteriorated. Repaint the entire surface if, in the opinion of the A/E, a uniform appearance cannot be accomplished by touch-up.
- E. Support Steel / Ferrous Metal: Clean, remove rust from, and paint with zinc-chromate primer any fire suppression support steel and bare ferrous metal that is not factory-painted, shop-painted, or galvanized, and that remains exposed to view in the finished areas of the building, including mechanical/electrical rooms and storage rooms. **Consult with the Lead Contractor to ensure** that the prime coat shall be compatible with the finish coat.
- F. Steel Hangers, etc.: Clean, remove rust from, and paint with zinc-chromate primer and aluminum-bronze paint all steel hangers, boxes, straps, and rods furnished under the Fire Suppression Work, that are not provided with rust-protective finish or are damaged in installation, and that remain exposed to view or are in unfinished and mechanical spaces.

- G. Exterior Ferrous Metal: Paint with a prime coat any ferrous metal installed outside the building that is not factory-painted, shop-painted, or galvanized.
- H. Paint exposed piping, valves, etc.
- I. Refer to Division 09 for additional requirements.

END OF SECTION 21 05 00.73

SECTION 21 05 00.74 - CLEANING AND WASTE MANAGEMENT FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."

1.02 CLEANING AND WASTE MANAGEMENT

- A. Contractor shall do its own cleanup, shall move materials that are in the way of constructing its work, shall repair and replace any damage it does, and shall do any other work of a similar nature which shall be done.

1.03 PROGRESS CLEANING

- A. Maintain all work areas in a neat and orderly manner, free of debris. Clean up all occupied travel areas at the end of each shift. Clean up material removed immediately after use.
- B. Clean up building and site thoroughly every day.
- C. Equipment cleaning and touch-up:
 - 1. Use only cleaning materials recommended by the Manufacturer of the surface to be cleaned.
 - 2. Clean interiors of all enclosures of dirt and debris before installing trim or covers.
 - 3. Brush-clean, prime, and paint-in-kind rust spots on any part.
 - 4. Clean galvanized piping in exposed areas with diluted acetic acid.
 - 5. Clean copper piping in exposed areas with fine emery cloth and solvent.
 - 6. Clean all gauges, thermometers, traps, dirt legs, strainers, and fittings.

1.04 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

- A. On-site rubbish collection containers shall be furnished and weekly rubbish removal service shall be provided or all rubbish and debris from the site shall be removed daily. Refer to Division 01.
- B. Do NOT use the Owner's dumpster for on-site rubbish and debris.
- C. Disposal of material by removal:

1. Prior to the completion of the work, remove from the construction area all material designated to be removed.
 2. All material to be removed shall become the property of the Contractor.
 3. Conduct cleaning and disposal operations in compliance with local ordinances and anti-pollution laws. Make all necessary arrangements with private parties and with local officials pertinent to locations and regulations for dumping. Contractor shall pay fees or charges required for dumping of materials.
 4. Contractor shall be responsible for locating a legal dump site and for all expenses due to length of haul.
 5. Do not burn or bury rubbish and waste materials on Project site. Do not dispose of waste into streams or waterways.
- D. Prior to completion of the work, remove from the vicinity of the work all temporary structures, unused materials, concrete forms, and other like materials. All work areas shall be graded and left in a neat manner conforming to the natural appearance of the landscape or as directed elsewhere in the Contract Documents.

1.05 FINAL CLEANING

- A. Repair, patch, and touch up all scratched or damaged surfaces to specified finish that matches adjacent surfaces, before final acceptance of the work. Repair dents and marred finishes to the satisfaction of the A/E. Prepare for finish painting, where painting is specified.
- B. Thoroughly clean all fixtures; material; finished surfaces of equipment; painted, enameled, or varnished work; and all other exposed finished surfaces. Remove all labels, stickers, marks, stains, fingerprints, spots, rust, oil, grease, dirt, dust, and other foreign materials, so that the Work is presented in a complete and finished condition, ready for acceptance and intended use.
- C. Verify that all drains are free of debris.
- D. Employ experienced workers or professional cleaners for final cleaning.
- E. Expedite the cleaning, washing, waxing, and polishing required within other Sections of these Specifications.
- F. Refer to Division 01 for additional requirements.

END OF SECTION 21 05 00.74

SECTION 21 05 00.77 - CLOSEOUT PROCEDURES FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."
- B. Section 21 05 00.33, "Shop Drawings Submittals for Fire Suppression."
- C. Section 21 05 00.45, "Testing and Inspecting Services for Fire Suppression."
- D. Section 21 05 00.73, "Execution of Fire Suppression."
- E. Section 21 05 00.74, "Cleaning and Waste Management for Fire Suppression."
- F. Section 21 05 00.78, "Closeout Submittals for Fire Suppression."
- G. Section 21 05 00.79, "Demonstration and Training for Fire Suppression."
- H. Section 21 05 53, "Identification for Fire Suppression Piping and Equipment."

1.02 DESCRIPTION

- A. Complete and submit the following items, which is a partial list of the items required prior to Contract Completion:

ITEM	REFER TO SECTION
Shop drawing submittals	21 05 00.33
Operational inspection, testing, adjusting, and balancing of all equipment and systems	21 05 00.45, as well as other Sections
Protection	21 05 00.73, as well as other Sections
Painting	21 05 00.73
Cleaning	21 05 00.74
Operating and Maintenance Manuals	21 05 00.78
All required certifications and labels	21 05 00.78, as well as other Sections
Guarantees and equipment warranties	21 05 00.78, as well as other Sections

ITEM	REFER TO SECTION
Spare parts	21 05 00.78, as well as other Sections
As-Built Documents	21 05 00.78
Instruction of the Owner's personnel	21 05 00.79, as well as other Sections
Equipment and piping identification; valve tagging	21 05 53

B. Provide or perform all of the above items before Contract Completion.

1.03 PUNCH LIST PROCEDURE

A. Refer to Division 01.

END OF SECTION 21 05 00.77

SECTION 21 05 00.78 - CLOSEOUT SUBMITTALS FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 00, "Procurement and Contracting Requirements."
- B. Division 01, "General Requirements."
- C. Section 21 05 00.77, "Closeout Procedures for Fire Suppression."
- D. Section 21 05 00.79, "Demonstration and Training for Fire Suppression."

1.02 OPERATING AND MAINTENANCE MANUALS

- A. An Operating and Maintenance Manual, approved by the A/E, is required for Contract Completion. Compile and submit a draft of the Operating and Maintenance Manual to the A/E (90) days before Contract Completion.
- B. Submit the final corrected Operating and Maintenance Manual (7) days before Contract Completion.
- C. Upon approval, provide:
 - 1. (1) Operating and Maintenance Manual in the physical format described below.
 - 2. A Portable Document Format (PDF) file containing a copy of the Operating and Maintenance Manual saved on a Universal Serial Bus (USB) flash drive.
- D. The following items, together with any other pertinent data, shall be included in each Operating and Maintenance Manual. This list is not necessarily complete and shall be used only as a guide. Format of manual is to be as follows:
 - 1. Operating and Maintenance Manual binders shall be heavy-duty, loose-leaf, 3-ring, vinyl-covered binders, no larger than 11 in. wide x 12 in. high, with pocket inside the covers to receive folded oversized sheets.
 - 2. Identify each binder on front and spine with the printed title "OPERATION AND MAINTENANCE MANUAL," project name, and subject matter of contents.
 - 3. Material shall be typewritten or printed and shall be fully legible.
 - 4. Inside cover:
 - a. Title of Project.
 - b. Date of Project completion.

- c. Name and address of the Owner.
 - d. Name and address of the Prime Contractor.
 - e. Date of submittal.
 - f. Name and address of the Contractor.
 - g. Name and address of the Architect.
 - h. Name and address of the Engineer.
5. Second Page: Index.
6. Each section shall be divided by labeled tabs.
7. First Section: A copy of each shop drawing and approved submittal with an index at the beginning of the section.
8. Second Section:
- a. A list of all equipment used on the job.
 - b. Complete list of all devices installed and cross-referenced to room number or location.
 - c. Parts list with numbers of replaceable items, including sources of supply.
 - d. Model and serial numbers of components of systems installed.
 - e. Suppliers and Manufacturers List:
 - 1) Typed statement of material to be furnished.
 - 2) Manufacturers' and nearest Factory Representatives' names, addresses, telephone numbers, and e-mail addresses.
 - 3) Routine and 24-hour emergency service/repair information:
 - a) Name, address, telephone number, and e-mail address of servicing agency.
 - b) Names of personnel to be contacted for service arrangements.
9. Third Section:
- a. Description of systems.
 - b. Manufacturer's literature describing each piece of equipment, including the following:
 - 1) Operating and maintenance instructions.
 - 2) Lubrication instructions.
 - 3) Start-up and shutdown procedures.

- 4) Routine and emergency servicing instructions.
 - c. Valve charts.
 - d. Special wrenches, keys, etc.
 - e. Copies of all testing, adjusting, and balancing reports.
 - f. Prints of all system wiring and control diagrams.
 - g. All certifications and related information.
 - 1) Typed verification of compliance with certification requirements.
 - h. Copies of all written warranties.
 - i. The Owner's receipt of spare parts.
 - j. The Owner's statement of completion of instruction. Refer to Section 21 05 00.79.
- E. Refer to Division 01 for additional requirements.

1.03 GUARANTEES AND WARRANTIES

- A. Guarantee: Contractor shall guarantee its equipment, workmanship, and materials for a period of (1) year from the date of Contract Completion. Should defects develop within this period, the Contractor shall, at no cost to the Owner, remedy the defects and reimburse the Owner for all damage to other Work caused either by the defects or as a result of the work of correcting the same.
- B. Extended Warranties: The table below contains a list of items with warranty requirements and their specification sections. These extended warranties are over and above the 1-year guarantee specified above. This list may not be a complete list of all required warranties. Include copies of all written warranties in the Operating and Maintenance Manual, whether listed or not.

ITEM WITH WARRANTY REQUIREMENTS	SPECIFICATION SECTION
Division 21 items	
Electric Bell	21 10 00

DESIGNER / TYPIST — NON-STATE PROJECTS (C):

- C. Post-Warranty Service:
- 1. The products and Manufacturers identified throughout these Specifications represent reputable suppliers that generally have an inventory of spare parts, and a network of authorized service providers that can support the products installed on this Project.

2. Contractor shall, for a reasonable charge, assist the Owner for a period of (7) years in obtaining parts and service on the products installed.
3. Once the Contractor's 1-year guarantee has expired, the Owner will pay the cost of these parts and service.
4. If the item cannot be repaired, the Contractor shall recommend an equivalent replacement item at a reasonable cost to the Owner.

DESIGNER / TYPIST — STATE OF OHIO PROJECTS (D):

- D. Provide a copy of a certificate of warranty commencement signed by the Contractor, the Owner, and the A/E in each Operating and Maintenance Manual. The A/E will prepare the correct warranty form for signatures.
- E. Refer also to Division 00, "General Conditions," Division 01, and other Specification Sections that define the starting date of the guarantee period, or that discuss either additional warranty requirements, or extended equipment warranties beyond the standard period.

1.04 PROJECT AS-BUILT DOCUMENTS

- A. Contractor shall maintain, at the job site, (1) copy of the As-Built Documents, including the following items related to its Trade:
 1. Contract Drawings.
 2. Specifications.
 3. Addenda.
 4. Reviewed Shop Drawings.
 5. Change Orders.
 6. Other modifications to Contract.
 7. Field Test Records and Start-Up Reports.
 8. Minutes of Coordination and Progress Meetings.
 9. (1) set of the Contract Drawings that will be used for As-Built Drawings. This set shall be used exclusively for documenting and recording the exact location of all installed Work.
 10. Option: With the approval of the A/E and Owner, As-Built Documents may be made available in an electronic format via an Internet-accessible server.
- B. Do not use As-Built Documents for construction purposes.

- C. Store As-Built Documents apart from documents used for construction at the Contractor's temporary field office, or in another location readily available to the A/E.
- D. Provide files and racks for storage of As-Built Documents, as appropriate, with As-Built Documents identified.
- E. Maintain As-Built Documents in clean, dry, legible condition.
- F. Use red ballpoint pen for all marking of As-Built Documents.
- G. As-Built Documents shall be available at all times for inspection by the A/E and Owner.
- H. As-Built Drawings:
 - 1. As-Built Drawings shall be legibly marked to record actual "as-built" construction with the following information (some of these items may not be applicable):
 - a. Horizontal and vertical location of above-ground and underground utilities and appurtenances referenced to permanent surface improvements.
 - b. Location of internal utilities and appurtenances concealed in construction referenced to visible and accessible features of structure.
 - c. Field changes of dimension and detail.
 - d. Changes made by Change Order.
 - e. Details not on the original Contract Drawings.
 - 2. Do not permanently conceal any work until the required information has been recorded.
 - 3. Record the location of all concealed utilities on the As-Built Drawings.
 - 4. Record deviations in locations of concealed piping, valves, equipment, and all buried or concealed utility lines and services, etc., dimensioned from a fixed control point, including the following:
 - a. Depth of bury.
 - b. At each change of direction.
 - c. At each change of slope.
 - d. Additional dimensions in order to fully define location.
 - 5. Minor piping variations need not be recorded.
 - 6. Record locations of abandoned piping, including exterior lines.
 - 7. Record deviations made necessary to incorporate equipment different from the Design Base equipment.
 - 8. Record all Addendum Items.

9. Mark locations of access panels on the As-Built Drawings.
10. Add valve tag numbers to the As-Built Drawings.
- I. Include a complete list of all devices installed and cross-referenced to room number or location.
- J. At completion of project, the Contractor shall deliver (1) printed copy of its As-Built Documents to the A/E.
- K. Contractor shall accompany its As-Built Documents with a transmittal letter, in duplicate, containing the following information:
 1. Date.
 2. Project title and number.
 3. Contractor's name and address.
 4. Title and number of each As-Built Document.
 5. Certification that each As-Built Document as submitted is complete and accurate.
 6. Signature of the Contractor or its Authorized Representative.

1.05 SPARE PARTS

- A. Provide a receipt, signed by the Owner, for (1) complete set of all the operating appurtenances and loose items related to the operation and maintenance of the completed Project for the Owner's use after the guarantee period expires. Insert the receipt in the Operating and Maintenance Manual. These appurtenances and items include, but not are limited to, the following:
 1. Keys to panels, door and window hardware, and other devices not directly provided to the Owner from the Manufacturer.
 2. Operating handles, levers, cranks, specialized wrenches or drivers, remote controls, and similar items.
 3. Extra materials (e.g., attic stock) listed below. This list may not be a complete list of all spare parts. Provide all spare parts, whether listed or not.

SPARE PART AND QUANTITY	SPECIFICATION SECTION
Division 21 items	
Fire suppression water pressure gauge — (1) spare gauge	21 10 00
Gaskets for each piece of equipment that requires gaskets	21 10 00

SPARE PART AND QUANTITY	SPECIFICATION SECTION
Sprinklers — (2) minimum of each type	21 13 13
Sprinklers — (2) minimum of each type	21 13 16
Sprinklers — (2) minimum of each type	21 13 19
Gaskets for each piece of equipment that requires gaskets	21 31 13
Mechanical seals — (1) set for each pump	21 31 13
Gaskets for each piece of equipment that requires gaskets	21 31 16
Mechanical seals — (1) set for each pump	21 31 16
Gaskets for each piece of equipment that requires gaskets	21 41 26
Any other parts mentioned elsewhere	

END OF SECTION 21 05 00.78

SECTION 21 05 00.79

DEMONSTRATION AND TRAINING FOR FIRE SUPPRESSION

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide training schedule and training outline for approval (45) days prior to Contract Completion.
- B. Provide a minimum of (4) hours of in-service training for the system's operators. Provide all applicable user manuals and related training documentation.
- C. After all system operational tests have been completed, schedule an instruction period with the Owner. Schedule well in advance, so that all the Owner's personnel may attend if they desire. Coordinate with the Owner and A/E for date and time of training sessions.
- D. Participate in training sessions. Instruct the Owner in the operation and maintenance of all systems and equipment. Use Operating and Maintenance Manuals to familiarize the Owner with equipment and procedures. Allow sufficient time for this instruction.
- E. Record a video of all Owner training, instructions, and equipment start-up demonstrations on a digital video disc (DVD). Turn over a copy of the DVD to the Owner.
- F. Instruction shall include the following:
 - 1. Location of equipment and explanation of what it does (function).
 - 2. Reference to operating instruction manuals for record and clarity.
 - 3. Coordination of written and verbal instructions, so that the operation of each system is fully understood by operating personnel.
 - 4. Complete review of items contained in Operating and Maintenance Manuals.
 - 5. Discussion of maintenance procedures that are required be followed by the Owner, including cleaning procedures and what items to check.
 - 6. Complete demonstration and explanation of each System.
- G. Provide a copy of the Owner's "Certificate of Equipment Demonstration," signed by the Contractor, the Owner, and the A/E, in each Operating and Maintenance Manual. If the Owner does not have a standard demonstration form, the A/E will provide one.

END OF SECTION 21 05 00.79

**DEMONSTRATION AND
TRAINING FOR FIRE
SUPPRESSION
21 05 00.79 - 1**

10/19/26

(Rev:)

SECTION 21 05 16 - EXPANSION FITTINGS AND LOOPS FOR FIRE-SUPPRESSION PIPING

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide thermal expansion control for all piping (not shown on the Drawings).
- B. Provide expansion joints and accompanying anchors and guides where expansion cannot be provided for with loops and offsets (not shown).
- C. Isolate all piping for both noise and vibration transmission.
- D. To prevent damage to building, equipment, joists, hangers, and piping, provide expansion loops, guides, anchors, and offsets in piping systems to accurately control pipe movement due to equipment operation or thermal gradients.
- E. Anchor points are not indicated on the Drawings. Install anchor points to properly control piping movement. Typically, piping shall be anchored at tees, elbows, and bends. Piping shall also be anchored in each direction.

PART 2 PRODUCTS

2.01 EXPANSION JOINTS

- A. Acceptable Manufacturers: Metraflex or approved equal by Flexicraft, Flexonics Products, Hyspan Precision Products Inc., Minnesota Flexible Corp., Twin City Hose Inc., or Uniroyal General Rubber.
- B. Metraflex. Stainless steel bellows and elements; cast iron equalizing rings, tie-rods, and pipe connections; 250 PSIG working pressure. Use packless, internally guided type for lines 2 in. and smaller.
- C. 2 in. and Smaller Lines: Metraflex HPMF. Stainless steel with copper sweat ends, packless, internally guided expansion compensator type.
- D. 3 in. and Larger Lines: Metraflex MC. All wetting parts stainless steel, including bellows and elements. Vanstone flanges, cast iron equalizing rings, steel cover over bellows. Pipe connections, 250 PSIG working pressure. Size each bellows end for 3 in. expansion.
- E. Seismic/Thermal Expansion Joints: Metraflex Metraloop®.

2.02 PIPE ALIGNMENT GUIDES

- A. Acceptable Manufacturers: Metraflex or approved equal by Flexicraft, Flexonics Products, Hyspan Precision Products Inc., Minnesota Flexible Corp., Twin City Hose Inc., or Uniroyal General Rubber.
- B. Metraflex. Painted steel cylinder guide assembly with welded mounting brackets, 2-piece pipe clamp "Spider" assembly.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Line Expansion:
 - 1. U-Bends: Install U-bends or loops in all piping subject to expansion and contraction.
 - a. Option: Provide Metraflex Metraloop, sized in accordance with the Manufacturer's recommendations.
 - 2. Expansion Joints: Where unable to provide U-bends, or where specifically shown or specified, provide expansion joints. Install according to the Manufacturer's instructions.
- B. Branch Connections: Branch connections to mains shall be made with a minimum of (2) 90-degree elbows, and shall incorporate at least (1) change of direction in the horizontal plane and (1) change of direction in the vertical plane, before connecting to equipment or fixtures or before dropping in or rising in a wall.
- C. Guides: Supplement all loops and expansion joints with adequate guides as close to loops and joints as possible, and at recommended intervals from loops and joints, to preserve alignment and pitch. Guides shall be rigidly secured to the structure and shall permit axial movement only. Follow the Manufacturer's instructions where applicable.
- D. Anchors: Install pipe anchors where required to secure the pipe and eliminate movement. Attach pipe anchors securely to the structure.

END OF SECTION 21 05 16

SECTION 21 05 17 - SLEEVES AND SLEEVE SEALS FOR FIRE-SUPPRESSION PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."
- B. Division 07, "Thermal and Moisture Protection."

1.02 DESCRIPTION

- A. Provide sleeves and sleeve seals, including all accessories required for a complete installation, at penetrations of exterior floor and wall assemblies. Sleeve where piping:
 - 1. Passes through equipment room walls.
 - 2. Passes exposed through non-fire-rated construction walls.
 - 3. Passes exposed through non-fire-rated construction above-grade floors.
 - 4. Passes through smoke-rated or fire-rated separations. Apply firestopping to fire-suppression penetrations of fire-rated floor and wall assemblies to maintain or restore original fire-resistance rating of assembly according to Division 07.

1.03 TERMINOLOGY

- A. Firestopping: Material or combination of materials used to retain integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, and hot gases through penetrations in fire-rated wall and floor assemblies.

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. American Society for Testing & Materials (ASTM):
 - a. ASTM E 84, Standard Test Method for Surface Burning Characteristics of Building Materials."
 - b. ASTM E 814, "Standard Method of Fire Tests of Through Penetration Fire Stops."
 - 2. Factory Mutual Global (FMG).
 - 3. National Fire Protection Association (NFPA):
 - a. 2023 NFPA 70, "National Electrical Code® (NEC)" (latest edition).

- b. 2021 NFPA 101, "Life Safety Code" (latest edition).
- 4. Underwriters Laboratories, Inc. (UL) 1479, "Fire Tests of Through-Penetration Firestops."

1.05 SUBMITTALS

- A. Submit Manufacturer's product data, installation instructions, and details for each required product.

PART 2 PRODUCTS

2.01 SLEEVE SEALS — NON-FIRE-RATED FIRE-SUPPRESSION PENETRATIONS

- A. Piping Sleeve Material:
 - 1. Up to 8 in. diameter: Machine-cut Schedule 40 black steel pipe.
 - 2. 8 in. and larger: Schedule 30 black steel pipe.
 - 3. Bare copper piping: Copper sleeves.
 - 4. Option: Schedule 40 plastic pipe may be used for piping sleeves, where adjacent ceiling space is not used for return air plenum. Refer to the Drawings.
- B. Sizing: Sleeves shall be large enough for sleeve seals to be installed. Size to allow 1/2 in. minimum clearance all around piping.

2.02 SLEEVE SEALS — FIRE-RATED FIRE-SUPPRESSION PENETRATIONS

- A. Acceptable Manufacturers: Either provide sleeves and sleeve seals from the Manufacturers listed in **Division 07 or provide sleeves** and sleeve seals from the following Manufacturers: 3M Fire Barrier caulking or putty, BP Fyre Putty, Carborundum Fibersil, Chase Foam, CSD Sealing Systems "FSP," Dow Corning 3-6548, Hilti Fire Stop, International Protective Coatings Corporation "Flame-Safe," Johns-Manville Cerafiber, KBS Mortar Seal, or Nelson Fire Stop.
- B. ASTM E119 and E814 Silicone RTV foam, UL-approved.
- C. Sleeves shall be large enough for fire-rated sleeve seals to be installed. Size to allow 1/2 in. minimum clearance all around.

2.03 FIRESTOP PILLOWS

- A. Acceptable Manufacturers: KBS or approved equal by 3M or Specified Technologies Inc.
- B. KBS SEALBAG™: UL-listed, FMG-approved.

**SLEEVES & SLEEVE SEALS
FOR FIRE SUPPRESSION**

PIPING

21 05 17 - 2

10/19/26

(Rev:)

2.04 SLEEVE SEALS — WATERTIGHT FIRE-SUPPRESSION PENETRATIONS

- A. Provide sleeves with waterstop anchor flange at midpoint where penetrating structure at or below grade.
 - 1. Acceptable Manufacturers: 3M, Calpico, or Hilti.
- B. Option: Compression Seals: GPT Industries Link-Seal. Stainless steel bolts and nuts. Provide correct size seal and coordinate with sleeve size.
 - 1. Acceptable Manufacturers: GPT Industries or approved equal by Calpico, Metraflex, Michigan, or Wayne.
- C. Sleeves shall be large enough for watertight sleeve seals to be installed. Size to allow 1/2 in. minimum clearance all around.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate location of sleeves, core-drilled holes, and other holes required with the Lead Contractor, Structural Engineer, equipment, and other Trades.

3.02 INSTALLATION — GENERAL

- A. Indicated sizes and locations are based on early Project information and should be verified by the Contractor.
- B. Chases and constructed openings that are not indicated on the Drawings shall be provided at the expense of the Contractor requiring the chase or constructed opening.
- C. If it becomes necessary to cut into the work because of the failure of the Contractor to notify the Lead Contractor, then the Lead Contractor shall do any cutting and patching required later, at the Contractor's expense.
- D. Provide dimensions and locations of openings required in walls for sleeves, piping, and similar items, and deliver sleeves, to the Lead Contractor, so as not to delay construction.
- E. BEFORE CONSTRUCTION OF MASONRY OR CONCRETE WALLS IS STARTED, give the Lead Contractor locations and sizes of all openings required for the installation of equipment or sleeves.
- F. If sleeves are not installed due to the fault of the Contractor, the Lead Contractor shall core-drill holes through masonry or concrete construction at the expense of the Contractor.

- G. In New Drywall, Masonry, or Concrete Wall Construction, as used in this Project:
1. Furnish sleeves for fire suppression items to the Lead Contractor for installation where directed by the Contractor.
 2. Carefully coordinate and check locations of sleeves immediately before and after each concrete pour and/or masonry installation.
 3. Option for above ceiling only: Lead Contractor shall omit concrete block(s) where required for penetrations and shall provide square-finished opening. Do not omit concrete blocks in sound barrier walls, under bearing points, or in fire-rated or smoke-rated walls.
 4. Coordinate with other Contractors, Structural Engineer, and A/E.
- H. Sleeves are not required in floor slabs on grade or in core-drilled openings not requiring firestopping or watertightness.
1. Exception: Sleeves are required at core-drilling through hollow-core pre-cast concrete slabs and through concrete block walls to facilitate containment of required firestopping material.
- I. Furnish all piping sleeves and caulking for openings in new walls and turn them over to the Lead Contractor for installation and caulking.
- J. Piping shall not bear on sleeves. Install sleeves plumb and true to line, grade, and position.
- K. Cutting:
1. Cut off sleeves through walls flush with each surface.
 2. Cut off sleeves 1/8 in. above finished floors and 3 in. above floors in kitchens, utility rooms, equipment rooms, and shafts. Cut off bottom of sleeve flush with surface.
 3. Refer to Division 01 for specific requirements regarding approved cutting methods.
- L. All piping sleeves and escutcheons shall have ample clearance for pipe and covering and shall have chrome-plated wall and floor escutcheons over the pipe in finished areas. Provide chrome-plated or prime-coated escutcheons in other rooms.
- M. Unused sleeves shall be plugged and fire-packed. Provide finish that matches finish of adjacent surface.

3.03 INSTALLATION — FIRE-RATED FIRE-SUPPRESSION PENETRATIONS

- A. Verify which walls, ceilings, or floors are fire-rated, if any, and provide approved fire-blocking or fire-protective devices.

- B. Use approved, UL-listed, fire-retardant sealants, backing, and packing to maintain fire rating of the structure penetrated.
- C. Spray UL-listed foam sealant around exposed piping entering and leaving fire-rated wall or floor structures.
- D. To ensure fire blocking, close space around piping and other fire suppression items passing through walls and floors. Seal space up to a 1/2 in. gap with sealant or caulking. Close off space greater than a 1/2 in. gap with sheet metal and seal airtight.
- E. For larger openings, provide firestop pillows.
- F. Pack all fire-rated or sound-rated separation sleeves with glass fiber, high-temperature mineral wool, aluminum-silica fiber, fire-retardant rope, calcium silicate, or other noncombustible material to maintain fire rating of structure, and finish with fire-rated sleeve seals.
- G. Fill space around all sleeves extending into exposed areas with material compatible with adjacent construction and finish.
- H. All openings shall have been UL-tested utilizing the proper rated penetrations to be equal to, or greater than, the barrier assembly in which the penetration occurs.
- I. Install all penetrations and openings in accordance with the Manufacturers' instructions.
- J. The fire-blocking assembly shall maintain the required fire-resistance rating of the wall or floor in which it is placed. Apply a further sealant, if necessary, to attain a smoke-tight condition. Close openings without sleeves in the same manner.
- K. Refer to **Division 07 for additional requirements.**

3.04 INSTALLATION — WATERTIGHT FIRE-SUPPRESSION PENETRATIONS

- A. Sleeving with absolutely watertight seal is required for fire suppression work passing through **exterior walls above grade**, and **underground** foundation walls and **other below-grade** penetrations into building.
- B. Piping sleeves penetrating outside wall or slabs on grade shall have welded intermediate flange imbedded in masonry. Space around piping shall be clamped, packed, and caulked to make an absolutely watertight seal within sleeve. Caulk around outside of sleeve.
 - 1. Option: Install compression seals.

END OF SECTION 21 05 17

SECTION 21 05 23 - GENERAL-DUTY VALVES FOR WATER-BASED FIRE-SUPPRESSION PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 21 05 53, "Identification for Fire-Suppression Piping and Equipment."

1.02 DESCRIPTION

- A. Provide valves and accessories for fire-suppression piping system(s):
 - 1. To facilitate the maintenance, operation, isolation, and servicing of all water-based fire-suppression piping systems.
 - 2. In locations accessible to qualified personnel for operation, removal, maintenance, or servicing.
 - 3. As shown on the Drawings and as discussed in other Sections of these Specifications.
 - 4. In accordance with 2022 NFPA 13 and 2019 NFPA 14.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. American National Standards Institute (ANSI).
 - 2. American Water Works Association (AWWA).
 - 3. Manufacturer's Standardization Society of the Valve and Fittings Industry (MSS).
 - 4. National Fire Protection Association (NFPA):
 - a. 2022 NFPA 13, "Standard for the Installation of Sprinkler Systems."
 - b. 2019 NFPA 14, "Standard for the Installation of Standpipes and Hose Systems."

PART 2 PRODUCTS

2.01 MANUFACTURERS AND MODEL NUMBERS

- A. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number. Other Acceptable Manufacturers will be permitted, unless specifically excluded.

- B. All valves of the same type used on the Project shall be by the same Manufacturer.

2.02 ANGLE VALVES (BRONZE BODY)

- A. Acceptable Manufacturers: Kennedy Valve Division of McWane Inc. or approved equal by Hammond, Crane, Milwaukee, Nibco, Watts, Reliable Automatic Sprinkler Co., Tyco Fire Products, Automatic Sprinkler, Mueller, Powell, Victaulic, American-Darling, or United Brass Works.
- B. Kennedy Valve Division Figure 98-SD. 175 lb. WOG, Buna-N seat disc, and bronze handwheel.
- C. Hose-End Valves: Kennedy Valve Division Figure 936. 175 lb. working pressure, bronze bonnet, renewable composition disc, iron handwheel, cap, and chain.

2.03 BALL VALVES (BRONZE BODY, UP TO 2 IN.)

- A. Acceptable Manufacturers: Nibco or approved equal by American-Darling, Automatic Sprinkler, Crane, Hammond, Kennedy Valve Division of McWane Inc., Milwaukee, Mueller, Powell, Reliable Automatic Sprinkler Co., Tyco Fire Products, United Brass Works, Victaulic, or Watts.
- B. 1/4 in. to 1 in. (full port): Nibco Model KT-585-70-UL. MSS SP 110-conforming, 175 lb. WWP, threaded 2-piece, chrome-plated ball, blow-out-proof stem, and reinforced TFE seats.
- C. 1-1/4 in. to 2 in. (standard port): Nibco Model KT-580-70-UL. MSS SP 110-conforming, 175 lb. WWP, threaded 2-piece, chrome-plated ball, blow-out-proof stem, and reinforced TFE seats.

2.04 CHECK VALVES

- A. Acceptable Manufacturers: Kennedy Valve Division of McWane Inc. or approved equal by American-Darling, Automatic Sprinkler, Crane, Hammond, Milwaukee, Mueller, Nibco, Powell, Reliable Automatic Sprinkler Co., Tyco Fire Products, United Brass Works, Victaulic, or Watts.
- B. 2 in. and Smaller (bronze body swing): Kennedy Valve Division Figure 440-SD. MSS SP 80, 200 PSI (WOG) bronze cap, threaded ends, Buna-N disc, and bronze hinge, pin, and disc assembly.
- C. 2-1/2 in. and Larger (cast iron body swing): Kennedy Valve Division Figure 1126A. 175 lb. working pressure, 3/4 in. drain, stainless steel hinge pin with bronze bushings, malleable iron disc holder, rubber-faced disc, Buna-N "O" ring, and flanged connections.
1. Do NOT provide double-disc and spring-assisted check valves.

2.05 DRAIN VALVES

- A. Acceptable Manufacturers: Milwaukee or approved equal by HCI, Nibco, or Walworth.
- B. Milwaukee BA-100H or BA150H. 3/4 in. bronze hose-end ball valve, with cap and chain (where concealed). Provide 3/4 in. garden hose thread adapter on outlet of drain valve, with cap.

2.06 FIRE PUMP SUCTION CONTROL VALVE

- A. Acceptable Manufacturer: CLA-VAL Co. or approved equal by Watts.
- B. CLA-VAL Co. Model 50B-5KG. 125 PSI SWP, cast iron body, brass, bronze, and stainless steel trim, hydraulically operated, pilot-controlled, diaphragm-type, globe-type or angle-type valve, single removable seat and renewable resilient disc, flanged ends, direct-acting adjustable pilot control, spring-loaded, designed for modulating service to permit flow when controlling pressure exceeds spring setting, open or closed indicating device, sediment evacuator, and dampening device.

2.07 GATE VALVES

- A. Acceptable Manufacturers: Kennedy Valve Division of McWane Inc. or approved equal by American-Darling, Automatic Sprinkler, Crane, Hammond, Milwaukee, Mueller, Nibco, Powell, Reliable Automatic Sprinkler Co., Tyco Fire Products, United Brass Works, Victaulic, or Watts.
- B. 2 in. and Smaller (bronze body): Kennedy Valve Division Figure 66. 175 lb. working pressure, bronze bonnet, wedge and yoke bushing, malleable iron handwheel, OS&Y rising bronze stem, non-asbestos packing, and threaded ends.
- C. 2-1/2 in. and Larger (iron body): Kennedy Valve Division Figure 4068A. AWWA Standard C509, 200 lb. working pressure, 125 lb. ANSI B16.1 flanges, OS&Y rising stem, cast iron rubber-coated wedge disc, repackable under pressure, and bronze stem assembly. Standard tap sizes:
 - 1. 1/2 in.: 2-1/2 in. to 4 in.
 - 2. 3/4 in.: 6 in.
 - 3. Option: Butterfly valve. Kennedy Valve Division. 175 lb. (WWP), ductile iron body, aluminum-bronze disc, stainless steel stem, Buna-N seat, and phenolic resin ring.
 - a. Valves 6 in. and smaller: Furnish lever lock handle.
 - b. Valves 8 in. and larger: Furnish gear operator.

- D. Hose-End Valves (bronze body): Kennedy Valve Division Figure 39. 175 lb. working pressure, bronze bonnet and disc, non-rising stem, iron handwheel, cap, and chain.

2.08 GLOBE VALVES (BRONZE BODY, UP TO 2 IN.)

- A. Acceptable Manufacturers: Kennedy Valve Division of McWane Inc. or approved equal by American-Darling, Automatic Sprinkler, Crane, Hammond, Milwaukee, Mueller, Nibco, Powell, Reliable Automatic Sprinkler Co., Tyco Fire Products, United Brass Works, Victaulic, or Watts.
- B. Kennedy Valve Division Figure 97-SD. 125 lb. SWP, 200 lb. WOG, rising stem, bonnet packing gland, disc assembly, stem and wheel nut, threaded ends, renewable composition soft disc, non-asbestos packing, and aluminum handwheel.

2.09 WALL POST INDICATOR VALVES

- A. Acceptable Manufacturers: Kennedy Valve Division of McWane Inc. or approved equal by American-Darling, Automatic Sprinkler, Crane, Hammond, Milwaukee, Mueller, Nibco, Powell, Reliable Automatic Sprinkler Co., Tyco Fire Products, United Brass Works, Victaulic, or Watts.
- B. Valve: Kennedy Valve Division Figure C701. 175 lb. (WWP), high-strength alloy iron body and bonnet, non-rising bronze alloy stem and seat ring, bronze bushings, solid taper wedge, 125 lb. (ANSI) flanges, and 2 in. square operating nut.
- C. Wall Indicator: Kennedy Valve Division Figure 641S or 651. High-strength alloy iron cap and body, bronze operating stem, cast iron handwheel, wall flange, extension barrels, wall plate, valve position indicator, straight or angle type, and gear kit.

2.10 SEALANT TAPE

- A. Polytetrafluoroethylene (PTFE) tape.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate valves and accessories with structure and with mechanical and electrical equipment.
- B. Coordinate the exact location of valves, etc., and the proper number, size, and type threads required, with the Local Fire Department.
- C. Coordinate all access panel locations, if required, with the Lead Contractor.

3.02 INSTALLATION

- A. Install valves:
 - 1. At service connections to equipment not provided with valves.
 - 2. At branch lines from main lines.
 - 3. At low points for draining.
- B. Install valves with handwheel at or above centerline of pipe.
- C. Install supervisory switches on:
 - 1. Ball valves.
 - 2. Butterfly valves.
 - 3. Gate valves.
 - 4. Wall post indicator valves.
- D. Drain Valves: Install at low points in piping to provide complete drainage of the fire-suppression system if other means of draining are not provided.
- E. Fire Pump Suction Control Valve: Valve shall be serviceable for repair without removal from line. Provide remote sensing line from valve to suction line. Install valve in accordance with the Manufacturer's recommendations.
- F. Wall Post Indicator Valve: Install each valve at 4 ft.-0 in. from centerline of valve operator to finished grade.
- G. To facilitate replacement of devices, install isolation valves upstream of pressure alarm switch, automatic drain device, and flow control valves.
- H. Open up valves fully before soldering to piping.
- I. During soldering or brazing, remove valve internals and protect the remaining element of sweat end valves against heat damage.
- J. Do not make welded connections to valves.
- K. Do NOT install any valves where the fluid operating pressure exceeds 80% of the valve's pressure rating.
- L. Do NOT use bronze valves on iron fittings or pipe without dielectric unions.
- M. Tighten packing glands before placing valves in service.

N. Tag all valves. Refer to Section 21 05 53.

O. Chain all fire water supply valves open. Provide locks. Give keys to the Owner.

3.03 TESTS

A. Test all valves for tightness.

B. Test-operate all valves at least once from closed-to-open-to-closed positions while valve is under pressure.

END OF SECTION 21 05 23

SECTION 21 05 29 - HANGERS AND SUPPORTS FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 03, "Concrete."
- B. Section 21 05 16, "Expansion Fittings and Loops for Fire-Suppression Piping."
- C. Section 21 05 48, "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."

1.02 DESCRIPTION

- A. Hangers and Supports for Fire-Suppression Equipment and Systems: Provide for equipment and piping installed under the Fire-Suppression Work.
- B. Supports for Equipment:
 - 1. Lead Contractor shall provide all structural supports between building structural framing members to support major fire-suppression equipment.
 - 2. Attach all items of equipment securely to structure, floor, or concrete pad. Means of attachment shall be approved by the A/E and shall not degrade the structure.
 - 3. For any variation in supports needed for equipment actually purchased, the Contractor shall bear all costs of modifications, including redesign costs.
- C. Support Steel: Attach to building structure to support piping and smaller fire-suppression equipment items. Provide supports between structural members, except where specifically called out to be provided as part of the General Trades Work. Obtain prior written approval of the A/E.
- D. Hangers for Piping: Install all necessary pipe hangers, supports, clamps, wall brackets, and attachments in order to:
 - 1. Properly support piping from building structure.
 - 2. Allow controlled thermal movement of piping systems.
 - 3. Facilitate action of expansion joints, loops, and bends, where used.
 - 4. Provide slope where indicated.
- E. Concrete Work: A qualified Tradesperson shall provide all concrete work.

- F. Provide all necessary inserts, expansion shields, beam clamps, piping floor supports, auxiliary steel, angles, clips, mechanical fasteners, reinforcement steel, structure drilling, welding to structure, etc.
- G. Components shall be selected and matched to the load imposed on them.
- H. Refer to Structural and Architectural Drawings and Specifications to determine what supports are provided **as part of the General Trades Work**.

1.03 QUALITY ASSURANCE

A. Standards:

1. **American Concrete Institute (ACI).**
2. American National Standards Institute (ANSI)
3. American Society for Testing and Materials (ASTM).
4. American Society of Mechanical Engineers (ASME), including:
 - a. ASME B31.1, "Code for Power Piping."
 - b. ASME B31.9, "Building Services Piping."
5. Factory Mutual Global (FMG).
6. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS).
 - a. Standard Practice 58.
 - b. Standard Practice 69.
 - c. Standard Practice 89.
7. National Fire Protection Association (NFPA):
 - a. **2022 NFPA 13**, "Standard for the Installation of Sprinkler Systems."
8. Underwriters Laboratories, Inc. (UL).

PART 2 PRODUCTS

2.01 STRUCTURAL STEEL

- A. Steel plates, shapes, and bars; steel beams; strut channels; standard-weight black steel pipe; or steel angle. ASTM A36 / ASTM A36M, black and galvanized, of adequate size and strength to safely carry the weight of equipment item.

2.02 HANGERS FOR PIPING

- A. Acceptable Manufacturers: Anvil International, B-Line Systems Inc., Carpenter & Patterson Inc., ERICO, Globe Pipe Hanger Products Inc., National Pipe Hanger Corp., or PHD Manufacturing Inc.
- B. Uninsulated Piping:
1. Steel, cast iron, or plastic; 1/2 in. to 8 in.: Anvil International Fig. 69. Galvanized carbon steel, adjustable swivel ring.
 2. Steel, cast iron, or plastic; 10 in. and larger: Anvil International Fig. 260. Adjustable clevis, carbon steel yoke, U-strap, rod, and hex nuts.
 3. Copper, 1/2 in. to 4 in.: Anvil International Fig. CT-99. Copper-plated hangers, adjustable tubing ring, carbon steel ring, and malleable iron adjusting nut.
- C. For Piping Indicated to be Installed "Tight to Ceiling or Wall" (uninsulated, 1/2 in. to 4 in.): Anvil International Fig. 262. Carbon steel.
1. Option: PowerStrut channels with PowerStrut Model PS 3126 hold-down clamp. Use with rubber cushion for copper pipe.
- D. Vertical Piping:
1. Anvil International Fig. 261. Friction riser clamp with 2-point bearing.
 2. Use Anvil International Fig. CT-121 copper-plated riser clamp for copper pipe.

2.03 HANGER RODS, CLAMPS, AND BRACKETS FOR PIPING

- A. Acceptable Manufacturers: Anvil International, B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, Miro, National Pipe Hanger Corp., or PHD Manufacturing Inc.
- B. Hanger Rods: Solid steel, threaded ends or all-thread rods of diameter listed below, with double-nut attachment to the hanger and at the hanger attachment. Sizes are as follows:

PIPE SIZES	ROD DIAMETER
Up to and including 4 in.	3/8 in.
5 in. to 8 in.	1/2 in.
10 in. and 12 in.	5/8 in.

C. Hanger Clamps:

1. Beam clamps (up to 8 in. diameter pipe): Anvil International Fig. 218, Fig. 228, or Fig. 292. Top beam clamp, steel jaw, hook rod with nut, and spring washer steel eye-bolt, with Anvil International Fig. 157 or Fig. 290 extension piece for appropriate rod diameter.
 - a. Option: Anvil International Fig. 86, Fig. 93, or Fig. 94. Malleable iron clamp, hardened steel cup point set screw with Anvil International Fig. 89 steel retaining clip.
2. ~~Purlin clamps: PHD Manufacturing Fig. 290. Malleable iron with jam nut and pointed steel set screw. Confirm type to be used with the Lead Contractor and Metal Building Manufacturer.~~

D. Hanger Brackets:

1. Acceptable Manufacturers: ABB (Thomas & Betts) Superstrut®, Anvil International, B-Line Systems Inc., Carpenter & Patterson Inc., ERICO, Globe Pipe Hanger Products Inc., National Pipe Hanger Corp., PHD Manufacturing Inc., or Unistrut Corp.
2. Wall brackets: Carbon steel, back plates, and bolts.
 - a. Anvil International Fig. 194. Lightweight 750 lb. load.
 - b. Anvil International Fig. 195. Medium-weight 1,500 lb. load.
 - c. Option: Unistrut or ABB Superstrut® A1200.

~~keep or delete 2.04? (Mike Y removed from Part 3)~~

2.04 SPRING HANGERS FOR PIPING

- A. Acceptable Manufacturers: Anvil International, or Kinetics Noise Control, or approved equal by B-Line Systems Inc., Carpenter & Patterson Inc., ERICO, Globe Pipe Hanger Products Inc., National Pipe Hanger Corp., or PHD Manufacturing Inc.
- B. Anvil International Fig. B268, Type A or B. Steel, adjustable pre-compressed variable spring.

2.05 TRAPEZE HANGERS FOR PIPING

- A. Acceptable Manufacturers: Unistrut Corp., or ABB (Thomas & Betts) Superstrut®, or approved equal by Anvil International, B-Line Systems Inc., ERICO, GS Metals Corp., National Pipe Hanger Corp., Powerstrut, or Wesanco Inc.
- B. Unistrut P1000 or ABB Superstrut® A1200. Angle iron, or channel of sufficient length to support pipes and insulation on individual hangers or roller supports.

2.06 HANGER COVERS FOR PIPING

- A. Acceptable Manufacturers: Buckaroos Inc. or approved equal by Pipe Shields Inc. or PHS Industries.
- B. Buckaroos Wraparoos. 2-piece PVC cover. Snap-on design, adhesive seals, to 205 degrees F. temperature. Includes 18 in. long PVC hanger rod cover. Available in multiple colors. The A/E shall approve all color selections for fire suppression equipment.

2.07 PIPE ROLL STANDS

- A. Acceptable Manufacturers: Anvil International or approved equal by B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, National Pipe Hanger Corp., or PHD Manufacturing Inc.
- B. Anvil International Figure 271. Cast iron roll, stand with holes for anchoring.

2.08 FINISH

- A. Unless otherwise noted, all steel hangers and supports shall be standard black.
 - 1. Exception: Hangers and supports for exposed exterior applications shall be galvanized.

2.09 CONCRETE

- A. All concrete shall be 5,000 PSI minimum and placed in accordance with applicable ACI Standards. Refer to Division 03.
- B. Class II with a 28-day compressive strength of 5,000 lbs., and a maximum water-to-cement ratio of 0.50.
- C. Concrete bases and pads: Granular material conforming to Division 03 requirements.
- D. Concrete Encasement of Underground Piping: Material conforming to Division 03 requirements. Minimum of 3 in. thick and containing red marker dye. Concrete shall be pre-mixed by the concrete supplier; do NOT trowel in color additive by hand.

2.10 INSERTS

- A. In Masonry with Voids: Master-Carr #94255A100. Steel plated with yellow chromated zinc, female threaded toggle.
 - 1. Acceptable Manufacturers: Master-Carr or approved equal by Vertical Solutions or Hohmann & Barnard.

- B. **In Poured Concrete:** Anvil International Fig. 282. Malleable iron, universal mounting, lateral adjusting.
1. Acceptable Manufacturers: Anvil International Inc. or approved equal by B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, Klo-Shure, National Pipe Hanger Corp., PHD Manufacturing Inc., or Unistrut.

PART 3 EXECUTION

3.01 INSTALLATION

- A. All piping supports, parts, and installation shall conform to the latest requirements of **2022 NFPA 13**, ASME B31.1, and MSS SP-58, SP-69, and SP-89, and shall not exceed the maximum pipe deflections allowed by ASME B31.9, except as supplemented or modified by the requirements of this Specification.
- B. Install all piping systems with adequate provisions made for expansion and to prevent stresses on valves and equipment. Provide adequate pipe anchors, guides, and supports from building structure to properly control pipe movement. Method shall suit job conditions. **Refer to Section 21 05 16.**
- C. Provide proper steel angles and channel systems, etc., between structural members for hanger supports where necessary. Weld to steel structural members where noted. Consult with the A/E regarding this procedure.
- D. All hangers installed shall be appropriate for type of building structure indicated.
- E. **Do not support hangers from roof deck or floor above.** Span from structural members with supplementary steel where direct attachment to structural members is not practical.
- F. Install hangers and supports with all necessary inserts, bolts, rods, nuts, washers, and other accessories, according to the Manufacturer's recommendations. Hangers shall be double-nutted.
- G. Use a separate hanger for each pipe. Do not hang one pipe from another.
- H. Provide additional supports required to maintain piping alignment.

- I. Spacing of Piping Supports: Locate and space piping supports according to NFPA requirements. Maximum spacing of piping supports shall be as follows:

STEEL PIPE SIZE	THREADED LIGHT WALL	STANDARD WEIGHT
1 in. to 3 in.	12 ft.	_____
1 in. to 1-1/4 in.	_____	12 ft.
1-1/2 in. and larger	_____	15 ft.
Exception: Locate hangers on exposed horizontal piping, mounted at less than 12 ft. above floor, at 8 ft. on center.		

- J. Securely support all piping from structure with all necessary hangers, rods, brackets, and accessories of approved type and size in accordance with 2022 NFPA 13.
- K. Securely fasten pipe and hang with NFPA-approved hangers spaced not less than 12 in. or more than 18 in. from a sprinkler. (1) hanger is required on each length of pipe and on cross mains between each branch line.
- L. All hangers shall support (5) times the total load of the water-filled pipe in addition to a minimum of 250 lb. load applied at the hanger per 2022 NFPA 13.
- M. Support piping near elbows, such that the total cantilevered length for both sides of the elbow does not exceed 3 ft.
- N. Support piping at pumps and equipment from floor, structure, or walls, so that piping weight is not supported by pumps or by equipment.

O. Structural Steel: Verify size required with the Structural Engineer.

- P. Option: Trapeze hangers may be used where parallel runs of pipe occur.

Q. Vertical Piping Hangers:

1. Support vertical risers at the floor with friction riser pipe clamps.
2. Shorten legs to conceal riser clamp within pipe chase and still provide adequate support.

R. Hanger Clamps:

1. DO NOT USE C-CLAMPS WITHOUT RETAINING CLIPS ON PIPING 1-1/2 IN. AND LARGER.
2. ~~Beam Clamps:~~ Use top flange beam clamps to avoid burning metal deck.

3. Purlin Clamps:

- a. ~~Provide additional auxiliary steel and fasteners to hang from purlins.~~
- b. ~~Where hanging from Z-purlins, use purlin clamps.~~
 - 1) ~~Option: Drill vertical portion of purlin and provide angle bolted to purlin and threaded for hanger rod. Obtain approval of the Lead Contractor, Metal Building Manufacturer, and A/E for attachment method before proceeding. Comply with the Metal Building Manufacturer's recommendations.~~

S. Metal Fabrication:

- 1. Fit connections together to form hairline joints.
- 2. Field-weld connections that cannot be shop-welded because of shipping size, space limitations, and field conditions.
- 3. When field-welding, comply with AWS D1.1 procedures for shielded metal arc welding, appearance and quality of welds, and methods used in correcting welding work.

T. Clean Room and Food Processing Areas: Provide hanger covers to cover and protect steel clevis type pipe insulation hangers and shields from insects, bacteria, moisture, dust, and dirt.

U. Ceilings:

- 1. Do NOT support piping from ceiling and ceiling grid systems.
- 2. Piping, and each ceiling and ceiling grid system, shall be a separate installation. Independently support each from the building structure.
- 3. Where interferences occur, in order to support piping, install trapeze-type hangers or supports. Locate them where they do not interfere with access to valves and other fire-suppression equipment items.

V. Painting:

- 1. Touch-up:
 - a. Clean field welds, bolted connections, and abraded areas of all shop paint.
 - b. Paint exposed areas immediately after erecting hangers and supports.
 - c. Use same paint materials as used for shop painting.
 - d. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.

2. Galvanized surfaces: Clean and apply galvanizing repair paint to comply with ASTM A780.

3.02 INSTALLATION — TRAPEZE HANGERS FOR PIPING

- A. Design support system. Provide support capacity, support spacing, trapeze hanger rod diameter, and quantity to support total piping load.
 1. All rods on trapeze hangers shall be 1/2 in. minimum.
 2. Clamp each pipe individually.
 3. Provide individual piping attachment to each strut channel hanger, angle, or channel.
- B. Obtain approval of a **Structural Engineer** licensed in **Ohio**. Submit details to the A/E for approval.

3.03 INSTALLATION — CONCRETE

- A. Provide timely notification and carefully coordinate locations, sizes, and extent of all concrete foundations and supports for fire-suppression equipment with the Lead Contractor **and Structural Engineer**.
- B. For any equipment item to be supported, provide dimensions for rough-in to the Lead Contractor.

DESIGNER — EDIT TABLE. Add “furnish,” “install,” or “provide”:

are all the items in the table applicable for fire suppression systems?

Item	Lead Contractor	Fire Suppression Contractor
Anchor bolts		
Concrete bases for		
Concrete curbs around pipe sleeves		
Concrete encasement of underground piping under footings and foundations		
Concrete equipment pads for		
Concrete fill for inertia base frames		

Item	Lead Contractor	Fire Suppression Contractor
Concrete-filled protection posts for [REDACTED]		
Concrete pads for [REDACTED]		
Concrete pad, reinforced, for [REDACTED]		
Concrete slabs with mitered edges		
Concrete surrounding [REDACTED]		
Concrete thrust blocks and supports for underground piping installation		
Steel reinforcing bars and mesh		

- C. Differences in elevation between (2) points shall not exceed 0.16 in. in 1 ft. and 0.38 in. in 5 ft. Slab area shall be moist-cured to prevent excessive moisture loss.

Markus — keep or delete D? Mike Y questions whether it is needed since the Site Contractor usually does underground fire suppression piping

- D. Concrete Encasement of Underground Piping:
- Clean around underground piping prior to encasement.
 - Where required, concrete encasement shall extend 6 in. around piping or sleeve for 1 ft. each side of footings or foundations, and up to footing bearing elevation. If piping is less than 2 ft. below grade to top of piping, provide 6 in. concrete encasement.
 - Backfill around piping after concrete has properly hardened.
- E. A Testing Laboratory Representative shall be present during the placement of all concrete unless this requirement is waived by the A/E. Provide ample notice to the Testing Laboratory and maintain records of all tests, indicating exact location of the structure represented by each test.
- Obtain concrete for required tests at point of placement.
 - Perform (1) strength test for each 50 yards, or fraction thereof, placed in any (1) day.
 - Determine slump for each strength test.
 - Determine air content for each strength test.

5. Determine concrete temperature for each strength test when air temperature is less than 40 degrees F. or will be within (24) hours.
 6. Do not place concrete if slump, air content, or air temperature vary from required limits.
- F. Curing:
1. Concrete may be either moist-cured or receive an application of curing compound.
 - a. Exception: When concrete above grade is placed in the open and the air temperature exceeds 60 degrees F., the concrete shall be moist-cured for the first (24) hours.
 2. Whichever curing method is used, it shall commence immediately after disappearance of water sheen and continue for at least (7) days. Do not allow curing to be delayed overnight.
 3. Prevent excessive moisture loss from formed surfaces. If forms are removed before (7) days have elapsed, cure the formed surfaces by moist-curing or application of curing compound for the remainder of the curing period.
- G. (Lead Contractor shall keep) (Keep) all concrete pads moist for (5) days.
- H. Concrete Testing: When concrete fails to meet the acceptance criteria specified in ACI 301, Section 17.2, the A/E may order further testing of in-place concrete, in accordance with Section 117.3. When such tests are ordered, the Lead Contractor shall pay cost of testing.
- I. Lead Contractor shall bear all costs of correcting rejected work.
- J. Refer to Division 03 for additional requirements regarding concrete work.

3.04 INSTALLATION — INSERTS

- A. Avoid cutting concrete or masonry by using inserts.
- B. Refer to the Manufacturer's recommendations for correct selection as to piping size and loading.
- C. Confirm with the Lead Contractor that the concrete used is of a sufficient strength to hold the insert.
- D. Provide the correct size and type of inserts, along with information on their correct location, to the Lead Contractor, so as not to delay construction.
- E. **Masonry Construction with Voids:** Install appropriately sized all-thread rods within a 2-component polyester resin adhesive used in conjunction with a galvanized metal screen. Install rods in holes drilled with carbide-tipped drill bits.

- F. **New Poured Concrete:** Before pouring, the Contractor shall furnish inserts (when used or required) for its Work to the Lead Contractor, who shall install where directed by the Contractor.

END OF SECTION 21 05 29

SECTION 21 05 43 - UNDERGROUND FIRE-SUPPRESSION SYSTEMS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 21 05 00.35, "Special Procedures for Fire Suppression."
- B. Section 21 05 00.73, "Execution of Fire Suppression."
- C. Section 21 05 29, "Hangers and Supports for Fire Suppression Piping and Equipment."

1.02 DESCRIPTION

- A. Provide:
 - 1. Excavation and backfill.
 - 2. Underground service identification tape.
 - 3. Paving.
- B. Concrete Work (pads, bases, slabs, encasement of piping, etc.): Refer to Section 21 05 29.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. Associated General Contractors of America (AGCA) "Manual of Accident Prevention in Construction".
 - 2. Occupational Safety and Health Administration (OSHA) requirements.
 - 3. State Department of Transportation (DOT) construction specifications.
 - 4. State and Local requirements.

PART 2 PRODUCTS

2.01 BACKFILL

- A. Typical backfill conditions:
 - 1. Piping where bedding is not specified or required by Code: Previously excavated material.

2. All underground piping: Crushed stone (DOT Item 304), pea gravel, or coarse sand from 4 in. below piping to slab or 4 in. below grade.
3. Other piping outside building: Pea gravel from 4 in. below piping to 12 in. minimum above pipe, then previously excavated material.
4. Piping under drives, streets, and parking lots: Pea gravel from 4 in. below piping to 12 in. minimum above pipe, then previously excavated material. Piping less than 2 ft. below grade to top of pipe shall be encased in 6 in. of concrete.
5. Piping in trenches cut through rock: Tamped backfill shall be used for at least 6 in. under and around pipe and for at least 2 ft. above pipe.
6. Piping under footings and foundations: Concrete encasement.

B. Materials:

1. Crushed stone: (DOT Item 304).
2. Pea gravel: 1/8 in. minimum to 3/4 in. maximum diameter.
3. Sand: Clean, dry, coarse or medium.
4. Washed gravel: 3/4 in. size.

C. Materials Prohibited from Being Used for Backfill:

1. Material containing large rocks (over 2 in.), building materials, masonry debris, cinders, rubbish, wood, or other material subject to decay.
2. Any material which may cause damage to piping.
3. Frozen earth.

2.02 UNDERGROUND SERVICE IDENTIFICATION TAPE

- A. Acceptable Manufacturers: Seton or approved equal by Brady, Ideal, or Emedco.
- B. Seton. 6 in. wide metallic underground warning/detection tape detectable with metal detector.

2.03 PAVING

A. Asphalt Paving:

1. Aggregate base: DOT Item 304, 8 in. total, compacted in (2) 4 in. layers.
2. Base course: DOT Item 402, 1-3/4 in. thickness.
3. Surface course: DOT Item 404, 1-1/4 in. thickness.
4. Tack coat: DOT Item 407.

5. Prime coat: DOT Item 408.

B. Sealer: DOT Item 409.

PART 3 EXECUTION

3.01 EXCAVATION

A. Existing Soil Conditions:

1. Bidders are strongly urged to visit the site and investigate existing soil conditions themselves.
2. Rock excavation is not anticipated. Contractor will be entitled to additional compensation for excavating any rock that is not indicated on the Contract Drawings.
 - a. Rock excavation is defined as the excavation of any buried boulders and rock, in excess of 1/2 cu. yd. in volume, which requires systematic drilling with jackhammer or use of other special equipment.
 - b. If rock or boulders are encountered, immediately stop all excavation. Notify the A/E, and submit a cost per cubic yard, over and above Base Bid, for excavation and backfill of rock areas.
 - c. All other excavation, including soft shale and similar sedimentary soil formations, shall be regarded as "earth" excavation, and shall be included in the Contractor's Bid.
 - d. Do not proceed with rock excavation without prior written authorization of the A/E.
 - e. DO NOT USE EXPLOSIVES.
3. If available, refer to Soils Report.
4. Whenever, in the opinion of the A/E, the soil is unsuitable for supporting piping, provide proper foundation after receiving written approval of the A/E. The Contract price will be adjusted accordingly.

B. Obtain approval from the A/E for bearing conditions.

C. Existing Underground Utilities:

1. Contact the Julie Before You Dig Utility Service (800-892-0123) well in advance of the start of any excavation to determine if any of the Utility Companies have underground utilities in, or near, the project area.
2. Contact the Owner and the Local Water and Sewer Department, Gas Company, Electric Company, Telephone Company, etc., regarding the possibility of encountering existing utilities. Maintain the integrity of all existing utilities.

3. Locate and identify existing underground utilities, if any, in the areas of new Work. If utilities are to remain in place, provide adequate means of protection during earthwork operations. Should uncharted, or incorrectly charted, piping or other utilities be encountered during excavation, notify the A/E immediately for instructions before proceeding. Cooperate with other Contractors, Utility Companies, and the Owner in keeping services and the facilities in operation.
 4. Protect existing utilities encountered during excavation work in a manner acceptable to the Owner of the Utility. Excavation damage to, or interruption of service of, any underground utility shall be a liability of the Contractor, whether or not plotted on the Drawings. Contractor shall promptly repair or replace, at its expense, all damage to any existing utility to the full satisfaction of the Utility Company, the Owner of the Utility, and the A/E.
 5. All existing utility service piping, conduits, wiring, etc., which are uncovered, whether within or beyond the property lines, shall be suitably supported, protected, and maintained in operation, and shall be protected against settlement when excavations are refilled.
- D. Coordinate timing of excavations in advance with other Contractors, the A/E, and the Owner, where required by AHJ, or where otherwise specified herein.
- E. Excavations shall be open-cut from the surface. No undercuts will be permitted, except where specifically directed by the A/E.
- F. Remove surplus and/or unsuitable-for-fill excavated material from the site. Deposit any usable excavated material where directed by the A/E.
- G. Excavation Safety:
1. ALL EXCAVATION WORK SHALL COMPLY WITH REQUIREMENTS OF THE LATEST OSHA STANDARDS.
 2. Excavation and trench wall supporting, shoring, sloping, cribbing, stepping of excavations, and other steps required for safety shall be in strict accordance with OSHA and Local Code requirements.
 3. Slope sides of excavations to comply with requirements of OSHA and AHJ. Shore and brace where sloping is not possible because of space restrictions or instability of material exposed.
 4. Maintain sides and slope of all excavations in a safe condition, until completion of backfilling.
 5. Provide materials for shoring and bracing, such as sheet piling or sheathing, uprights, stringers, and cross-braces, in good serviceable condition required to safely support walls of the excavation. Maintain shoring and bracing in excavations regardless of time period excavations will be open. Carry down shoring and bracing as excavation progresses.

6. Barricade open excavations occurring as part of this Work, and post with warning lights. Erect warning lights in accordance with OSHA and AHJ requirements. Consult with the A/E regarding additional requirements.
 7. Maintain in place adequate barricades, guards, planking, plating, signage, warning lights, etc., at and around excavations. Maintain in a safe condition until backfilling is completed.
 8. Install steel cover plates, safety barricades, and coverings for holes in walks, pavements, streets, or grassy areas.
 9. Protect existing structures that are to remain, such as utilities, sidewalks, pavements, and other facilities, from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
 10. Refer to AGCA's "Manual of Accident Prevention in Construction" and to the safety regulations of the appropriate State Agency.
 11. Refer to Section 21 05 00.35 for additional safety requirements.
- H. Provide and operate pumping equipment to keep excavation free of water at all times.
- I. Protect excavations from frost by covering and heating.
- J. Trenches:
1. Coordinate the use of a joint trench with all Utility Companies sharing the trench.
 2. Do not excavate utility trenches closer than 5 ft. parallel to building or column footings, except with prior written approval of the A/E. When parallel trenches deeper than the building footings are required, the horizontal distance from the footing shall be equal to, or greater than, 1-1/2 times the vertical distance below the footing, but in no case shall the horizontal distance be less than 5 ft., except with the written approval of the A/E.
 3. Coordinate routing of underground lines with the A/E to minimize disruption and damage to trees, shrubs, walks, drives, and other landscape features.
 4. Mark trench location. Hold trench width to a minimum. Cover as soon as possible.
 5. Minimize time trenches are kept open. Fill as soon as possible.
- K. Roadways and Public Right-of-Way:
1. **Lead Contractor shall saw-cut** any street, drive, sidewalk, curb, parking lot paving, or other permanent hard surface that is encountered in the path of the excavation.
 2. Avoid the need to cut concrete or paving by roughing in piping BEFORE concrete or paving is installed.
 3. Cut and replace any exterior paving or sidewalks to install underground work.

4. Excavation, backfill, surface repair, and traffic control within the public right-of-way shall be in accordance with governing agency rules and regulations.
 5. Contractor shall pay any fee associated with alteration of a roadway.
- L. Refer to Section 21 05 00.73 for cutting and patching responsibilities.
- M. Backfilling:
1. The bedding and backfill requirements listed in this Section represent a minimum requirement. If State or Local requirements are in excess of listed requirements, such requirements shall supersede.
 - a. Reference datum: All dimensions or elevations are relative to finished grade (not existing grade).
 2. ALL BACKFILL WORK SHALL COMPLY WITH REQUIREMENTS OF THE LATEST OSHA STANDARDS.
 3. Deposit fill in layers of thickness required by the nature of the soil, or as directed, but not exceeding 6 in. compacted thickness, to a point 24 in. above pipe, and 12 in. thickness above this point. Compact each layer to a uniform solid mass. Place fill in horizontal layers, beginning with lowest areas and building up until entire area to be filled is at a uniform elevation. Compaction shall be a minimum of 90 lbs. per cu. ft., laboratory dry weight.
 4. Do not place fill material on frozen ground, or use fill in a frozen condition.
 5. Backfill only when exact locations of lines and equipment have been recorded, and all tests and inspections have been completed.
 6. Control moisture content of compacted fill to ensure maximum density by adding water and working soil prior to compacting.
 7. Use machine tampers around perimeter of foundation walls or areas inaccessible to large equipment and rollers. Do not use rolling equipment in areas adjacent to foundations.
 8. Provide clay bulkheads, minimum 3 ft. long, across the trench at 100 ft. intervals to impede the natural movement of ground water. Top of bulkheads shall extend 6 in. above top of pipe.
 9. Backfill to finish grade or to levels suitable for finish grading.
 10. Lead Contractor shall provide finish grading.
 11. Where excavations have not been properly filled or where settlement occurs, refill, compact, smooth off, and finish excavations to a final condition acceptable to the Owner and A/E.
 12. Install continuous underground service identification tape above buried outside piping service at 18 in. above pipe elevation on pipe centerline to identify all the underground Fire Suppression Work.

**UNDERGROUND FIRE
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(Rev:)

13. Mark location of trenches and lines on As-Built Drawings.
14. (11) months after Contract Completion, review the excavated areas and add any backfill required due to settling, etc. Re-seed or re-sod.
15. **Lead Contractor shall provide** seed grass cover to reestablish landscape to previous condition where disturbed by Fire Suppression Work. This includes replacement of existing plants, shrubs, and related items.
16. **Lead Contractor shall repair** and restore plants, shrubs, trees, sodded areas, paving, streets, curbs, and walks. Match existing in the area where excavations are made.
17. **Lead Contractor shall maintain** seeded lawns not less than (11) months after Contract Completion and add any backfill required due to settling, etc. If seeded in non-growing season and not given full (11) months of maintenance, or if not considered acceptable at that time, continue the following growing season until an acceptable lawn is established, as determined by the A/E.

3.02 INSTALLATION — UNDERGROUND FIRE-SUPPRESSION PIPING

- A. Provide excavation and backfilling or directional boring for the Fire Suppression Work.
- B. Bedded Piping:
 1. Provide pipe bedding for all water lines located outside of the building.
 2. Provide pipe bedding for all water lines located below the floor slab.
 3. Mechanical excavation shall extend 4 in. to 6 in. below final elevation of pipe.
 4. In trenches cut through rock, mechanical excavation shall extend 6 in. below and around final elevation of pipe.
 5. Bed pipes in pea gravel, minimum 4 in. below pipe, 6 in. above pipe.
- C. Unbedded Piping:
 1. Mechanical excavation shall be held to 4 in. above final elevation. The remainder shall be trenched by manual excavation, so that piping is fully supported on undisturbed soil.
 2. DO NOT SHORE PIPING IN TRENCH.
 3. If excavation should go deeper than required, fill with bank run gravel and tamp firmly to achieve final elevation. Wherever material is encountered in the bottom of the excavation that is not capable of supporting piping properly, remove such material to depth required. Backfill with crushed stone and tamp firmly to achieve final elevation.
- D. Place no backfill until underground lines have been tested. Compact backfill in 6 in. deep layers.
- E. Contractor shall be responsible for any other repairing of excavated areas.

- F. Coordinate with the Lead Contractor.

3.03 PAVING

- A. **Lead Contractor shall repave** all areas affected by fire-suppression construction work. Paving shall either match existing pavement or comply with the requirements listed below.
- B. Preparation:
1. Proof-roll the sub-grade. Roll and compact to obtain an acceptable, firm, even sub-grade surface. Fill and consolidate dented or depressed areas. Remove incompatible material, replace with clean fill, and compact solidly.
 2. Coordinate junction of new and existing pavement. Meet existing surface levels and maintain drainage slopes.
- C. Installation:
1. Base Materials: Install 8 in. aggregate base material and compact in accordance with DOT Item 304 until a uniformly smooth, hard surface, conforming to the lines, grades, elevations, and cross-section shown, has been established. Moisture may be added at job site to aid compaction.
 2. Surface Materials:
 - a. Apply prime coat uniformly to aggregate base in accordance with DOT Item 408, at rate of 0.25 gallons to 0.30 gallons per sq. yd. Allow to dry and cure.
 - b. Install asphalt materials in (2) layers and compact each layer in accordance with DOT Item 401. Provide a hard, smooth, unyielding surface, true to thickness and elevation shown. Roll until no roller marks are visible.
 - c. After final rolling, do not permit vehicular traffic on the pavement until it has cooled and hardened, i.e., for at least (8) hours.
- D. Final Sealing:
1. Prior to sealing, thoroughly clean and sweep area. Block all traffic to areas to be sealed. Provide striping as shown on the Drawings. Maintain barriers until sealing and striping is complete.
 2. Apply (2) coats of sealer to all asphalt surfaces (30) days after paving installation.

END OF SECTION 21 05 43

SECTION 21 05 48 - VIBRATION AND SEISMIC CONTROLS FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 21 05 16, "Expansion Fittings and Loops for Fire-Suppression Piping."
- B. Section 21 05 29, "Hangers and Supports for Fire-Suppression Piping and Equipment."

1.02 DESCRIPTION

- A. Provide a complete vibration isolation system to isolate motorized equipment, piping, and appurtenances from the building structure and ceiling construction.
- B. Refer to the Structural Drawings for the Seismic Design Category.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. American Society for Testing & Materials (ASTM).
 - 2. International Code Council, Inc. (ICC):
 - a. International Building Code (IBC).
 - b. Evaluation Service (ES).
 - 3. National Fire Protection Association (NFPA): 2023 NFPA 70, "National Electrical Code® (NEC)."
 - 4. Occupational Safety and Health Administration (OSHA) 29 CFR 1910.7.
- B. Testing Agency Qualifications: An independent agency, with the experience and capability to conduct the testing indicated, that is a nationally recognized testing laboratory as defined in 29 CFR 1910.7, and that is acceptable to the AHJ.

1.04 SUBMITTALS

- A. The Isolator Manufacturer's submittal shall include drawings showing the complete design for the supplementary bases and a tabulation of the design data on the isolators, including the following:
 - 1. Outside diameter of the springs.
 - 2. Free, operating, and solid heights of the springs.
 - 3. Free and operating heights of the neoprene or fiberglass isolators.
 - 4. Isolation efficiency based on the lowest operating speed of the equipment supported.

PART 2 PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

- A. Kinetics Noise Control Inc. (KNC) or approved equal by Vibration Mountings and Controls Inc., Mason Industries Inc., or Vibro-Acoustics.
- B. A single Manufacturer or Supplier shall design and furnish vibration isolation devices, including auxiliary steel bases and pouring forms.
- C. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number.
- D. Other Acceptable Manufacturers will be permitted, unless specifically excluded.

2.02 ISOLATOR SCHEDULE

- A. Furnish isolator types as follows:

EQUIPMENT ITEM	SPECIFICATION SECTION	ISOLATOR
Piping	21 10 00	KNC Model SFH

2.03 VIBRATION ISOLATORS

- A. Spring "S" Mounts: KNC Model FDS. Freestanding, unboxed stable spring mounts incorporating leveling bolts, and a means of bolting to the support. Mounts shall have 1/2 in. neoprene-jacketed fiberglass noise isolation pads bonded to the bottom of the spring mounts. To ensure stability, springs shall be sized according to Manufacturer requirements. The springs shall have a minimum additional travel of 50% between the design height and the solid height.
- B. Hanger "H" Mounts: KNC Model SFH or Model SH. Combination spring and fiberglass isolation hangers incorporating 2 in. thick neoprene-jacketed fiberglass inserts in series with spring, all encased in welded steel brackets. The spring shall have a minimum additional travel of 50% between the design height and the solid height.
- C. Inertia "I" Bases: KNC Model CIB. Prefabricated base frame with primer steel perimeter members, recessed isolator brackets, and welded in and tied reinforcing bars. Combination spring and neoprene isolation hangers in series with top and bottom load plate assembly.

2.04 SEISMIC-RESTRAINT DEVICES

- A. Acceptable Manufacturers: Amber/Booth Company, California Dynamics Corporation, Cooper B-Line, Hilti, Loos & Co., Mason Industries, TOLCO, and Unistrut.
- B. Seismic-restraint devices shall have horizontal and vertical load testing and analysis and shall bear preapproval by ICC-ES, or preapproval by another agency acceptable to AHJ,

showing maximum seismic-restraint ratings. Ratings based on independent testing are preferred to ratings based on calculations. If preapproved ratings are not available, submittals based on independent testing are preferred. Calculations (including combining shear and tensile loads) to support seismic-restraint designs shall be signed and sealed by a qualified professional engineer.

- C. Rated strengths, features, and application requirements shall be as defined in reports by an evaluation service member of ICC-ES or another agency acceptable to AHJ.
- D. Structural Safety Factor: Allowable strength in tension, shear, and pullout force of components shall be at least (4) times the maximum seismic forces to which they will be subjected.
- E. Channel Support System: MFMA-3, shop-fabricated or field-fabricated support assembly made of slotted steel channels with accessories for attachment to braced component at one end and to building structure at the other end and other matching components and with corrosion-resistant coating; and rated in tension, compression, and torsion forces.
- F. Restraint Cables: (ASTM A 603 galvanized) (ASTM A 492 stainless)-steel cables with end connections made of steel assemblies with thimbles, brackets, swivels, and bolts designed for restraining cable service; and with a minimum of (2) clamping bolts for cable engagement.
- G. Hanger Rod Stiffener: (Steel tube or steel slotted-support-system sleeve with internally bolted connections) (Reinforcing steel angle clamped) to hanger rod. Do not weld stiffeners to rods.
- H. Bushings for Floor-Mounted Equipment Anchor: Neoprene bushings designed for rigid equipment mountings, and matched to type and size of anchors and studs.
- I. Bushing Assemblies for Wall-Mounted Equipment Anchorage: Assemblies of neoprene elements and steel sleeves designed for rigid equipment mountings, and matched to type and size of attachment devices.
- J. Resilient Isolation Washers and Bushings: 1-piece, molded, oil-resistant and water-resistant neoprene, with a flat washer face.
- K. Mechanical Anchor: Drilled-in and stud-wedge or female-wedge type in zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchors with strength required for anchor and as tested according to ASTM E 488. Minimum length of (8) times diameter.
- L. Adhesive Anchor: Drilled-in and capsule anchor system containing polyvinyl or urethane methacrylate-based resin and accelerator, or injected polymer or hybrid mortar adhesive. Provide anchor bolts and hardware with zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

PART 3 EXECUTION

3.01 PREPARATION

- A. Steel components shall be phosphatized and painted. All nuts, bolts, and washers shall be zinc-electroplated.
- B. Structural steel bases shall be thoroughly cleaned of welding slag and primed with zinc-chromate or metal etching primer.
- C. All isolators exposed to the weather shall have steel parts PVC-coated or hot-dip-galvanized.
- D. Aluminum components shall be etched and painted.

3.02 INSTALLATION

- A. Coordinate installation with the Lead Contractor.
- B. Equipment: Mount on, or suspend with, vibration isolators all motorized equipment, including motors, pumps, etc.
- C. Piping:
 - 1. Isolate all piping, within 50 ft. of mechanical equipment mounted on vibration isolators, from the structure via vibration and noise isolators. Isolate suspended piping by incorporating combination spring and fiberglass hangers in the supporting rods. Install hangers at regular intervals. **Refer to Section 21 05 29.**
 - 2. Isolate vertical pipe risers from the structure via vibration-isolating and noise-isolating expansion hangers. The hangers shall have a minimum rated deflection of (4) times the anticipated pipe expansion and shall be enclosed in a housing for fail-safe operation.
 - 3. Provide acoustic seals at wall openings through which piping runs from the equipment room into adjoining spaces.
- D. Concrete Fill for Inertia Base Frames: Refer to Section 21 05 29.

3.03 INSTALLATION — SEISMIC-RESTRAINT DEVICE

- A. Comply with seismic-restraint requirements in the IBC unless requirements in this Section are more stringent.
- B. Equipment and Hanger Restraints:
 - 1. Install restrained isolators on fire suppression equipment.
 - 2. Install resilient, bolt-isolation washers on equipment anchor bolts where clearance between anchor and adjacent surface exceeds 0.125 in.
 - 3. Install seismic-restraint devices using methods approved by an evaluation service member of ICC-ES or another agency acceptable to AHJ providing required submittals for component.

- C. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.
- D. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure at flanges of beams, at ~~upper truss chords or bar joists,~~ or at concrete members.
- E. Drilled-in Anchors:
 - 1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not damage existing reinforcing or embedded items during coring or drilling. Notify the **Structural Engineer** if reinforcing steel or other embedded items are encountered during drilling. Locate and avoid ~~prestressed tendons,~~ electrical and telecommunications conduit, and gas lines.
 - 2. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.
 - 3. Wedge anchors: Protect threads from damage during anchor installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.
 - 4. Adhesive anchors: Clean holes to remove loose material and drilling dust prior to installation of adhesive. Place adhesive in holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive.
 - 5. Set anchors to the Manufacturer's recommended torque, using a torque wrench.
 - 6. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

END OF SECTION 21 05 48

SECTION 21 05 53 - IDENTIFICATION FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT

PART 1 GENERAL

1.01 DESCRIPTION

- A. Identify the following fire-suppression items by labels, tags, or stencils:
1. **New** equipment, control cabinets, and similar items.
 2. **New** equipment located above an accessible ceiling: Every location of a , etc.
 3. **New** piping:
 - a. Exposed in equipment rooms, **trenches, crawl spaces, tunnels**, accessible service areas, or accessible chases.
 - b. Running above accessible ceiling construction.
 4. All **new** valves.
 5. **New** valves located above an accessible ceiling: Every location of a shut-off valve, etc.
- B. Do NOT use "Dymo" or similar tape type labels.

1.02 QUALITY ASSURANCE

- A. Standards:
1. American National Standards Institute / American Society of Mechanical Engineers (ANSI/ASME):
 - a. ANSI/ASME A13.1, "Scheme for the Identification of Piping Systems."
 - b. ANSI Z535.1, "Safety Color Code."
 2. Occupational Safety and Health Administration (OSHA).

PART 2 PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

- A. Seton, Brady, Emedco, or MSI.

2.02 EQUIPMENT IDENTIFICATION

- A. Label: Aluminum, brass, laminated multi-color rigid plastic, or stainless steel, with holes for attachment to equipment.
 - 1. Size: Sufficient for text, minimum 2-1/2 in. wide by 3/4 in. high.
 - 2. Text characters: Engraved or stamped. Equipment tag text characters sized for normal viewing distance, minimum 1/4 in. high, maximum 3/4 in. high. Equipment specification number text characters approximately 3/4 size of tag characters.
 - 3. For laminated plastic labels, match background and character colors in system identification schedules in Part 3. If exact matches are not available, match background colors from schedules, and provide character colors in contrasting white on black.

2.03 PIPING IDENTIFICATION

- A. 1 in. wide, gloss enamel color band. Stencils, same color as band.
- B. For first color, 1 in. wide color band. For second color, 1/2 in. wide color band. Stencils.
- C. Labels: Preprinted, color-coded plastic, with text characters indicating pipe service and arrows indicating flow direction. **Include pressure rating where applicable.**
 - 1. Pre-tensioned: Pre-coiled and semi-rigid label formed to cover full circumference of pipe (including insulation and jacket) and to adhere to pipe without fasteners or adhesives.
 - 2. Self-adhesive: Flexible label long enough to cover full circumference of pipe (including insulation and jacket) with permanent contact adhesive backing.

2.04 VALVE IDENTIFICATION

- A. Tags: Aluminum, brass, laminated multi-color rigid plastic, or stainless steel, with hole for attachment to valve.
 - 1. Size: Minimum 2 in. diameter round or 1-1/2 in. square.
 - 2. Text characters: Engraved or stamped. 1/4 in. high indicating pipe service and 1/2 in. high indicating valve number.
 - 3. Fasteners: Metal bead chain, metal "S" hook, or nylon tie.
- B. Brass or aluminum tags with brass or aluminum chain.

- C. Tags: Aluminum, brass, laminated multi-color rigid plastic, or stainless steel, with hole for attachment to valve.
 - 1. Size: Minimum 2 in. diameter round or 1-1/2 in. square.
 - 2. Text characters: Engraved or stamped. 1/4 in. high indicating pipe service and 1/2 in. high indicating valve number.
 - 3. Fasteners: Metal bead chain, metal "S" hook, or nylon tie.

2.05 STENCILS

- A. Card stock or metal pre-cut stencil with equipment tag, or piping service and flow direction arrow.
- B. Paint: Exterior grade, gloss alkyd enamel. Colors in accordance with schedules listed in Part 3.

PART 3 EXECUTION

3.01 PREPARATION

- A. Clean surfaces of substances which could impair bond of identification device adhesive, where applicable.

3.02 COORDINATION

- A. Coordinate to ensure that the identification used by all Trades is uniform in type, style, and appearance.
- B. Equipment: Coordinate the exact nomenclature to be used on labels with the Owner and A/E.
- C. Piping: Coordinate with the Owner regarding piping identification system. If piping identification system does not exist, refer to Identification Schedule.
- D. Valves: Coordinate with the Owner regarding valve numbering system. If valve-tagging system does not exist, refer to Identification Schedule.
- E. Coordinate label, tag, and fastener materials with expected conditions. Do not use plastic materials in locations where expected temperatures are outside of the Manufacturer's recommended limits.

3.03 INSTALLATION

A. Equipment:

1. Provide a label for each equipment item that has a unique tag. Do not provide a label if multiple items of equipment have the same tag.
2. Locate equipment label where it will be accessible and visible when project is complete. For equipment that is installed exposed in finished space, locate tag in an accessible concealed location, such as behind an access door.
3. Attach label with stainless steel screws.
 - a. Exception: Use compatible adhesive where screws might damage equipment.

B. Piping:

1. Apply piping identification only after any finish painting is completed.
2. Provide labels for piping at MAXIMUM 25 FT. intervals.
3. Identify piping at connections to equipment, at valves, at branches from main, and at each riser.
4. Where a pipe passes through a wall, label the pipe where it enters and exits the wall.
5. Option: Stencils may be used instead of manufactured labels. Apply stencil over background, in colors indicated. When paint is dry, provide a clear coat finish over stencil. Stencils shall be legible from a standing position.

C. Valves:

1. Provide a unique tag on each valve. Tags shall be stamped with service designation and numbered consecutively for each service. Attach tags to valves in such a manner that valve shall be operable without removing or damaging tag. Tags shall be readable from a standing position when valve is in normal position. Add valve tag numbers to As-Built Drawings.
2. Prepare a valve tag directory with charts showing locations, designations, and sizes of valves. Laminate under plastic and mount as directed by the A/E. Include additional copies of valve charts in the Operating and Maintenance Manuals.

3.04 PIPING IDENTIFICATION SCHEDULE

A. Text character size shall be as follows:

Overall Pipe Size	Minimum Character Height
Up to 1-1/4 in. diameter	1/2 in. high

Overall Pipe Size	Minimum Character Height
1-1/2 in. to 2 in. diameter	3/4 in. high
2-1/2 in. to 6 in. diameter	1-1/4 in. high
8 in. to 10 in. diameter	2-1/2 in. high
12 in. and larger diameter	3-1/2 in. high

B. Identify piping as follows:

Type of Service	Band Color	Designation
Fire Line	Red	FL
Drain	(Room Color)	—

C. Identify piping as follows:

Type of Service	Background Color	Striping Color	Designation
Wet Sprinkler	Red	White	FIRE-WET
Dry Sprinkler	Red	White	FIRE-DRY
Drain	Green	White	FIRE DRAIN
Compressed Air	Blue	White	FIRE CA
Dry Chemical Suppression	Orange	Black	DRY CHEM

D. Identify piping as follows:

Type of Service	Background Color	Character Color	Designation
Wet Sprinkler	Red	White	FIRE-WET
Dry Sprinkler	Red	White	FIRE-DRY
Drain	Green	White	FIRE DRAIN

Type of Service	Background Color	Character Color	Designation
Compressed Air	Blue	White	FIRE CA
Dry Chemical Suppression	Orange	Black	DRY CHEM

3.05 VALVE IDENTIFICATION SCHEDULE

A. Identify valves as follows:

Type of Service	Valve Tag Designation
Water Service	F
Standpipe	SP
Combination Standpipe and Sprinkler	F
Sprinkler	FIRE
Drain	D
Compressed Air	CA
Dry Chemical Suppression	DRY CHEM

END OF SECTION 21 05 53

SECTION 21 10 00- WATER-BASED FIRE-SUPPRESSION SYSTEMS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 10 44 00, "Fire Protection Specialties."
- B. Section 21 05 23, "General-Duty Valves for Water-Based Fire-Suppression Piping."
- C. Section 21 05 29, "Hangers and Supports for Fire-Suppression Piping and Equipment."
- D. Section 21 05 48, "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."
- E. Section 21 05 53, "Identification for Fire-Suppression Piping and Equipment."
- F. Section 21 12 00, "Fire-Suppression Standpipes."
- G. Section 21 13 13, "Wet-Pipe Sprinkler Systems."
- H. Section 21 13 16, "Dry-Pipe Sprinkler Systems."

1.02 DESCRIPTION

- A. Provide piping and fittings as shown or required to completely install all water-based fire-suppression systems on this Project.
- B. Valves: Refer to Section 21 05 23 and to the Drawings.
- C. Provide water-based fire-suppression piping specialties or accessories as shown or required by the Equipment Manufacturer or AHJ to complete the installation of all fire-suppression systems installed on this Project.
 - 1. Automatic ball drip.
 - 2. Automatic drain device.
 - 3. Backflow preventers (BFPs).
 - 4. Drains.
 - 5. Electric bell.
 - 6. Extinguishers, dry chemical: Refer to Section 10 44 00.
 - 7. Fire department connection.

8. Fire department (siamese) connection.
9. Fire hose rack.
10. Fire hose reel.
11. Floor control valve sectional test and drain device.
12. Gauges, pressure.
13. Hydrants, roof.
14. Identification signs.
15. Inspector's test and drain assembly.
16. Pump test connection.
17. Switches, pressure alarm for dry-pipe sprinkler piping system.
18. Switches, valve supervisory.
19. Switches, water flow.
20. Water motor gong.

D. All products of the same type shall be by the same Manufacturer.

1.03 QUALITY ASSURANCE

A. Standards:

1. American National Standards Institute (ANSI).
2. American Society for Testing and Materials (ASTM).
3. American Society of Mechanical Engineers (ASME):
 - a. ASME/ANSI B31.1, "Code for Power Piping."
 - b. ASME Boiler and Pressure Vessel Code, Section IX, "Welding and Brazing Qualifications."
4. American Society of Sanitary Engineers (ASSE).
5. American Welding Society (AWS).
6. Factory Mutual Global (FMG).
7. Food and Drug Administration (FDA).
8. National Electrical Manufacturers Association (NEMA).
9. National Fire Protection Association (NFPA):
 - a. 2022 NFPA 13, "Standard for the Installation of Sprinkler Systems."

- b. 2019 NFPA 14, "Standard for the Installation of Standpipes and Hose Systems."
 - c. 2025 NFPA 24, "Standard for the Installation of Private Fire Service Mains and Their Appurtenances."
 - d. 2023 NFPA 70, "National Electrical Code® (NEC)," Articles 110 and 408.
- 10. Underwriters Laboratories, Inc. (UL), including UL 393
 - 11. Code of Federal Regulations (CFR) Title 40, "Protection of Environment," Part 59, "National Volatile Organic Compound Emission Standards for Consumer and Commercial Products," Subpart D, "National Volatile Organic Compound Emission Standards for Architectural Coatings."
 - 12. California Department of Health Services (CDHS) "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."
 - 13. Illinois Administrative Code (IAC):
 - a. IAC Title 41: Fire Protection
 - b. IAC Title 71: Public Buildings, Facilities, and Real Properties

PART 2 PRODUCTS

2.01 PIPING AND FITTINGS FOR SPRINKLERS AND STANDPIPES

- A. Threaded, Welded, Or Grooved Piping: Schedule 30 or Schedule 40 steel as listed in 2022 NFPA 13.
- B. Mains, Larger than 2 in.: Schedule 10 roll-grooved pipe shall be permitted. SCHEDULE 5 SHALL NOT BE USED.
- C. Piping, 2 in. and Smaller: ANY WALL THICKNESS LESS THAN SCHEDULE 30 SHALL NOT BE USED.
- D. Black steel (ASTM A53).
- E. Wrought steel (ANSI B36.10M).
- F. Seamless welded steel (ASTM A53, ASTM A135, or ASTM A795).
- G. Type "K," "L," or "M" seamless copper tubing (ASTM B88), equivalent in strength and pressure rating to respective steel pipe specified.
- H. Fittings:

1. Cast iron threaded or flanged fittings: 125 lb. SWP or 250 lb. SWP (ANSI B16.4 and ANSI B16.1).
2. Malleable iron threaded fittings: 150 lb. SWP or 300 lb. SWP (ANSI B16.3).
3. Ductile iron grooved fittings (ASTM A536, Grade 65-45-12).
4. Factory-made wrought steel, butt-weld fittings (ANSI B16.9).
5. Butt-welding ends for pipe, valves, flanges, and fittings (ANSI B16.25).
6. Steel pipe flanges and flanged fittings (ANSI B16.5).
7. Forged steel fittings, socket-welded and threaded (ANSI B16.11).
8. Wrought copper and bronze solder-joint pressure fittings (ANSI B16.22).

Marcus — Mike Y says T-drill seldom used. Suggests using Pro-Press fittings.

- a. Option: "T-drill" mechanically formed tee connections, brazed joints.
- I. Nipples: Same weight and material as connected pipe, except shoulder nipples shall be extra-heavy. Do not use short nipples.
- J. CPVC Pipe:
1. ASTM F 442/F 442M and UL 1821, SDR 13.5, for 175-psig rated pressure at 150 degrees F., with plain ends. Include "LISTED" and "CPVC SPRINKLER PIPE" markings.
 2. Option: Lubrizol "Blazemaster." CPVC pipe/fittings, SDR 13.5, meeting cell class 23547/24447 in accordance with ASTM D1784, with pipe in accordance with ASTM F442, and listed for automatic fire sprinkler systems.
 - a. Acceptable Manufacturers: Lubrizol or approved equal Spears "FlameGuard."
 3. CPVC Fittings: UL-listed, FMG-approved, 175-PSIG rated pressure at 150 degrees F, socket type. Include "LISTED" and "CPVC SPRINKLER FITTING" markings.
 - a. NPS 3/4 to NPS 1-1/2: ASTM F 438 and UL 1821, Schedule 40, socket type.
 - b. NPS 2 to NPS 3: ASTM F 439 and UL 1821, Schedule 80, socket type.
 - c. CPVC-to-metal transition fittings: CPVC, 1-piece, with dimensions equivalent to pipe; (1) end with threaded brass insert, and (1) socket end.
 - d. CPVC-to-metal transition unions: CPVC, with dimensions equivalent to pipe; (1) end with threaded brass insert, and (1) socket end.
 - e. Flanges: CPVC, (1) or (2) pieces.

2.02 SOLDERING, BRAZING, AND WELDING

- A. Solder: ASTM B32 alloy Sn94 and Sn95. 95/5 tin-antimony, Grade 95, lead free.
 - 1. Acceptable Manufacturers and Models: Engelhard "Silvabrite 100," J.W. Harris Co. "Bridgit," or Oatey.
- B. Brazing: AWS 5.8. Classification BCUP-3 or BCUP-4.
- C. Welding: AWS D10.12 for sizes 8 in. or less.

2.03 SOLVENT CEMENTS AND ADHESIVE PRIMERS

- A. For Joining CPVC Piping and Tubing: ASTM F493. Compliant with CDHS.
 - 1. Use CPVC solvent cement with a VOC content of 490 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 - 2. Use adhesive primer with a VOC content of 550 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.04 JOINTS, UNIONS, AND CONNECTIONS

- A. Threaded Piping Joints:
 - 1. National Pipe Thread (NPT) taper, general purpose (ASME B1.20.1).
 - 2. TFE tape, applied to male thread only.
 - 3. Option: Thread compounds by Permatex.
- B. Copper Pipe Unions: Cast bronze, Class 150, ground joint, solder end. Do NOT use wrought copper unions.
 - 1. Acceptable Manufacturers: Chase, Crane, Dart, Epco, Flagg, Mueller, Nibco, or Rockwell.
- C. Steel Pipe Unions: Black malleable iron, Class 250, bronze ground ball joint.
 - 1. Acceptable Manufacturers: Chase, Crane, Epco, Fairbanks, Grinnell, Mueller, or Stockham.
- D. Dielectric Unions: Metal union with dissimilar materials electrically separated by gasket. Union materials, end connections, and pressure rating compatible with piping materials, system fluid, system pressures, and system temperatures.
 - 1. Acceptable Manufacturers: Capitol, Dart, Epco, Vogt, or Watts.
- E. Flanges: Cast iron or cast bronze, 125 lb. or higher. ASME B16.5 flat-face or raised-face metal flange plates with matching bolt holes, gasket, bolts, nuts, and washers. Flange

materials, end connections, and pressure rating compatible with piping materials, system fluid type, system pressures, and system temperatures.

- F. Dielectric Flanges: Flange with electrically-isolating gasket, bolt sleeves, and bolt washers. Flange materials, end connections, and pressure rating compatible with piping materials, system fluid type, system pressures, and system temperatures.

1. Acceptable Manufacturers: Capitol, Crane, Dart, Epco, or Vogt.

- G. Gaskets: Nonmetallic. ASME B16.21, 1/8 in. minimum thickness, fibrous type with double coat of graphite, asbestos-free.

ADD 1?

1. Exception: Use Type 304 stainless steel with carbon steel guide on high temperature piping systems.

- H. Bolts and Other Hardware:

1. Bolts for steel, cast iron, brass, and bronze, for 250 PSI working pressure and for 450 degrees F. or below: ASME B18.2.1 carbon steel with American Standard regular, square heads. ASME B18.2.2 American Standard heavy-hexagon, semi-finished nuts. ASME B16.22.1 plain washers and ASME B18.21.1 lock washers.

- a. Option: ASME B18.2.1 tee-head, high-tensile steel bolts and nuts may be used in mechanical joint pipelines.

2. Where equipment is installed outside, all hardware and bolts shall be stainless steel.

2.05 ROLL GROOVED COUPLINGS AND FITTINGS

- A. Acceptable Manufacturers: Victaulic Co., Anvil International "Gruvlok", or Tyco Grinnell.

- B. Rigid couplings with 750 PSI pressure rating and 150 degree F. temperature rating, sizes 2 in. through 6 in.:

1. Victaulic Style 009 (slant bolt pad) with bolt pads metal-to-metal.
2. Tyco-Grinnell Figure 772 with even-gap bolt pads.
3. Anvil-Gruvlok Figure 7401 with even-gap bolt pads.
4. Option: Grooved couplings may be installed on 2 in. to 8 in. roll grooved standard weight pipe.

- C. Flexible couplings with 750 PSI pressure rating and 150 degrees F. rating:

1. Victaulic Style 77.
2. Tyco-Grinnell Figure 705.

3. Anvil-Gruvlok Figure 7001.
- D. Install Schedule 40 piping in accordance with Coupling Manufacturer's instructions.
- E. DO NOT USE ANY OTHER TYPE OF ROLL GROOVED COUPLINGS.
- F. Option for Branch: Installing Contractor may use a ductile iron bolted branch outlet with locating collar, Grade E pressure-responsive gasket seal, and carbon steel bolts and nuts.
- G. Grooved valves and accessories are acceptable.

2.06 WATER-BASED FIRE-SUPPRESSION PIPING SPECIALTIES

- A. Automatic Ball Drip:
 1. Acceptable Manufacturers: Croker or approved equal by Elkhart Brass, Guardian, Potter Roemer, Tyco Fire Products, Victaulic, or Viking.
 2. Croker Figure No. 6781 or 6783. Cast brass, 3/4 in. size, and threaded inlet and outlet.
- B. Automatic Drain Device:
 1. Acceptable Manufacturers: Potter Roemer or approved equal by Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Electric Signal, Reliable Automatic Sprinkler Co., System Sensor, Tyco Fire Products, Victaulic, or Viking.
 2. Potter Roemer Model 5982 or Model 5984. Cast brass, 3/4 in. size, and threaded inlet and outlet.
- C. Backflow Preventers (BFPs):
 1. Acceptable Manufacturers: Zurn Wilkins or approved equal by Ames Co., Conbraco Industries, Febco, Mifab-Beeco, or Watts.
 - a. BFP Manufacturer and Model shall be on the list of accepted equipment published by the (AHJ).
 2. BFP-~~XXXX~~, Double Check, 3/4 in. to 2 in., (ASSE Standard 1015): Zurn Wilkins Model 950XLUS. Suitable for continuous pressure, inlet and outlet full port resilient seated ball valves, inlet strainer, test cocks, union connections between shut-off valves and device.

need "FDA-approved" for fire suppression??

3. BFP-~~XXXX~~, Double Check, 2-1/2 in. and above, (ASSE Standard 1015): Zurn Wilkins Model 350OSYFSC. Suitable for continuous pressure, inlet and outlet OS&Y gate valves, inlet strainer, test cocks, (2) independently operated check valves. Entire assembly coated inside and outside with FDA-approved epoxy material.

4. **BFP**, Double Check Detector (ASSE Standard 1048): Zurn Wilkins Model 350DA. FMG-approved or UL-listed, suitable for continuous pressure application, 175 PSI maximum pressure, 33 degrees F. to 140 degrees F. temperature range, epoxy-coated cast iron body, replaceable seats and seat discs, stainless steel and bronze internal parts, spring-loaded center guided check valves, inlet and outlet OS&Y gate valves with test cocks, and flanged connections. Bypass assembly shall consist of water meter, valves, and double check BFP.

Mike Y deleted 5 and 6

- ~~5. **BFP**, Reduced Pressure Principle, 3/4 in. to 2 in., (ASSE Standard 1013): Zurn Wilkins Model 975XLUS. Suitable for continuous pressure, inlet and outlet full port resilient seated ball valves, bronze Y-type inlet strainer, test cocks, differential pressure relief valve with ASME A112.1.2 air gap fitting between (2) independently operated check valves, union connections between shut-off valves and device.~~
- ~~6. **BFP**, Reduced Pressure Principle, 2 1/2 in. and above, (ASSE Standard 1013): Zurn Wilkins Model 375OSYFSC. Suitable for continuous pressure, inlet and outlet OS&Y gate valves, inlet strainer, test cocks, differential pressure relief valve with ASME A112.1.2 air gap fitting between (2) independently operated check valves. Entire assembly coated inside and outside with FDA approved epoxy material.~~
7. Refer to Fixture and Equipment Branch Size Schedule on the Drawings for sizes and applications.
8. Note: Before ordering, obtain approval from the **AHJ & Local Water Utility** for the specific type and model of BFP to be used.
9. Each BFP shall be serviceable without requiring removal from piping.
10. Each BFP shall have a permanently attached plate indicating type and listing approvals.

D. Drains:

1. Acceptable Manufacturers: Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Electric Signal, Potter Roemer, Reliable Automatic Sprinkler Co., System Sensor, Tyco Fire Products, Victaulic, or Viking.
2. Furnish drains using Schedule 80 black steel pipe (ASTM A53).
 - a. Exception: Through exterior walls, the piping shall be galvanized or non-ferrous.

E. Electric Bell:

1. Acceptable Manufacturers: Potter Electric Signal Co. or approved equal by Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Roemer, Reliable Automatic Sprinkler Co., System Sensor, Tyco Fire Products, Victaulic, or Viking.

2. Potter Electric Signal Co. Model PBA1208. 8 in. diameter, red powder coating finish, weatherproof backbox, 120-volt, 5-year warranty.

DESIGNER — IF USE THIS UNIT, SPECIFY AUTOMATIC BALL DRIP

F. Fire Department Connection:

1. Potter Roemer Model 5795-01-90: Hard-coated aluminum with 4 in. Storz connection on (1) end, 4 in. NPT on other.
 - a. Acceptable Manufacturers: Potter Roemer or approved equal by Croker, Elkhart Brass, Firematic, or Guardian.
 - b. Exact location of the connection shall comply with the requirements of the Local Fire Department. Verify all information before ordering.
 - c. Provide Potter Roemer Model 5799-01 Storz blind cap with chain, 90-degree elbow, and metal escutcheon plate lettered "AUTO SPKR," with raised letters at least 1 in. in height. Torque caps shall prevent removal without tools.
2. Croker Model 6352: Hard-coated aluminum with 5 in. Storz connection on (1) end, 4 in. NPT on other with 90-degree elbow and 30-degree down angle.
 - a. Acceptable Manufacturers: Croker or approved equal by Elkhart Brass, Firematic, Guardian, or Potter-Roemer.
 - b. Exact location of the connection shall comply with the requirements of the Local Fire Department. Verify all information before ordering.
 - c. Provide with Storz cap, attachment cable, and metal escutcheon plate lettered "AUTO SPKR," with raised letters at least 1 in. in height. Torque caps shall prevent removal without tools.

DESIGNER — IF USE THIS UNIT, SPECIFY AUTOMATIC DRAIN DEVICE

G. Fire Department (Siamese) Connection:

1. Acceptable Manufacturers: Potter Roemer or approved equal by Croker, Elkhart Brass, Firematic, or Guardian.
2. Potter Roemer Model 5021-D: Flush-mounted, back outlet, polished chrome-plated finish, 2-way connection, 2-1/2 in. x 2-1/2 in. inlets, complete with (2) clapper valves, 2-1/2 in. nipples, snoots, swivels, plugs, and chains.
 - a. Type of threads and exact location of the connection shall comply with the requirements of the Local Fire Department. Verify all information before ordering.
 - b. Provide metal escutcheon plate that matches pump test connection escutcheon plate; plate shall be lettered "AUTO SPKR," with raised letters at least 1 in. in height. Torque caps shall prevent removal without tools.

H. **Fire Hose Rack:**

1. Acceptable Manufacturers: Croker or approved equal by Akron Brass, Elkhart Brass, Guardian, or Potter Roemer.
2. **FHR-** Croker 3010 Series. Semiautomatic hose rack assembly shall consist of the following:
 - a. Croker Model 3351. 1-1/2 in. red enamel, 16-gauge steel, pin-type, with **(wall bracket) (pipe clamp)** and 180-degree swing.
 - b. Croker Model 3365. 1-1/2 in. rough brass rack nipple.
 - c. Croker Model 3401. 1-1/2 in. x 100 ft.-0 in. long, single jacket, lined, rack light hose with couplings, 500 lb. test.
 - d. Croker Model 3440. 1-1/2 in. rough brass pin lug coupling.
 - e. Croker Model 3460. 1-1/2 in. red polycarbonate resin, adjustable fog nozzle.
 - f. Croker Model 5040. 1-1/2 in. rough brass, 300 lb. rating, and double female angle valve.
 - g. Croker Figure 5370. 1-1/2 in. polished brass, adjustable pressure-restricting device. Provide when pressure exceeds 100 PSI.
 - h. Croker Figure 7171. 1-1/2 in. red nylon, wraparound cover lettered "FIRE HOSE."

I. **Fire Hose Reel:**

1. Acceptable Manufacturers: Croker or approved equal by Akron Brass, Elkhart Brass, Guardian, Larsen's Manufacturing Co., or Potter Roemer.
2. **FHR-** Croker Series 7000. Semiautomatic hose rack assembly shall consist of the following:
 - a. Croker Model 3401. 1-1/2 in. x 100 ft.-0 in. long, single jacket, lined, rack light hose with couplings, 500 lb. test.
 - b. Croker Model 3440. 1-1/2 in. rough brass pin lug coupling.
 - c. Croker Model 3460. 1-1/2 in. red polycarbonate resin, adjustable fog nozzle.
 - d. Croker Model 5030. 1-1/2 in. rough brass, 300 lb. rating, male x female angle valve.
 - e. Croker Figure 5370. 1-1/2 in. polished brass, adjustable pressure-restricting device. Provide when pressure exceeds 100 PSI.
 - f. Croker Model 7068. Red enamel, heavy-duty, welded steel, swing hose reel with **(wall bracket) (pipe clamp)**, and 180-degree swing.
 - g. Croker Figure 7171. Red vinyl, slipover cover, lettered "FIRE HOSE."

- J. Floor Control Valve Sectional Test and Drain Device:
1. Acceptable Manufacturers: AGF Manufacturing Inc. or approved equal by Tyco Fire Products or Viking.
 2. AGF Manufacturing Model 8000 "Commercial RiserPACK" Floor Control Assembly. 300 PSIG-rated, includes inspector test connection, flow switch, pressure gauge, and gauge valve.
- K. Gauges, Pressure:
1. Acceptable Manufacturers: Winters Instruments or approved equal by Ashcroft, Marsh, Marshalltown, Terice, Weiss, or Weksler.
 2. Winters Instruments Product Code Q1721. Aluminum dial, steel case, brass socket, 4 in. diameter, 1/4 in. NPT male connection, with range of 0 PSIG to 250 PSIG. UL 393.
 3. Provide spare gauge.
- L. Hydrant, Roof:
1. Acceptable Manufacturers: Croker or approved equal by Elkhart Brass, Firematic, or Potter Roemer.
 2. Croker Model 6840. Cast brass, 2-way, 2-1/2 in. x 2-1/2 in. x 4 in. vertical manifold, including 2-1/2 in. angle valves, (2) outlets, matching threads, brass caps, and drains, rough finish.
 3. Consult Local Fire Department for exact type and size of threads.
- M. Identification Signs:
1. Acceptable Manufacturers: Reliable Automatic Sprinkler Co. or approved equal by Globe Fire Sprinkler Corp. or Tyco Fire Products.
 2. Reliable Automatic Sprinkler Co. Model A:
 - a. 18-gauge steel, porcelain-enameled, with white lettering on red background.
 - 1) Valve control: Style A.
 - 2) Specific identification: Style B.
 - 3) Fire alarm: Style D.
 - 4) Hydraulic calculated system: Style E.
 - b. 20-gauge steel, porcelain-enameled, white lettering on red background.
 - 1) Cold weather: Style C.

N. Inspector's Test and Drain Assembly:

1. Acceptable Manufacturers: AGF Manufacturing Co. or approved equal by Reliable Automatic Sprinkler Co. or Tyco Fire Products.
2. AGF Manufacturing Co. Model 1011T. 1 in. test and drain ball valve, with locking kit and lock with (2) keys, pressure-relief valve, pressure gauge, tamper-resistant sight glass, and 1/4 in. 3-way globe valve, with flow equivalent to (1) sprinkler.

O. Pump Test Connection:

1. Acceptable Manufacturers: Potter Roemer or approved equal by Croker, Elkhart Brass, or Guardian.
2. Potter Roemer (Model 5866, freestanding) (Model , wall-mounted): Polished (brass) (chrome-plated) finish, 2-way connection, 2-1/2 in. x 2-1/2 in., complete with plugs, chains, (brass) (chrome-plated) 2 ft.-4 in. high pipe sleeve.
 - a. Type of threads and exact location of the connection shall comply with the requirements of the Local Fire Department.
 - b. Provide metal escutcheon plate that matches siamese connection escutcheon plate; plate shall be lettered "PUMP TEST CONNECTION," with raised letters at least 1 in. in height.
 - c. Provide (2) 2-1/2 in. full flow, rough brass gate valves with swivel inlets to be removed after each fire pump test and plugs re-installed.

P. Switches, Pressure Alarm:

1. Acceptable Manufacturers: Potter Electric Signal Co. or approved equal by Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Roemer, Reliable Automatic Sprinkler Co., System Sensor, Tyco Fire Products, Victaulic, or Viking.
2. Potter Electric Signal Co. Model PS10-1A. Factory-preset, electrical contacts, diaphragm, spring, and plunger enclosed in a metal base with a molded Bakelite cover.

Q. Switches, Valve Supervisory:

1. Acceptable Manufacturers: Potter Electric Signal Co. or approved equal by Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Roemer, Reliable Automatic Sprinkler Co., System Sensor, Tyco Fire Products, Victaulic, or Viking.
2. OS&Y valves: Potter Electric Signal Co. Model OSYSU-1. NEMA 4 and NEMA 6P, weather-resistant and submersible tamper-resistant switch on cover, single-pole double-throw switch, roller-type switch actuator, and spring-loaded plunger. Do NOT provide double-pole switches.

3. Post indicator or butterfly valves: Potter Electric Signal Co. Model PCVS-1. Weatherproof, tamper-resistant, (1) or (2) single-pole double-throw switches, constructed of die-cast aluminum, tamper switch cover, with 1/2 in. male fitting for attachment to post indicator or butterfly valve.

R. Switches, Water Flow:

1. Acceptable Manufacturers: Potter Electric Signal Co. or approved equal by Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Roemer, Reliable Automatic Sprinkler Co., System Sensor, Tyco Fire Products, Victaulic, or Viking.
2. Potter Electric Signal Co. Model VSR-F. Vane-type sectional, with (2) single-pole double-throw snap action switches, 40 degrees F. to 120 degrees F. operating temperature range, (250) (450) PSI service pressure, 8 GPM to 10 GPM minimum flow activation for alarm activation, and (die-cast aluminum) (NEMA 4-rated) enclosure.

DESIGNER — PROVIDE IF REQUIRED BY AHJ (S)

S. Water Motor Gong:

1. Acceptable Manufacturers: Reliable Automatic Sprinkler Co. or approved equal by Automatic Sprinkler, Croker, Firematic, Globe Fire Sprinkler Corp., Keles, Potter Electric Signal Co., Potter Roemer, System Sensor, Tyco Fire Products, Victaulic, or Viking.
2. Reliable Automatic Sprinkler Co. Model C. Cast iron body, non-corrosive impeller, 10 in. diameter cast aluminum gong, red finish, 3/4 in. inlet piping, and 1 in. drain outlet in body housing.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate pipe elevations and final locations with the work of all other Contractors before installation.
- B. Coordinate the exact location of valves, siamese connections, pump test connections, etc., and the proper number, size, and type valve or hose threads required, with the Local Fire Department. Install correct required threads.

3.02 INSTALLATION — PIPING AND FITTINGS

- A. The installation instructions apply to all water-based fire-suppression piping systems, except where otherwise noted.

- B. All piping materials and installation shall comply with State and Local Building Code requirements.
- C. Provide information on chases, sleeves, and openings required for Fire Suppression Work to other Contractors.
- D. Provide all fittings and accessories required to complete the installation.
- E. Personnel Passages and Exit Paths: Do not install piping where it will impede personnel using exit paths.
- F. Structural Members: Do not install piping where it will interfere with structural members. Do not cut structural members for piping installation without written approval from the **Structural Engineer**.
- G. ~~Equipment, Such as Cranes: Do not install piping where it will interfere with equipment operation.~~
- H. Roofs: Coordinate penetration sizes, locations, sleeves, and weatherproofing, where applicable.
- I. Ceilings: Piping subject to freezing may be installed in unheated ceiling spaces only where specifically shown, immediately above the ceiling of a heated space, with building insulation closely packed above the piping to protect it from freezing. Contractor providing the piping shall ensure that vulnerable piping is protected against freezing.
- J. Walls:
 - 1. Coordinate penetration sizes, locations, sleeves, and weatherproofing, where applicable.
 - 2. For masonry unit walls, locate piping penetrations at masonry joints where feasible.
 - 3. Piping and sprinklers located in, and passing through, exterior walls shall be subject to review by the A/E.
 - 4. Do not install piping containing fluids subject to freezing in exterior walls, regardless of wall insulation location. Where unavoidable, obtain prior approval of the A/E, and install piping so that the wall insulation is between the pipe and the outside wall.
- K. Doors: Do not install piping where it will restrict access to or operation of doors.
- L. ~~Skylights: Do not install piping above or below skylights.~~
- M. Windows: Do not install piping in front of windows or where it will restrict access to or operation of windows.

- N. Floors:
1. Coordinate penetration locations, sleeves, and weatherproofing (where applicable).
 2. **Lead Contractor shall provide** 3 in. high concrete curbs around pipe sleeves where pipes penetrate floors in **kitchens, utility rooms, equipment rooms, shafts, or other** areas potentially subject to flooding. Refer to **Section 21 05 29** regarding concrete work.
- O. Access Doors or Panels: Do not install piping where it will restrict access to or operation of access doors or panels.
- P. Shafts: Do not run piping through a shaft **unless it uses the shaft for vertical offset**.
- Q. Elevator Equipment Rooms: Do not extend piping through elevator equipment rooms.
- R. Piping, Conduit, Cable Trays, and Wiring: Coordinate all runs. Group runs to minimize space usage and maximize access.
- S. HVAC Ductwork: Do not penetrate ductwork with piping without written approval of A/E (unless the Drawings specifically indicate the penetration).
- T. HVAC Equipment: Do not install piping where it will restrict access to or operation of equipment.
- U. Electrical Equipment:
1. Do not install piping in an electrical equipment room unless it specifically serves that room.
 2. Do not install piping above, within 36 in. of front of, or in the immediate vicinity of, electrical panels, electrical switchgear, or other electrical equipment, as described in 2017 NEC Articles 110 and 408.
 - a. If compliance with these requirements is not possible, obtain approval in writing from the AHJ and A/E to provide a sheet metal drip pan under the piping to prevent water and condensation from running or dripping onto electrical items. Provide drip pan at no additional cost to the Owner.
- V. Luminaires: Do not install piping across, in front of, or directly below luminaires.
- W. Interferences: If any system piping is placed where it causes or contributes to interference, relocate it. Obtain instructions from the A/E. The A/E's decision shall be final in regard to the arrangement of equipment, ductwork, piping, etc. Contractor causing the conflict shall assume all costs and responsibilities resulting from improper coordination of work for the corrective work of all Contractors.

3.03 PIPING LAYOUT

- A. The Drawings are schematic and are not intended to precisely show piping locations.
- B. Arrange all piping in accordance with the best standards of the Trade.
- C. Install piping in a neat and workmanlike manner. Install horizontal piping parallel or perpendicular to building lines, unless indicated otherwise. Install vertical piping plumb, unless indicated otherwise.
- D. Securely support piping from structure with approved hangers, rods, anchors, guides, brackets, and accessories.
 - 1. Independently support piping from building structural members. Do not support piping from ceilings, ducts, other pipes, cable trays, conduit, wiring, etc.
 - 2. Refer to Section 21 05 29.
- E. Install piping systems with adequate provisions for expansion, contraction, vibration, and prevention of stresses on valves and equipment.
- F. Adjust locations and provide additional offsets, drains, and air vents needed to properly install piping system without interferences.
- G. Exercise due and particular caution to determine that all parts of the work are quickly and easily accessible.
- H. Where special conditions are encountered in the field, the A/E shall decide arrangement and alignment of piping.
- I. Changes in Piping Direction:
 - 1. Make changes in piping direction with fittings. Do not bend or spring piping.
 - 2. Offset or reroute piping to resolve interferences that develop in the field.
 - 3. Provide offsets for the proper operation of the system.
 - 4. Offset piping around columns, beams, and other obstructions.
 - 5. In pipe spaces to be entered for servicing, offset piping so that, where possible:
 - a. All lateral runs are located either near floor or 6 ft.-0 in. or higher above floor.
 - b. All vertical piping is held close to the wall through 6 ft.-0 in. or higher above floor to permit access.
 - 6. Keep all piping to side of chases wherever possible.
- J. Conceal pipes, except in equipment rooms, in rooms without ceilings, or where specifically indicated otherwise.

- K. Install unions and valves in accessible locations. Use removable ceiling pads or provide access panels in ceilings, walls, and floors for service access. Coordinate installation of access panels.
- L. Keep piping level, except where a slope is required.
- M. Install piping in such a manner as to allow complete drainage of the water-based fire-suppression piping system, **toward alarm valve and/or riser**. Provide rigid support for drain piping. If other means of draining are not provided, install drain valves at all low points.

Markus — Mike Y suggests O be deleted, and just let NFPA 13 govern

- N. Slope piping to drainage points with minimum slope of 1 in. in 40 ft. (0.2%).
- O. Avoid trapping piping.
 - 1. Provide a 3/4 in. ball drain valve with capped hose end connection at each low point in the piping.
 - 2. Do not trap drain piping except at equipment connections.
- P. Make reductions in piping lines with reducing couplings or weld fitting reducers.
- Q. For horizontal piping, use eccentric reducer to keep top of pipe level.
- R. DO NOT USE BULLHEAD CONNECTIONS IN ANY PIPING SERVICE.
- S. Where connection size is smaller than piping, make reduction at final connection only. Do not reduce size of pipe drop.
- T. Valves:
 - 1. Install valves on each piece of equipment.
 - 2. Install valves at service connections to equipment not provided with valves.
 - 3. Install valves at branch lines from main lines.
 - 4. Refer to **Section 21 05 23** for valve requirements.
- U. Provide piping identification. Refer to Section 21 05 53.

3.04 PIPING AND FITTINGS FABRICATION

- A. Cut pipe accurately, and work into place without forcing or springing.
- B. Clean piping components of loose material.

- C. Protect all piping, fittings, and valves from excessive tool marks.
- D. Joints, Unions, and Connections:
 - 1. Ream pipe or tubing ends and remove burrs to full inside diameter after cutting.
 - 2. Threaded connections:
 - a. Threads shall conform to ASME B1.20.1.
 - b. Provide union adjacent to each threaded valve (unless union is integral to valve body).
 - 3. Soldered or brazed connections:
 - a. Soldering and brazing work shall comply with ASME/ANSI B31.1.
 - b. Mating surfaces of joint shall be smooth and clean.
 - c. Exterior of joint shall be smooth.
 - d. 100% lead-free solder shall be used on water piping, joints, and fittings. Certify that the solder and brazing are lead-free
 - e. Refer to **Section 21 05 23** for valve requirements.
 - 4. Welded connections:
 - a. An AWS-certified welder shall perform all welding and brazing in accordance with ASME/ANSI B31.1.
 - b. A qualified welder, certified in accordance with the State Building Code, shall install welded pressure piping.
 - c. Do not torch-cut pipe.
 - d. **Use backing rings on all welded joints.**
 - e. Butt-welded joints shall have substantially full penetration and recommended bead reinforcement.
 - f. Slip-on, socket, and fillet welds shall have geometry in accordance with ASME/ANSI B31.1.
 - g. Welds shall be stamped at each joint or fitting.
 - h. Remove slag, weld scale, and other welding residue as work proceeds and at completion.
 - 5. Dissimilar material connections:
 - a. Make joints using adapters compatible with materials of both piping systems.
 - b. Install dielectric unions **or dielectric flanges** on all connections between copper tubing and ferrous metal piping or equipment having ferrous metal connection.

- c. Isolate direct contact between pipe, fittings, valves, **flanges**, and hangers of dissimilar metal by use of dielectric unions, shims, gaskets, coatings, or other approved methods.
 - d. Use dielectric gaskets where joining dissimilar piping material.
 - 6. Flanged connections:
 - a. Flanged joints shall be faced, square, and true.
 - b. Install gaskets suitable for the operating temperature and pressure of the piped fluid or gaseous medium.
 - 7. Connections to equipment or devices:
 - a. Make connections to equipment as recommended by the Manufacturer or as detailed on the Drawings.
 - b. Provide union **or flange** near each connection arranged to permit removal of parts and equipment without disconnecting piping beyond the union **or flange**. **Provide isolation valve on piping system side of each equipment connection union or flange to permit equipment removal without shutting down piping system.**
 - c. Provide union **or flange** in final piping connection to each device or item of equipment, and elsewhere, to make-up or disconnect piping.
 - d. Do not make welded connections to equipment **or valves**.
- E. Grooved Connections:
 - 1. A factory-trained representative of the Coupling Manufacturer shall provide on-site training for the Contractor's field personnel in the use of grooving tools, application of groove, and installation of grooved joint products.
 - a. The Manufacturer's Representative shall periodically visit the job site and review installation.
 - b. Provide documentation of the site visits by the Manufacturer's Representative in the Operating and Maintenance Manuals.
 - c. Include verification of proper gasket lubricants, uniform gap spacing of couplings, proper torque requirements, and notation of any installation deficiencies.
 - 2. Grooved Piping Installing Contractor shall have prior experience installing at least (5) grooved mechanical piping systems.
 - 3. Install grooved joint couplings, gaskets, and fittings in accordance with the Coupling Manufacturer's written installation instructions.
 - 4. Each grooved pipe end shall be clean and free from indentations, projections, and roll marks in the area from pipe end to groove and on both sides of groove.

5. Provide elastomer gasket suitable for the intended service. The Coupling Manufacturer shall produce gasket.
6. Verify and use the Coupling Manufacturer's recommended gasket lubricants, including any specific lubricants to enhance the temperature rating of the coupling. Lubricate gasket both inside and out where required by the Coupling Manufacturer. Submit lubricant data prior to installation and provide proof that the lubricant is being used on the job site.
7. Alternate tightening bolts **on opposite sides of coupling** until properly torqued in accordance with the Coupling Manufacturer's specifications.
8. If the Coupling Manufacturer specifies torque requirements for its fittings, test random couplings, at the A/E's discretion, to verify that proper torque requirements were achieved during installation.
9. Verify that there are no bolt pad gaps on metal-to-metal type couplings, or that equal gaps are maintained on both sides of the coupling, in accordance with the Coupling Manufacturer's written installation instructions.
10. Remove and replace any joints deemed improperly installed.

F. Escutcheons:

1. Where exposed pipes pass through walls, floors, or ceilings of finished rooms, provide chrome-plated escutcheons.
 - a. Option: Prime-coated **(black iron) (steel) (rough brass)** escutcheons may be provided in unfinished rooms.
2. Protect escutcheons from tool marks.

G. At the end of each day's work, cap or plug open ends in installed piping for protection.

3.05 WATER-BASED FIRE-SUPPRESSION PIPING CLEANING, TESTING, AND STERILIZATION

A. Pipe Cleaning:

1. After erection, lines shall be blown or flushed free of loose materials. Clean strainer screens and sediment pockets prior to putting lines in service.
2. Flush fire suppression mains at GPM flows required by **2022 NFPA 13 and 2025 NFPA 24**.

B. Flush out all new water piping systems before systems are placed in operation.

C. Perform tests in accordance with AHJ requirements. Tests shall be witnessed by the A/E's Representative.

D. Take precautions to remove all air before performing hydrostatic tests.

Markus — Mike Y wants to know if E meets or exceeds NFPA 13

- E. Test piping at 125 PSIG for (6) hours with no pressure drop.
- F. If a leak occurs, replace the defective piece or joint with new materials at no cost to the Owner. No caulking will be permitted. Repeat test until no leaks occur.
- G. Contractor shall pay all fees for testing and use of testing equipment.

3.06 INSTALLATION — SPECIALTIES

- A. ALL specialties listed in this Section shall conform to the following Standards:
 - 1. ASTM.
 - 2. FMG.
 - 3. NEMA.
 - 4. NFPA.
 - 5. UL "Fire Protection Equipment List."
- B. Provide specialties in locations accessible to qualified personnel for operation, removal, maintenance, or servicing, as shown on the Drawings, in accordance with **2022 NFPA 13 and 2019 NFPA 14**, as discussed in other Sections of these Specifications, and in order to facilitate the maintenance, operation, and servicing of all fire-suppression piping systems. Coordinate with structure, and mechanical and electrical equipment.
- C. Automatic Ball Drip:
 - 1. Install in vertical position.
 - 2. Extend 3/4 in. drain from automatic ball drip to nearest floor drain.
- D. Automatic Drain Device: Install in a horizontal position where indicated on the Drawing.
- E. BFPs:
 - 1. BEFORE ORDERING, coordinate model and features of **double check** BFP with **State Plumbing Inspector and AHJ**.
 - 2. Provide approved BFP in accessible location in accordance with State Plumbing Code and **(State) (County) (Local)** Department of Health.
 - 3. Install each BFP per Manufacturer's recommendations.
 - 4. Install BFPs so as to be easily accessible, and at a maximum of 9 ft. above the floor.
 - 5. Provide shut-off valve ahead of each BFP assembly.

6. Install double check detector BFP in a horizontal position. Discharge from double check BFP shall have an open air gap. Indirect waste to nearest drain.
 7. Discharge from each reduced pressure BFP shall have an open air gap. Indirect waste full size to nearest floor drain or floor sink using steel or copper piping.
 8. Provide (ball) (gate) valve ahead of strainer on inlet side of BFP assembly.
 9. Install supervisory switch on OS&Y gate valves.
 10. After installation of BFP, flush water supply line to remove debris. Clean out BFP after flushing.
 11. Test double check BFP at time of installation. A person or firm certified by the AHJ shall perform the test and provide the testing equipment. Submit test report to the A/E. Provide BFP certification.
- F. Electric Bell:
1. Coordinate location and mounting height with the A/E and Electrical Contractor before installing.
 2. Provide electric bell.
 3. Electrical Contractor shall provide wiring.
- G. Fire Department and Pump Test Connection:
1. Coordinate the exact location of the Fire Department connection with the A/E and the Local Fire Department. Obtain approval from the Local Fire Department for pump test connection locations BEFORE installing piping.
 2. Install fire pump test connection at not less than 18 in. and not more than 42 in. to finished grade
 3. Install automatic ball drip on Fire Department connection line.
- H. Fire Department (Siamese) and Pump Test Connection:
1. Coordinate the exact location of the Fire Department connection with the A/E and the Local Fire Department. Obtain approval from Local Fire Department for siamese and pump test connection locations BEFORE installing piping.
 2. Install siamese and fire pump test connection at not less than 18 in. and not more than 42 in. to finished grade.
 3. Install the automatic drain device inline to drain low point of the system between the Fire Department connection and the swing check valve.
- I. Fire Hose Racks: Install at not more than 6 ft. from floor to top.
- J. Fire Hose Reels: Install at not more than 6 ft. from floor to top.

- K. Floor Control Valve Sectional Test and Drain Device: Install in accordance with the Manufacturer's instructions.
- L. Gauges, Pressure:
 - 1. Provide pressure gauge with shut-off cock:
 - a. On each discharge from fire pump.
 - b. On top of each standpipe.
 - c. On top of each sprinkler riser.
 - d. At each test connection.
 - 2. Locate pressure gauges where water will not freeze.
 - 3. Provide each pressure gauge with a shut-off valve and with provision for draining.
 - 4. Install in accessible secure areas.
 - 5. Install a 3/8 in. ball valve at each pressure gauge.
- M. Hydrant, Roof:
 - 1. Install roof manifold as shown on the Drawings.
 - 2. Seal weather-tight.
 - 3. Install piping and valves to prevent hydrant freeze-up.
- N. Identification Signs: Coordinate installation with the Lead Contractor and A/E.
- O. Inspector's Test and Drain Assembly:
 - 1. For each riser, provide assembly at end of branch most remote from the alarm valve.
 - 2. Extend drain from assembly to the exterior, unless requested otherwise.
 - 3. The A/E will review and approve all locations.
 - 4. Torque plugs after fire pump test is completed.
- P. Pump Test Connection:
 - 1. Coordinate the exact location of pump test connection with the Local Fire Department.
 - 2. Install automatic ball drip on pump test connection riser.
- Q. Switch, Pressure Alarm:
 - 1. Coordinate location and electrical requirements, including type and rating of switch contacts required for alarm circuit, with the Electrical Contractor.

2. Provide on main riser.
 3. Electrical Contractor shall provide wiring to the fire alarm panel.
- R. Switches, Post-Indicator Valve Supervisory:
1. Coordinate location and electrical requirements with the Electrical Contractor.
 2. Provide where indicated on the Drawings.
 3. Provide post-indicator valves with supervisory switches.
 4. Provide a supervisory switch on each operable valve on supply lines.
 5. Test switches after installation.
 6. Electrical Contractor shall provide wiring to the fire alarm panel.
- S. Switches, Waterflow on limited area sprinkler systems:
1. Coordinate location and electrical requirements with the Electrical Contractor.
 2. Provide a waterflow switch:
 - a. On each main riser.
 - b. On each floor.
 - c. On outlet of fire alarm valve.
 - d. At other locations indicated on the Drawings.
 3. Do not install a flow switch within 12 in. of a tee or elbow connection or within 18 in. of a gate valve.
 4. Electrical Contractor shall provide wiring to the fire alarm panel.
- T. Water Motor Gong:
1. Coordinate location with the A/E.
 2. Install as close to alarm valve as possible.
 3. Install 3/4 in. strainer at outlet of retarding chamber on alarm valve.

END OF SECTION 21 10 00

SECTION 21 12 00 - FIRE-SUPPRESSION STANDPIPES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 21 05 00.73, "Execution of Fire Suppression."
- B. Section 21 05 23, "General-Duty Valves for Water-Based Fire-Suppression Piping."
- C. Section 21 05 29, "Hangers and Supports for Fire-Suppression Piping and Equipment."
- D. Section 21 05 48, "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."
- E. Section 21 05 53, "Identification for Fire-Suppression Piping and Equipment."
- F. Section 21 10 00, "Water-Based Fire-Suppression Systems."
- G. Section 21 13 13, "Wet-Pipe Sprinkler Systems."
- H. Section 22 11 16, "Domestic Water Piping."

1.02 DESCRIPTION

- A. Provide piping and fittings as shown or required to completely install a Class (I) (II) service-rated (wet) (dry) standpipe system combined with sprinkler system where indicated on the Drawings.
 - 1. Piping and fittings: Refer to Section 21 10 00 for additional requirements.
 - 2. Valves: Refer to Section 21 05 23.
 - 3. Hangers and supports: Refer to Section 21 05 29.
 - 4. Access panels. Refer to Section 21 05 00.73.
 - 5. Specialties: Refer to Section 21 10 00.
 - 6. Wet-pipe sprinkler system: Refer to Section 21 13 13.
- B. Design, install, and test complete system in accordance with 2022 NFPA 13 and 2019 NFPA 14, perform hydraulic calculations, prepare drawings, and submit to AHJ for approval.
- C. Test complete system.

1.03 QUALITY ASSURANCE

A. Standards:

1. American National Standards Institute (ANSI).
2. American Society for Testing and Materials (ASTM).
3. American Society of Mechanical Engineers (ASME).
4. American Welding Society (AWS).
5. Factory Mutual Global (FMG).
6. National Fire Protection Association (NFPA):
 - a. 2021 NFPA 10, "Standard for Portable Fire Extinguishers."
 - b. 2022 NFPA 13, "Standard for the Installation of Sprinkler Systems."
 - c. 2019 NFPA 14, "Standard for the Installation of Standpipes and Hose Systems."
7. Underwriters Laboratories, Inc. (UL).

PART 2 PRODUCTS

2.01 PIPING AND FITTINGS

- A. Unless noted otherwise, the piping between Fire Department connection and check valve, and the drain lines, shall be galvanized Schedule 30 or Schedule 40 black steel (ASTM A53) as listed in 2022 NFPA 13.

2.02 FIRE VALVE CABINETS

- A. Acceptable Manufacturers: Larsen's Manufacturing Co. or approved equal by Amerex, Croker, Elkhart Brass, Guardian, J. L. Industries, or Potter-Roemer.
- B. 18-gauge steel body, 20-gauge door, and 16-gauge frame. Welded, semi-concealed hinge, white enamel inside, prime coat outside. Steel door with (steel) (wire glass) (double-strength glass) panel. (Recessed) (Semi-recessed) (Surface) mounting.
- C. The Owner shall approve keying. Keying shall be the same for all types of cabinets. Furnish number of keys required.
- D. Detention Fire Valve Cabinet: Larsen's Manufacturing Co. Model DVC1818-SM. Heavy-duty, detention-grade, surface-mounted, 21-1/2 in. high x 21-1/2 in. wide x 8-1/2 in. deep, 12-gauge steel body, welded, semi-concealed hinge, baked red enamel finish, 12-gauge solid steel door with Model HSL II deadlock with malleable iron case and steel cover, high-

strength 1-1/2 in. x 3/4 in. bronze deadbolt, (5) level tumblers. Manufacturer shall apply white decal "FIRE DEPT VALVE."

- E. Fire Department Valve: 2-1/2 in., male/female, angle valve, wheel handle, chain and cap, (rough) (polished) brass finish, with threads matching Local Fire Department Standards.
- F. Cabinets, valves, and related equipment shall also conform to 2021 NFPA 10.

2.03 FIRE HOSE CABINETS

- A. Acceptable Manufacturers: Larsen's Manufacturing Co. or approved equal by Amerex, Croker, Elkhart Brass, Guardian, J. L. Industries, Kidde, Potter-Roemer, or Safety-First.
- B. [38 in. high x 32 in. wide x 8 in. deep (fully recessed)] [41-1/2 in. high x 35-1/2 in. wide x 8-1/2 in. deep (surface-mounted)], 18-gauge steel, baked white enamel inside, prime coat outside, space for (1) fire extinguisher, hose rack, and 2 1/2 in. Fire Department valve. Heavy-gauge, baked enamel painted steel door frame with (steel panel) (wire glass panel) (double-strength glass panel) ("Gemini" 1/4 in. thick black textured acrylic) door with horizontal white letters "FIRE HOSE." Mounting:
 - 1. Recessed:
 - a. In block wall: FHC- : Fully recessed, Larsen's Manufacturing Co. Model HC3238-R.
 - b. In dry wall: FHC- : Trimless, Larsen's Manufacturing Co. Model HC3238-RT.
 - 2. Semi-recessed.
 - 3. Surface: FHC- : Larsen's Manufacturing Co. Model 3238-SM.
- C. Detention Fire Hose Cabinet: Larsen's Manufacturing Co. Model DHC3238-SM. Heavy-duty, detention-grade, surface-mounted, 38 in. high x 32 in. wide x 8-1/2 in. deep, 12-gauge steel tub, welded, semi-concealed hinge, baked red enamel finish, 12-gauge solid steel door with Model HSL II deadlock with malleable iron case and steel cover, high-strength 1-1/2 in. x 3/4 in. bronze deadbolt, (5) lever tumblers, space for fire hose and rack, 2-1/2 in. Fire Department valve, and (1) fire extinguisher. Manufacturer shall apply white "FIRE HOSE" and "FIRE EXTINGUISHER" decals. Provide Larsen's Manufacturing Co. "Vigilante" alarm, 9 volt, (10) second alarm delay, and on/off jack. Provide 9 volt battery.
- D. Fire Department Valve: 2-1/2 in. (rough) (polished) brass Fire Department valve with brass chain and cap, with threads matching Local Fire Department Standards.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate with the Local Fire Department for the proper number, size, and type valve or hose threads required. Install correct required threads.
- B. Coordinate each cabinet installation with wall depth and wall construction, building equipment, fixtures, and the Lead Contractor.
- C. Coordinate each cabinet location with the A/E, HVAC Contractor, Electrical Contractor, and Owner.

3.02 INSTALLATION

- A. Entire installation and testing shall comply with the requirements of 2019 NFPA 14.
- B. Piping:
 - 1. Do not install wet piping in areas where space temperature may drop below 40 degrees F.
 - 2. Install drain pipes at base of sprinkler riser. Locate Inspector's Test at remote end of system. Pipe to nearest drain or outside to splash block. Discharge from drain/test pipes shall be at location approved by the A/E and Owner.
 - 3. All drain and test piping passing through an exterior wall shall be galvanized or non-ferrous to prevent staining the exterior wall by rust.
 - 4. Install a pressure gauge at top of each standpipe riser. Refer to Section 21 10 00.
 - 5. Install indicating valves to permit isolating one standpipe from another without interrupting the water supply.
 - 6. Securely anchor each mechanical joint and bend, using pipe clamps and tie-rods.
 - 7. Joints: Threaded pipe and fittings. Steel pipe with wall thicknesses less than Schedule 30 (in sizes 8 in. and larger) or Schedule 40 (in sizes less than 8 in.) shall not be joined by threaded fittings.
 - 8. Tag and identify all valves, piping, and equipment installed under this Work. Refer to Section 21 05 53.
 - 9. Refer to Section 21 10 00 for additional requirements.
- C. Fire (Hose) (Valve) Cabinets:
 - 1. Provide fire valve cabinets where shown on the Drawings.

2. Provide fire hose cabinets where shown on the Drawings complete with the following:
 - a. Hose, rack, and valve.
 - b. 2-1/2 in. Fire Department valve.
 - c. Fire extinguisher.
 3. Lead Contractor shall install cabinets so that they are not more than 5 ft. above floor to centerline of valve.
 4. Identify each cabinet with an engraved, laminated plastic identification nameplate.
- D. Connect to outlet of wet pipe fire alarm valve.
- E. Connect to fire pump discharge flange.
- F. Downstream of the fire pump, provide tee connections for the siamese and pump test connection. After the siamese and pump test connection, extend the fire waterline as shown on the Drawings and connect to the standpipe system.
- G. Provide outlets at each floor for sprinkler system where indicated on the Drawings.
- H. Connect standpipe to fire hose valves, fire hose cabinets, and fire valve cabinets.
- I. Sterilize water line from a point of connection to site water line to inlets of (check valves) (backflow preventer). Refer to (Section 21 10 00) (Section 22 11 16). Coordinate with the Plumbing Contractor.

3.03 TESTING

- A. Perform all tests before any piping is covered or concealed.
- B. Arrange and schedule all inspections required by the AHJ.
- C. Test the standpipe system hydrostatically at 225 PSIG for (2) hours with no drop in pressure. Measure the hydrostatic test pressure at the low point of the individual system or zone being tested. There shall be no visible leakage during the hydrostatic testing.
- D. Should leaks occur, remove each defective section or defective fitting and replace with new materials at no cost to the Owner. Repeat tests until no leaks occur.
- E. Do not use brine or other corrosive chemicals for testing systems.
- F. Make provision for the disposal of water issuing from test outlets to avoid property damage.

END OF SECTION 21 12 00

SECTION 21 13 13
WET-PIPE SPRINKLER SYSTEMs
(PART OF PLUMBING WORK)

Is this true?



PART 1 - GENERAL

1.1 PREFACE

- A. ALTHOUGH SPECIFIED IN DIVISION 21, THIS WORK IS TO BE A PART OF THE PLUMBING WORK. Delete 1.1 A; revise B to Fire Suppression Contractor
- B. Terminology: In this Section, the term "Contractor" shall mean the Contractor performing Plumbing Work on this Project, unless otherwise noted.
- C. All provisions and requirements of Division 21, "Fire Suppression," and Division 22, "Plumbing," apply to this Section.

1.2 RELATED DOCUMENTS

- A. Section 21 05 00.73, "Execution of Fire Suppression."
- B. Section 21 05 00.78, "Closeout Submittals for Fire Suppression."
- C. Section 21 05 23, "General-Duty Valves for Water-Based Fire-Suppression Piping."
- D. Section 21 05 29, "Hangers and Supports for Fire-Suppression Piping and Equipment."
- E. Section 21 05 48, "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."
- F. Section 21 10 00, "Water-Based Fire-Suppression Systems."
- G. Section 21 12 00, "Fire-Suppression Standpipes."
- H. Section 21 13 16, "Dry-Pipe Sprinkler Systems."

- I. Section 26 29 13, "Enclosed Controllers."

1.3 DESCRIPTION

- A. Provide a hydraulically designed wet-pipe sprinkler piping system where indicated on the Drawings, including piping, sprinklers, valves, and specialties, to provide coverage of all areas in the building indicated on the Drawings, including overhangs, areas under stairs, mezzanines, overhead doors (when open), etc. Hazard classification shall be in accordance with Code.
 - 1. Piping and fittings: Refer to Section 21 10 00 for additional requirements.
 - 2. Valves: Refer to Section 21 05 23.
 - 3. Access panels. Refer to Section 21 05 00.73.
 - 4. Specialties: Refer to Section 21 10 00.
- B. Design, install, and test complete system in accordance with 2022 NFPA 13; perform hydraulic calculations, prepare drawings, and submit to AHJ for approval.
- C. Lead Contractor shall provide site fire suppression water service line.
- D. Lead Contractor shall extend site fire suppression water service line into building.
- E. Connect to flanged site fire suppression water service line at 1 ft. above building floor slab as shown on the Drawings.
- F. Test complete system.

1.4 QUALITY ASSURANCE

- A. Standards:
 - 1. American National Standards Institute (ANSI).
 - 2. American Society for Testing and Materials (ASTM):
 - a. ASTM F 438.
 - b. ASTM F 439.
 - c. ASTM F 442/F 442 M.
 - d. ASTM F 493.
 - 3. American Society of Mechanical Engineers (ASME).
 - 4. American Welding Society (AWS).

5. Factory Mutual Global (FMG).
6. National Electrical Manufacturers Association (NEMA).
7. National Fire Protection Association (NFPA):
 - a. 2022 NFPA 13, "Standard for the Installation of Sprinkler Systems."
 - b. 2019 NFPA 14, "Standard for the Installation of Standpipes and Hose Systems."
 - c. 2025 NFPA 24, "Standard for the Installation of Private Fire Service Mains and Their Appurtenances."
8. Underwriters Laboratories, Inc. (UL) , including UL 1821.

1.5 SPARE PARTS

- A. Spare Sprinklers: Provide a sprinkler replacement wrench and a stock of spare sprinklers, (2) minimum of each type, for future use for each type and temperature rating of sprinkler installed, and mount in a metal sprinkler cabinet. Furnish minimum total of spare sprinklers in accordance with the following schedule:

NUMBER OF SPRINKLERS	EXTRA SPRINKLERS
Up to 300	6
301–1,000	12
More than 1,000	24

- B. Refer to Section 21 05 00.78 for additional information regarding spare parts.

PART 2 - PRODUCTS

2.1 PIPING AND FITTINGS

- A. Unless noted otherwise, the piping between Fire Department connection and check valve, and the drain lines, shall be galvanized Schedule 30 or Schedule 40 black steel (ASTM A53) as listed in 2022 NFPA 13.

- B. Option: Flexible sprinkler drops. In lieu of rigid pipe offsets or return bends for sprinkler drops, a flexible stainless-steel sprinkler drop system may be used to locate sprinklers where required by final finished ceiling tiles and walls.
1. FMG 1637-approved and UL 2443-approved.
 2. Drop shall include an FMG-approved and UL-approved braided hose.
 3. Drop system shall consist of a braided Type 304 stainless-steel flexible hose, zinc-plated steel male threaded nipple or groove style coupling for connection to branch-line piping, and a compatible connection to the sprinkler head.
 4. Flexible drop shall attach to ceiling using a bracket. Bracket shall allow installation before the ceiling tile is in place. Braided drop system is UL-listed for sprinkler services to 175 PSI and FMG-approved to 200 PSI.
 5. Maximum hose length is 6 ft.
 6. All hoses shall be factory-pressure-tested to 400 PSI.

2.2 VALVES

- A. Acceptable Manufacturers: Victaulic or approved equal by Reliable Automatic Sprinkler Co., Tyco Fire Products, or Viking.
- B. Alarm Check Valve and Trim: Victaulic Model 751 Groove X Groove. 4 in., 300 lb. maximum service pressure, ductile iron grooved body and handhole cover, bronze seat ring, aluminum bronze clapper, EPDM rubber clapper facing, retarding chamber with pressure switch, strainers, gauges, fittings, piping, drains, controls, and test valves, for installation in vertical position. Refer to Section 21 10 00 for required pressure switch.

DESIGNER — USE (C) ONLY WHEN A WATER MOTOR GONG IS NOT REQUIRED:

- C. Riser Check Valves and Trim: Victaulic Model 717R. Grooved ends, 4 in., UL-listed/FMG-approved to 200 PSI maximum service pressure, ductile iron body, welded-in nickel seat, EPDM rubber-encapsulated ductile iron clapper, gauges, fittings, piping, drains, and test valves, for installation in vertical position.

2.3 DISCONNECT SWITCHES

- A. NEMA Type 3R, factory-installed, heavy-duty, disconnect switch within the housing of the equipment, for exterior equipment or other NEMA Type required for wet and dusty environments.

2.4 MOTOR STARTERS

- A. NEMA Size 0 minimum, manual or magnetic factory-installed starters as shown on the Drawings, with selector switch or pushbutton as noted, pilot lights where noted, overloads, auxiliary contacts, fuses, and control power transformer, unless unit being controlled includes integral 24-volt control system. **Refer to Section 26 29 13 for additional information.**

2.5 SPRINKLERS

- A. Acceptable Manufacturers: Automatic Sprinkler, Firematic, Globe, Reliable Automatic Sprinkler Co., Tyco Fire Products, Victaulic, or Viking.
- B. Automatic spray, wet or dry, glass bulb, upright, pendent, sidewall, and concealed, directional deflectors, nominal 1/2 in. orifice, and 1/2 in. NPT threaded connection. Die cast brass, with hex-shaped wrench boss integrally cast into body.
- C. Sprinkler types:
- Rooms with ceilings, lobby area, and all public areas:** Victaulic Model V3802. Quick response, 5.6 K-factor, standard coverage, pendent, concealed, 1/2 in. adjustment, and electroplated white finish cover plate.
 - Rooms without ceilings, mechanical rooms, storage rooms, attics, and gymnasiums:** Victaulic Model V2704. Quick response, 5.6 K-factor, standard coverage, chrome-plated, upright, with guard.
 - Bottom of elevator hoistway** (2 ft. above pit floor): Victaulic Model V2710. Quick response, 5.6 K-factor, standard coverage, horizontal sidewall, brass.
 - Option: Extended coverage sprinklers may be provided in lieu of the above sprinkler types, provided that hydraulic calculations establish adequate system flow and pressure.

Mike Y deleted strike-thru in 5 and 6

- Exterior overhangs** (more than 4 ft. wide and more than 55 sq. ft. total area): ~~Victaulic Model V3610~~. Dry barrel type, quick response, 5.6 K-factor, standard coverage, horizontal sidewall, brass.
- Freezers and coolers:** ~~Victaulic Model V3606~~. Dry barrel type, quick response, 5.6 K-factor, standard coverage, recessed pendant, brass.
- Dry barrel type:**
 - Victaulic Model V3610. Quick response, 5.6 K-factor, standard coverage, horizontal sidewall, brass.

- b. Victaulic Model V3606. Quick response, 5.6 K-factor, standard coverage, recessed pendant, brass.
- 8. **Institutional-type**: Tyco pendent Model PH-2, Style B1, or horizontal Model PH-5. Specifically designed for detention facilities and has no projections: Chrome finish with deep cone escutcheon.

2.6 ESCUTCHEON PLATES

- A. **(Semi-recessed) (Flat)**, 1-piece or 2-piece, round, **(chrome-plated) (white) (prime-coated)** metal, unless noted otherwise.

Mike Y deleted B

- ~~B. Flat, round, 2-piece plastic, off white finish: "Ceiling Tite-200."~~

PART 3 - EXECUTION

3.1 COORDINATION

- A. Coordinate water line size requirements for the **limited area** sprinkler system with **(domestic water demand) (Plumbing Work)**. Increase water piping size if required for the **limited area** sprinkler system in accordance with the hydraulic calculations.

3.2 INSTALLATION

- A. Verify that the domestic cold water piping system is adequate to supply the **new** sprinkler system.
- B. Increase piping sizes if required to meet fire suppression needs.
- C. Provide proper size tee and extend outlet of tee as shown on the Drawings.
- D. Do not install wet piping in areas where space temperature may drop below 40 degrees F. Refer to Section 21 10 00 for additional piping requirements.

- E. Option: Install flexible drops only in Manufacturer-approved ceiling types.
1. Where installing in a ceiling grid, grid shall be a suspended medium or heavy load grid, shall meet ASTM C635/C635M construction, and shall be installed in accordance with ASTM C635/C635M requirements.
 2. Any flexible drop hoses kinked or bent beyond the FMG-recommended 8 in. bend radius shall be replaced with new hoses.
- F. Where any wet sprinkler piping is installed above ceilings, and below ceiling insulation, ensure that the ceiling insulation has no insulation voids above the piping.
- G. Install piping and sprinklers as high as possible exposed in areas without ceilings.
- H. Provide shielding to prevent water from reaching electrical switchgear, computer equipment, electrical panels, or other equipment.
- I. Install drainpipes at base of sprinkler riser. Locate Inspector's Test and Drain Assembly at remote end of system. Pipe to nearest drain or outside to splash block. Discharge from drain/test pipes shall be at location approved by the A/E and Owner.
- J. All drain and test piping passing through an exterior wall shall be galvanized or non-ferrous to prevent staining the exterior wall by rust.
- K. Valves:
1. Provide wet-pipe fire alarm valve.
 2. Coordinate location of wet-pipe fire alarm valve risers and piping with equipment.
- change to a lead-free seal?**
3. Chain valves open or provide a **lead** seal.
 4. Provide access panels where necessary to permit access to check valves.
- L. Motor Starters:
1. Provide factory-installed starters only where specifically listed in Division 21 Specification Sections.
 2. Electrical Contractor shall provide all other starters.
 3. On all starters provided by the Fire Suppression Contractor, if motor starter **heater** element's designated amperage is more than 110% of the full load current after balancing, the Fire Suppression Contractor shall provide new starter **heater** elements.

M. Sprinklers:

1. Sprinkler layouts shall be subject to review and approval by the A/E.
2. Density requirements:
 - a. Verify Occupancy Hazard for all areas, and minimum density requirements and sprinkler spacing of each area based on stated occupancies, in accordance with **2022 NFPA 13**.
 - b. **New Office Area (Light Hazard Occupancy)**: Minimum density of 0.10 GPM per sq. ft. over the most remote 1,500 sq. ft. In hydraulically calculated systems, each sprinkler shall have a maximum allowable protection area of 225 sq. ft. per sprinkler; maximum spacing of 15 ft. between sprinklers.
 - c. **Storage, Mechanical, and Factory Area (Ordinary Hazard Occupancy Group 1)**: Minimum density of 0.15 GPM per sq. ft. over the most remote 1,500 sq. ft. Protection area of 130 sq. ft. per sprinkler. Maximum spacing of 15 ft. between sprinklers.
 - d. **Garage and Shop Area**: Provide sprinkler coverage to meet the minimum density requirements of **2022 NFPA 13** for Ordinary Hazard, Group 2 occupancies.
 - e. **Special Storage Areas**: To provide sprinkler coverage for high-piled (over 12 ft.) storage areas and lower, depending on the medium, comply with all NFPA requirements.
 - f. Higher densities and closer spacing shall be in accordance with Code; refer to **2022 NFPA 13**, Tables 8.6.2.2.1(a) and (b).

DESIGNER — NOT REQUIRED FOR POTABLE WATER SYSTEMS (3):

3. Install return bends to all pendent-type sprinklers where required by **2022 NFPA 13** to prevent collection of sediment in sprinklers.
4. Install additional sprinklers below platforms, piping, ductwork, bus ducts, or similar items that obstruct an area below the ceiling or roof deck larger than 4 ft. x 4 ft.
5. Ceiling temperature coordination: Install sprinklers having temperature ratings in accordance with **2022 NFPA 13**. Select the proper sprinkler temperature ratings for the expected ambient temperature of the rooms or areas.
6. **Locate sprinklers generally in center of ceiling tiles (and in a symmetrical pattern related to room dimensions and to ceiling features such as beams, lights, diffusers, etc.) (at 1 ft. increments, and centered between rows of lighting fixtures whenever possible.)**
7. Do not install sprinklers that have been dropped or damaged, or that show a visible loss of fluid. DO NOT USE SPRINKLERS WITH CRACKED BULBS.
8. Sprinkler bulb protector shall remain in place until the sprinkler is completely installed. Remove bulb protector by hand, after installation is complete and prior to system being placed in service. DO NOT USE TOOLS TO REMOVE THE BULB PROTECTOR.

9. Dry-type sprinklers: Install:

- a. In coolers and freezers.
- b. In overhangs or other exterior areas.
- c. In areas shown on the Drawings.
- d. Where ambient temperature could be less than 40 degrees F.
- e. Option: Dry system.

N. Escutcheon Plates: Install where sprinklers protrude from ceiling, wall, or soffit. Match existing escutcheon plates.

O. Sprinkler Cabinet: Provide on Mechanical Room wall, 5 ft.-0 in. above floor (near sprinkler riser. Confirm exact location with) (in location designated by) the Owner and A/E.

P. Electrical Contractor shall complete the applicable sections of start-up checklists for all Fire Suppression equipment.

3.3 TESTING

A. Perform all tests before any piping is covered or concealed.

B. Arrange and schedule all inspections required by the AHJ.

C. Purge limited area sprinkler system of air.

D. Test the limited area sprinkler system hydrostatically at 225 PSIG for (2) hours with no drop in pressure. This test shall occur before final tie into existing system. Measure the hydrostatic test pressure at the low point of the individual system or zone being tested. There shall be no visible leakage during the hydrostatic testing.

E. Should leaks occur, remove each defective section or defective fitting and replace with new materials at no cost to the Owner. Repeat tests until no leaks occur.

F. Do not use brine or other corrosive chemicals for testing systems.

G. Make provision for the disposal of water issuing from test outlets to avoid property damage.

3.4 CERTIFICATION

- A. After all tests have been completed, complete and forward the Contractor's Material and Test Certificate, required by 2022 NFPA 13, to the AHJ.

END OF SECTION 21 13 13

SECTION 21 13 16 - DRY-PIPE SPRINKLER SYSTEMS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 21 05 00.73, "Execution of Fire Suppression."
- B. Section 21 05 00.78, "Closeout Submittals for Fire Suppression."
- C. Section 21 05 23, "General-Duty Valves for Water-Based Fire-Suppression Piping."
- D. Section 21 05 29, "Hangers and Supports for Fire-Suppression Piping and Equipment."
- E. Section 21 05 48, "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."
- F. Section 21 10 00, "Water-Based Fire-Suppression Systems."
- G. Section 21 13 13, "Wet-Pipe Sprinkler Systems."
- H. Section 26 29 13, "Enclosed Controllers."

1.02 DESCRIPTION

- A. Provide a hydraulically designed dry-pipe sprinkler piping system where indicated on the Drawings, including piping, sprinklers, valves, air compressor, and specialties, to provide coverage in all areas of the building indicated on the Drawings, including areas under stairs, mezzanines, overhead doors (when open), etc. Hazard classification shall be in accordance with Code.
 - 1. Piping and fittings: Refer to Section 21 10 00 for additional requirements.
 - 2. Valves: Refer to Section 21 05 23.
 - 3. Access panels. Refer to Section 21 05 00.73.
 - 4. Specialties: Refer to Section 21 10 00.
- B. Design, install, and test complete system in accordance with 2022 NFPA 13; perform hydraulic calculations, prepare drawings, and submit to AHJ for approval.
- C. Lead Contractor shall provide site fire suppression water service line.
- D. Connect to flanged site fire suppression water service line at 1 ft. above building floor slab where shown on the Drawings.

- E. Test complete system.

1.03 QUALITY ASSURANCE

A. Standards:

1. American National Standards Institute (ANSI).
2. American Society for Testing and Materials (ASTM).
3. American Society of Mechanical Engineers (ASME).
4. American Welding Society (AWS).
5. Factory Mutual Global (FMG).
6. National Electrical Manufacturers Association (NEMA).
7. National Fire Protection Association (NFPA): 2022 NFPA 13, "Standard for the Installation of Sprinkler Systems."
8. Underwriters Laboratories, Inc. (UL).

1.04 SPARE PARTS

- A. Spare Sprinklers: Provide a sprinkler replacement wrench and a stock of spare sprinklers, (2) minimum of each type, for future use for each type and temperature rating of sprinkler installed, and mount in a metal sprinkler cabinet. Furnish minimum total of spare sprinklers in accordance with the following schedule:

NUMBER OF SPRINKLERS	EXTRA SPRINKLERS
Up to 300	6
301–1,000	12
More than 1,000	24

- B. Refer to Section 21 05 00.78 for additional information regarding spare parts.

PART 2 PRODUCTS

2.01 PIPING AND FITTINGS

- A. Unless noted otherwise, the piping between Fire Department connection and check valve, and the drain lines, shall be galvanized Schedule 30 or Schedule 40 black steel (ASTM A53) as listed in 2022 NFPA 13.

2.02 DRY-PIPE ALARM VALVE AND TRIM

- A. Acceptable Manufacturer: Viking or approved equal by Victaulic.
- B. 175 PSI SWP, cast iron body, malleable iron clapper, Neoprene rubber gasket, brass pins and springs, bronze air plate, Buna N "O" ring, replaceable rubber air seat, 2 in. boss.
 - 1. 4 in. and smaller: Viking Model C-2.
 - 2. 6 in.: Viking Model D-1.
 - 3. 8 in.: Viking Model C-4.
- C. Trim: Include air pressure gauge; 3-way valve; air piping and unions; pressure switch; water pressure gauge; check, gate, globe, and OS&Y valves; drain cup; and all fittings.
- D. Refer to Section 21 10 00 for required pressure switch.
- E. Provide accelerator to meet Code requirements for sprinkler activation times.

2.03 DRY-PIPE VALVE ACCELERATOR

- A. Acceptable Manufacturer: Viking.
- B. Viking Series D-2. Brass body, EPDM diaphragm, stainless steel interior parts, and anti-flooding device.

2.04 DISCONNECT SWITCHES

- A. NEMA Type 3R, factory-installed, heavy-duty, disconnect switch within the housing of the equipment, for exterior equipment or other NEMA Type required for wet and dusty environments.

2.05 MOTOR STARTERS

- A. NEMA Size 0 minimum, manual or magnetic factory-installed starters as shown on the Drawings, with selector switch or pushbutton as noted, pilot lights where noted, overloads, auxiliary contacts, fuses, and control power transformer, unless unit being controlled includes integral 24-volt control system. **Refer to Section 26 29 13 for additional information.**

2.06 SPRINKLERS

- A. Acceptable Manufacturers: Automatic Sprinkler, Firematic, Globe, Reliable Automatic Sprinkler Co., Tyco Fire Products, Victaulic, or Viking.

- B. Automatic spray, wet or dry, glass bulb, upright, pendent, sidewall, and concealed, directional deflectors, nominal 1/2 in. orifice, and 1/2 in. NPT threaded connection.
- C. Sprinkler Types:
 - 1. Rooms with ceilings, lobby area, and all public areas: Victaulic Model V3802. Quick response, 5.6 K-factor, standard coverage, pendent, concealed, 1/2 in. adjustment, and electroplated white finish cover plate.
 - 2. Rooms without ceilings, mechanical rooms, storage rooms, attics, and gymnasiums: Victaulic Model V2704. Quick response, 5.6 K-factor, standard coverage, chrome-plated, upright, with guard.
 - 3. Option: Extended coverage sprinklers may be provided in lieu of the above sprinkler types, provided that hydraulic calculations establish adequate system flow and pressure.

Mike Y says conflicts with 21 13 13, 2.05 C 7

- 4. Exterior overhangs (more than 4 ft. wide with combustible storage below): Victaulic Model V2710. Dry type, quick response, 5.6 K-factor, recessed horizontal sidewall with up to 1/2 in. adjustment.

2.07 ESCUTCHEON PLATES

- A. Flat, 1-piece, round, chrome-plated metal, electroplated white finish, unless noted otherwise.

2.08 AIR COMPRESSOR AND MAINTENANCE DEVICE

- A. Acceptable Manufacturer: Viking.
- B. Viking Model C. Electric motor-driven, air-cooled, single-stage oil-less compressor, complete with check valve, pressure and moisture unloader, SPST pressure switch, motor starter, safety relief valve, cast iron pipe mounting bracket, and filter. Furnish air pressure maintenance device.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Provide sprinklers and piping to provide coverage of entire area indicated.
- B. Determine if the fire suppression water line is adequate to supply the new system.
- C. Provide proper size tee and extend outlet of tee as shown on the Drawings.

- D. Install piping and sprinklers as high as possible exposed in areas without ceilings.
- E. Provide shielding to prevent water from reaching electrical switchgear, computer equipment, electrical panels, or other equipment.
- F. Install drainpipes at base of sprinkler riser. Locate Inspector's Test and Drain Assembly at remote end of system. Pipe to nearest drain or outside to splash block. Discharge from drain/test pipes shall be at location approved by the A/E and Owner.
- G. All drain and test piping passing through an exterior wall shall be galvanized or non-ferrous to prevent staining the exterior wall by rust.
- H. Valves:
 - 1. Provide dry-pipe fire alarm valve.
 - 2. Coordinate location of dry-pipe alarm valve risers and piping with equipment.
- I. Install air piping, valves, air gauges, and specialties associated with the dry-pipe alarm valve, air compressors, and accelerator.
- J. Motor Starters:
 - 1. Provide factory-installed starters only where specifically listed in Division 21 Specification Sections.
 - 2. Electrical Contractor shall provide all other starters.
 - 3. On all starters provided by the Fire Suppression Contractor, if motor starter heater element's designated amperage is more than 110% of the full load current after balancing, the Fire Suppression Contractor shall provide new starter heater elements.
- K. Sprinklers:
 - 1. Sprinkler layouts shall be subject to review and approval by the A/E.
 - 2. Density requirements:
 - a. Verify Occupancy Hazard for all areas, and minimum density requirements and sprinkler spacing of each area based on stated occupancies, in accordance with 2022 NFPA 13.
 - b. New Office Area (Light Hazard Occupancy): Minimum density of 0.10 GPM per sq. ft. over the most remote 1,500 sq. ft. In hydraulically calculated systems, each sprinkler shall have a maximum allowable protection area of 225 sq. ft. per sprinkler; maximum spacing of 15 ft. between sprinklers.
 - c. Factory Area (Ordinary Hazard Occupancy Group 1): Minimum density of 0.15 GPM per sq. ft. over the most remote 1,500 sq. ft. Protection area of 130 sq. ft. per sprinkler. Maximum spacing of 15 ft. between sprinklers.

DRY-PIPE SPRINKLER

SYSTEMS

- d. **Garage and Shop Area:** Provide sprinkler coverage to meet the minimum density requirements of **2022 NFPA 13** for Ordinary Hazard, Group 2 occupancies.
 - e. **Special Storage Areas:** For sprinklering high-piled (over 12 ft.) storage areas and lower, depending on the medium, comply with all NFPA requirements.
 - f. Higher densities and closer spacing shall be in accordance with Code; refer to **2022 NFPA 13**, Tables 8.6.2.2.1(a) and (b).
- 3. Ceiling temperature coordination: Install sprinklers having temperature ratings in accordance with **2022 NFPA 13**. Select the proper sprinkler temperature ratings for the expected ambient temperature of the rooms or areas.
 - 4. **Locate sprinklers generally in center of ceiling tiles** and in a symmetrical pattern related to room dimensions and to ceiling features such as beams, lights, diffusers, etc.
 - 5. Do not install sprinklers that have been dropped or damaged, or that show a visible loss of fluid. **DO NOT USE SPRINKLERS WITH CRACKED BULBS.**
 - 6. Sprinkler bulb protector shall remain in place until the sprinkler is completely installed. Remove bulb protector by hand, after installation is complete and prior to system being placed in service. **DO NOT USE TOOLS TO REMOVE THE BULB PROTECTOR.**
- L. Escutcheon Plates: Install where sprinklers protrude from ceiling, wall, or soffit.
- M. Sprinkler Cabinet: Provide on the Mechanical Room wall, 5 ft.-0 in. above the floor near the fire riser. Confirm exact location with the Owner and A/E.
- N. Air Compressor and Maintenance Device:
- 1. Mount on floor or riser piping.
 - 2. If floor-mounted, install air compressor on 4 in. high concrete pad. Coordinate location and requirements with the Lead Contractor. Refer to **Section 21 05 29** regarding concrete requirements.
 - 3. Coordinate location, horsepower, and voltage requirements with the Electrical Contractor.
 - 4. Include low-pressure switch and motor starter.
 - 5. Electrical Contractor shall provide:
 - a. Fail-safe or emergency power.
 - b. Disconnect switch.
 - c. Control wiring.
 - d. Power wiring.

- O. Electrical Contractor shall complete the applicable sections of start-up checklists for all Fire Suppression equipment.

3.02 TESTING

- A. Test complete system in accordance with 2022 NFPA 13.
- B. Test air piping, compressor, and operation of dry-pipe valve.
- C. Perform all tests before any piping is covered or concealed.
- D. Arrange and schedule all inspections required by the AHJ.
- E. Test the sprinkler system hydrostatically at 225 PSIG for (2) hours with no drop in pressure. Measure the hydrostatic test pressure at the low point of the individual system or zone being tested. There shall be no visible leakage during the hydrostatic testing.
- F. Should leaks occur, remove each defective section or defective fitting and replace with new materials at no cost to the Owner. Repeat tests until no leaks occur.
- G. Do not use brine or other corrosive chemicals for testing systems.
- H. Make provisions for the complete draining of water from all sprinkler pipe drops after testing.
- I. Make provision for the disposal of water issuing from test outlets to avoid property damage.

3.03 CERTIFICATION

- A. After all tests have been completed, complete and forward the Contractor's Material and Test Certificate, required by 2022 NFPA 13, to the AHJ.

END OF SECTION 21 13 16